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Working Paper 20-013

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# Improving Customer Compatibility with Tradeoff Transparency

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**Abstract:**
Through a large-scale field experiment with 393,036 customers considering opening a credit card account with a nationwide retail bank, we investigate how providing transparency into an offering's tradeoffs affects subsequent rates of customer acquisition and long-run engagement. Although we find tradeoff transparency to have an insignificant effect on acquisition rates, customers who were shown each offering's tradeoffs selected different products than those who were not. Moreover, prospective customers who experienced transparency and subsequently chose to open an account went on to exhibit higher quality service relationships over time. Monthly spending was 9.9% higher and cancellation rates were 20.5% lower among those who experienced transparency into each offering's tradeoffs. Increased product usage and retention accrued disproportionately to customers with prior category experience – more-experienced customers who were provided transparency spent 19.2% more on a monthly basis and were 33.7% less likely to defect after nine months. Importantly, we find that these gains in engagement and retention do not come at the expense of customers' financial wellbeing – the probability of making late payments was reduced among customers who experienced transparency. We further find that the positive effects of tradeoff transparency on engagement and retention were attenuated in the presence of a promotion that provided financial incentives to choose particular offerings. Taken together, these results suggest that providing transparency into an offering's tradeoffs may be an effective strategy for informing customer choices, leading to better outcomes for customers and firms alike.

*[Keywords: Behavioral operations, tradeoff transparency, customer compatibility, customer behavior]*

## 1. Introduction

It is a common practice that when service companies market their offerings to prospective customers, they emphasize the advantages of each offering and downplay its tradeoffs. This strategy seems quite rational to the extent its objective is to convert browsers into buyers, as accentuating the downsides of an offering might drive some customers away. However, since value in many service settings arises from the long-term engagement and retention of satisfied customers, marketing messages that provide transparency into a service offering's tradeoffs (both its advantages *and its corresponding disadvantages*) may help customers make better-informed choices, facilitating the acquisition of more compatible customers, and improving long-run outcomes for customers and firms alike. Although this proposition may have intuitive appeal, the effect of providing prospective customers with transparency into an offering's tradeoffs has never been empirically investigated. In this paper, we introduce the concept of *tradeoff transparency*, which we define as voluntarily marketing the downsides of an offering with similar emphasis as its corresponding advantages, and we present the results of a large-scale field experiment in which we tested the impact of tradeoff transparency on customer acquisition and long-term engagement.

In service contexts, customers, who have heterogeneous needs and preferences (Frei 2006), and who provide key inputs to operating processes (Roels 2014, Sampson and Froehle 2006), are typically responsible for selecting the companies and service offerings with which they engage. This selection process is fraught with information asymmetry (Buell et al. 2016, Schmidt and Buell 2017), so much so that services have been classified as "experience goods," in that customers are rarely able to accurately assess their characteristics prospectively (Israel 2005). Yet service outcomes, like satisfaction and profitability, have been shown to hinge largely on customer compatibility – the degree of fit between the customer's individual needs and preferences, and the service offering's attributes (Buell, Campbell, et al. 2020). Hence, initiatives that reduce information asymmetry by making the tradeoffs of service offerings clearer to prospective customers may help those customers make better-informed decisions, leading to better long-term customer engagement. On the other hand, emphasizing an offering's tradeoffs may drive customers away, reducing customer acquisition, and undermining company performance.

To investigate this tension, we collaborated with Commonwealth Bank of Australia (CBA), the largest bank in the southern hemisphere (Buell and John 2018), and conducted a field experiment with 393,036 customers who were considering applying for bank-issued credit cards. Bank-issued credit cards are a relevant setting for this research because 1) credit cards vary in their design parameters, such that some offerings may be a better fit for an individual customer's needs and preferences than others, 2) credit card marketing typically emphasizes the advantages of each offering, and 3) customer engagement with credit cards (e.g., spend, late payments, retention, etc.) is a meaningful driver of long-term performance. Similar dynamics exist in a broad array of service contexts, where long-term outcomes are influenced by the engagement dynamics of heterogeneous customers. We note that since banks like CBA are legally required to provide customers with disclosure statements that reveal the terms and conditions of their offerings, this setting may be a conservative domain in which to study these effects. Afterall, tradeoff transparency in this context merely increases the salience of information about each offering's downsides, rather than presenting previously-undisclosed information. However, we note that even in settings where such disclosures are not mandated, online reviews that reveal the otherwise-hidden downsides experienced by prior purchasers are often available to customers. Hence, in most relevant contexts, as in this study, tradeoff transparency increases the salience of an offering's downsides.

In our field experiment, each prospective credit card customer was randomly assigned to either a control condition, in which they received traditional marketing messaging that emphasized the advantages of each credit card (e.g., our lowest interest rate on purchases, includes travel insurance, etc.), or a treatment condition, in which they received messaging that emphasized the same advantages, while also providing transparency into each card's tradeoffs (e.g., carries a higher annual fee, does not earn awards points, etc.).

We subsequently tracked the progression of customers in both conditions through the acquisition funnel and documented the engagement of customers who chose to open accounts for nine months after activation, to assess the impact of tradeoff transparency on service performance.

Although we find that providing tradeoff transparency in this setting had no impact on overall rates of customer acquisition, we find that it had a meaningful effect on customers' choices, usage, and retention. Holding constant other factors, customers who experienced tradeoff transparency selected a different mix of cards, spent 9.9% more on a monthly basis, and after nine months, were 20.5% less likely to cancel their accounts. Importantly, this increased engagement did not come at the expense of customers' financial wellbeing – the probability of making late payments declined 10.8% in the initial six months of the service relationship among those who experienced tradeoff transparency. We further find evidence that the effects of tradeoff transparency on customer engagement are especially acute among customers with prior-category experience. Providing tradeoff transparency to below-median age prospects, who had limited, if any, previous experience with credit cards, had a de minimis effect on usage and retention. In contrast, providing tradeoff transparency to older, more-experienced customers, led to significantly higher levels of engagement – increasing monthly spending by 19.2% and decreasing the probability of cancellation after nine months by 33.7%. Finally, we observe that the effects of tradeoff transparency on subsequent usage and retention are attenuated among customers who were attracted by a promotion. Midway through the experimental period, the bank introduced a financial incentive for customers who opened specific types of credit cards, and customers who opened accounts during this promotion period exhibited a statistically indistinguishable response to the treatment. The results are consistent with the idea that customers who organically seek out an offering are more likely to consider and benefit from tradeoff transparency than those who are attracted or influenced by a promotion. By examining both the short-term marketing impact and the longer-run operational impact of providing prospective customers with tradeoff transparency, the present study lends support to prior research calling for an integrated approach to marketing and operations (Ho and Zheng 2004). Moreover, it builds on a growing literature that demonstrates how voluntarily revealing facets of an operation that are traditionally kept hidden may in some cases improve outcomes for customers and service providers alike (Buell et al. 2017, Mohan et al. 2020).

## 2. Tradeoff transparency, customer acquisition, and engagement

In this paper, we define tradeoff transparency as voluntarily marketing the downsides of an offering with similar emphasis as its corresponding advantages. As such, the concept of tradeoff transparency is closely related to prior research on different types of customer-facing transparency, information provision, and information salience, all of which have been closely studied in the operations and marketing literatures. In these contexts, transparency can be characterized as operating in a manner that makes it easy for others to observe what actions are being performed. When engaging customers, organizations may opt for transparency into operations (Buell et al. 2017, Buell, Porter, et al. 2020), costs (Mohan et al. 2020), sustainability initiatives (Buell and Kalkanci 2020, Kalkanci et al. 2018), governance practices (Kosack and Fung 2014) and more. Relatedly, information provision and information salience, both of which can be outcomes of transparency, are the acts of conveying new information (Cabral et al. 2010, Luca 2016), or increasing the prominence of disclosed information (Bollinger et al. 2011, Chetty et al. 2009, Luca and Smith 2013), respectively. In the bank-issued credit card context, where regulations stipulate that comprehensive information be made available about the terms and conditions of various offerings, tradeoff transparency may be most accurately characterized as increasing the salience of information about a product's tradeoffs, although numerous legal and online communication studies have demonstrated that very few customers read such statements (Ayres and Schwartz 2014, Obar and Oeldorf-Hirsch 2020). For example, in one study that tracked the behavior of 48,154 households across 90 websites, fewer than 0.2% accessed the terms of service, and most of those who access it read only a small portion (Bakos et al. 2014). Consequently, a service offering's downsides, promoted through tradeoff transparency, are more likely to be noticed, and in turn factored into a customer's purchase decision, than if those tradeoffs are solely presented in the text of the terms and conditions. In this section, we explore literature that informs how providing prospective customers with tradeoff transparency may affect customer acquisition and long-term engagement. Because of the conflicting findings among a portion of the relevant studies we cite, we have adopted the convention of stating non-directional hypotheses in null form and directional hypotheses in alternative form.

## 2.1 Tradeoff Transparency and Customer Acquisition

Research has highlighted how the revelation of unflattering information about a company's service offerings, such as poor quality or high prices, can reduce consumer demand. For example, research conducted in online platforms found that the first bad review posted about a firm's offerings can cost a company up to 13% of its revenue (Cabral et al. 2010) and that a one-star increase on Yelp can boost a company's revenue by 9% on average (Luca 2016). Chetty et al. (2009) found that making taxes salient in a retail context – that is, posting sales taxes along with the prices of goods – increases consumers' perceptions of the prices charged by the retailer, and reduces demand by 8%. Similarly, Finkelstein (2009) found that drivers who pay tolls electronically are substantially less aware of, and less sensitive to, toll rate increases, and Blake et al. (2017) illustrated that shrouding service fees on an online ticketing platform can substantially raise revenues. To the extent that disclosing sensitive information can adversely affect demand, these results suggest that firms intent on acquiring new customers may have powerful incentives to hide negative information from prospective buyers.

On the other hand, an extensive array of research in psychology and marketing has demonstrated the benefits of self-disclosure in fostering intimacy. Interpersonally, self-disclosure has been shown to be critical in developing social relationships (Jourard 1971) and in forging a heightened level of trust (Wheeless and Grotz 1977). People disclose in intimate relationships (Derlega 1984), as self-disclosure allows parties to gain knowledge about one another while reducing ambiguity about the other's intentions (Laurenceau et al. 1998, Perlman and Fehr 1987). Similar effects have been documented in non-interpersonal contexts. For example, research has shown how people become more attracted to, and more willing to engage with, computers that self-disclose sensitive information. If a computer shares that its "abilities are really limited…" in that it can "do word processing and spreadsheets, but it cannot do any kind of physical activity, like play sports or walk down the street," people report a higher level of attraction to it, and are more likely to engage with it (Moon 2000). Similarly, when companies self-disclose their costs of producing goods and services – a practice which implicitly reveals their profit margins – consumers report trusting the firm more, are more willing to engage with it, and sales increase (Mohan et al. 2020).

Furthermore, research on operational transparency has demonstrated how showing the otherwise-hidden work being performed behind the scenes in an operation – can enhance consumers' appreciation for the firm and their perceptions of the value it creates (Buell et al. 2017, Buell and Norton 2011). Moreover, such transparency can bolster trust in and engagement with the firm, even when it reveals imperfect performance (Buell, Porter, et al. 2020, Kalkanci et al. 2018). Common across all of these examples, however, is the notion of voluntary self-disclosure. Indeed, trust and engagement are not engendered when the revelation is involuntary (Hoffman-Graff 1977, Mohan et al. 2020). Taken together, the net effects on acquisition rates of a firm providing transparency into the tradeoffs inherent in its offerings are equivocal. The revelation of unflattering information may suppress demand, but voluntary self-disclosure may engender higher levels of trust and brand attraction. Due to these offsetting effects, we formulate the following hypothesis in null form:

*Hypothesis 1 (H1):* Providing prospective customers with tradeoff transparency will have no effect on the firm's rate of customer acquisition.

## 2.2 Tradeoff Transparency and Customer Engagement

It is well established that customer satisfaction plays a key role in a firm's long-term performance. Research has demonstrated that highly-satisfied customers are more loyal and more profitable over time

(Anderson 1994, Anderson et al. 1994, Heskett et al. 1997, LaBarbera and Mazursky 1983). Satisfaction levels, in turn, are largely persistent between firms and individual customers, and are influenced in part by customer compatibility – the degree of fit between the needs of the customer and the capabilities of the operation (Buell, Campbell, et al. 2020). Customer compatibility can be influenced through a firm's segmentation strategies, wherein offerings are designed around the shared needs of particular operating segments of customers (Guajardo and Cohen 2017). However, the efficacy of such strategies is likely influenced by the degree of transparency firms provide to prospective customers into how those offerings are designed. Customers that are misaligned with a segmented service offering will be less satisfied with their experiences (Buell et al. 2016, Buell, Campbell, et al. 2020), and will impose more variability on the operation (Frei 2006) which, in turn, will hinder the firm's service performance (Karmarkar and Pitbladdo 1995, Chase and Tansik 1983).

Providing prospective customers with tradeoff transparency could improve customer compatibility in two ways: 1) by reducing Type I errors (false positives); and, 2) by reducing Type II errors (false negatives). In the context of customers choosing among service offerings, a Type I error occurs when a customer chooses an offering that is poorly aligned with his or her needs. When firms communicate the positive attributes of their offerings and omit the negative attributes, research in psychology has demonstrated that consumers often fill in the gaps in biased and overly favorable ways, and are more likely to choose the offering (Kivetz and Simonson 2000). Since such customers enter a service relationship with expectations that surpass the capabilities of the operation, the gap between their service expectations and experiences will result in dissatisfaction over time (McDougall and Levesque 2000, Oliver 1993, Parasuraman et al. 1985, Tse and Wilton 1988). In contexts with high switching costs, dissatisfied customers may remain with the firm (E. Anderson 1994, Buell et al. 2010, Yang and Peterson 2004), but are likely to spend less money than more highly-satisfied customers and may be more difficult and costly to serve (Coyles and Gokey 2005, Jones and Sasser 1995, Xue and Harker 2002). To the extent that providing prospective customers with tradeoff transparency can help reduce Type I errors, it stands to reason that the customers who choose the offering may be more aligned and satisfied with it, and may be more loyal and profitable to the firm.

On the other hand, in the context of customers choosing among service offerings, a Type II error occurs when a customer fails to select an offering that is well-aligned with his or her needs. Services have been characterized as "experience goods," since empirical research has demonstrated how customers are unable to fully assess them until after they have been delivered. Although providing consumers with incomplete information can trigger a psychological bias causing them to assume the best (Kivetz and Simonson 2000), when the lack of transparency is assumed to be volitional, people may assume the worst, undermining their trust and engagement (John et al. 2016). In this way, tradeoff transparency may also result in fewer Type II errors, further promoting service relationships that are value producing and satisfying for customers, and long-lived and highly profitable for firms (Heskett et al. 1997). To the extent that the information afforded by tradeoff transparency helps reduce Type I and Type II errors, we hypothesize that prospective customers who are provided with tradeoff transparency will select different offerings than those who are not:

> *Hypothesis 2 (H2):* Providing prospective customers with tradeoff transparency will result in the selection of different offerings than would be selected in the absence of such transparency.

We note that testing whether customers make *different* choices in the presence of tradeoff transparency, as hypothesized in H2, is not the same thing as testing whether customers make *better* choices in the presence of tradeoff transparency (i.e., whether customers select offerings that are better aligned with their needs and preferences). However, testing H2 is an important interim step, since if customer choices were not affected in any way by the presence of tradeoff transparency, then it could not follow that their choices were improved in its presence. Indeed, to the extent that the provision of tradeoff transparency reduces Type I and Type II errors, we would expect customer choices to not only be different, but to also be better in its presence than in its absence. Assessing the quality of individual customer choices on the basis of which offerings they choose could be compromised by the fact that customers may be making well-reasoned choices based on unobservable criteria. Indeed, part of what managers found attractive about the idea of tradeoff transparency, relative to other more prescriptive methods of influencing customer choices, is that it had the potential to empower customers, who have a much richer understanding of their own needs and preferences, to make better-informed decisions. In our experiment, we are able to directly observe how customers in the treatment and control conditions engage with the credit cards they choose over time, which serve as behavioral indicators of the long-run quality of their choices. We hypothesize that usage and retention will increase among customers who experience tradeoff transparency, since we predict that the offerings they will choose in the presence of tradeoff transparency will be better-aligned with their needs and preferences than the offerings they will choose in its absence:

> *Hypothesis 3 (H3):* Providing prospective customers with tradeoff transparency will increase product usage among those who choose to select the offering, relative to not providing such transparency.

> *Hypothesis 4 (H4):* Providing prospective customers with tradeoff transparency will increase product retention among those who choose to select the offering, relative to not providing such transparency.

Although we anticipate the main effects on usage and retention described above, prior research in the marketing domain has demonstrated that customers who are more familiar with a product category have the capacity to learn more from technical advertising (R. Anderson and Jolson 1980), and that more-experienced consumers, by virtue of their familiarity, are better able to select information about attributes that are more predictive of product performance, in turn leading to better decisions that are more aligned with their needs (Johnson and Russo 1984). Hence, we predict that the long-run benefits of providing prospective customers with transparency into the tradeoffs of a service offering will accrue disproportionately to those with more prior category experience, who can learn from and make better use of the information transparency affords. If transparency can be best leveraged by more-experienced customers to make choices that are better aligned with their needs, we would by extension predict that their levels of engagement would be disproportionately impacted. Accordingly, we hypothesize:

*Hypothesis 5 (H5):* The positive effect of tradeoff transparency on product usage will be most acute for customers who have more prior experience with the service category.

*Hypothesis 6 (H6):* The positive effect of tradeoff transparency on product retention will be most acute for customers who have more prior experience with the service category.

Lastly, a rich theoretical literature in economics and operations has modelled how customers sort among competing offerings, trading off price and service quality (Cachon and Harker 2002, Cohen and Whang 1997, Gabszewicz and Thisse 1979, Gans 2002, Hall and Porteus 2000, Karmarkar and Pitbladdo 1997, Li and Lee 1994, Shaked and Sutton 1982, Sutton 1986, Tirole 1990, Tsay and Agarwal 2000). Subsequent empirical work has documented that when service offerings are differentiated against the competitive set on the basis of quality, they attract more quality-sensitive customers, who are more likely to defect to higher-quality offerings. When offerings are instead differentiated on the basis of price, firms attract price-sensitive customers, who are less sensitive to quality, and more likely to defect to lower-priced offerings (Buell et al. 2016). Consistently, we predict that the effect of tradeoff transparency on customer engagement and retention will be negatively influenced by the presence of a promotion. Customers attracted to an offering on the basis of a discount or financial incentive may be less sensitive to the quality of the service experience than they are to the incentive itself, and hence, the benefits of increased alignment between their needs and the capabilities of the offering they select afforded by transparency are less likely to accrue to them. Moreover, to the extent that conflicts arise between the offering that's the best fit for a customer, and the promotional incentives provided to select it, we would further expect the presence of a promotion to diminish the impact of transparency on engagement and retention. Thus, we hypothesize:

*Hypothesis 7 (H7):* The positive effect of tradeoff transparency on product usage will be diminished among customers attracted to the offering by a promotion.

*Hypothesis 8 (H8):* The positive effect of tradeoff transparency on product retention will be diminished among customers attracted to the offering by a promotion.

## 3. Presentation of field experiment

To conduct this research, we partnered with Commonwealth Bank of Australia (CBA), which at the time of the experiment, had more than 1,000 branches, more than 45,000 employees, and more than 10 million retail banking customers. CBA was the largest issuer of credit cards in its market, with over 3 million credit card holders and annual transaction volume exceeding $50 billion USD. It offered nine types of personal credit cards, spread across three different families – awards cards, low rate cards, and low fee cards (**Figure 1**). We focus our analysis on customers who shopped for personal credit cards.

### 3.1 Data and methods

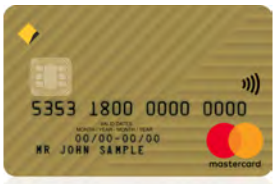
*3.1.1 Participants, design, and procedure.* From September 6, 2017 through February 4, 2018, we collaborated with CBA to conduct a field experiment, engaging all customers who visited the bank's credit card marketing website. 466,322 customers were randomly assigned to one of two experimental conditions.

Customers randomly assigned to the control condition, $TREAT_i$ =0, observed a version of the website that was consistent with the bank's traditional marketing efforts – emphasizing the features and benefits of each credit card in its primary copy. Customers randomly assigned to the treatment condition, $TREAT_i$ =1, observed an augmented version of the website, in which the primary copy additionally revealed the tradeoffs inherent in each offering (**Figure 2**). For example, customers in the control condition who shopped the low fee credit card would be able to view and learn about the card's benefits: $0 annual fee in the first year, free additional card holders, and up to 55 days interest-free on purchases. Furthermore, they would be able to view information about rates and fees, and about the security features of the card.

Customers randomly assigned to the treatment condition would additionally be able to view the tradeoffs inherent in the offering: (*i*) the purchase interest rate of 19.74% p.a. is higher than that of the low rate card, (*ii*) the low fee credit card does not include travel insurance, and (*iii*) there is an annual fee of $29 if the cardholder does not spend at least $1,000 on the card. The tradeoffs that were identified for display on the website were selected based on reviewing how the credit cards compared with one another, and finding the disadvantages of each, in terms of fees, limits, interest rates, and other features that had been identified by customers as important determinants of their service experiences interacting with the cards. We worked with the bank's copywriters, product, and compliance teams to arrive at the final copy that was used on the website, because it was important that the tradeoff transparency language complied with financial regulations, was devoid of jargon, was simple, and was brief.

**Low Rate Cards**

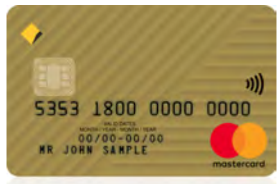

Low rate card
$59 annual fee
13.24% interest rate
21.24% cash advance interest rate




Low rate gold card
$89 annual fee
13.24% interest rate
21.24% cash advance interest rate
Includes international travel insurance

**Low Fee Cards**


Low fee card
$0 annual fee first year
$29 annual fee if spending<$1,000
19.74% interest rate
21.24% cash advance interest rate




Low fee gold card
$0 annual fee first year
$89 annual fee if spending<$10,000
19.74% interest rate
21.24% cash advance interest rate
Includes international travel insurance

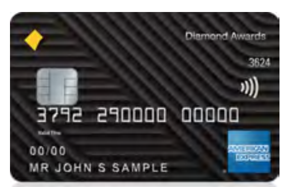

Student card
$0 annual fee first year
$29 annual fee if spending<$1,000
19.74% interest rate
21.24% cash advance interest rate
Interest free for 55 days

**Awards Cards**


Awards card
$59 annual fee first year
20.24% interest rate
21.24% cash advance interest rate
Earn 1.5 points per dollar on purchases
Includes Master Card and American Express


Gold awards card
$119 annual fee first year
20.24% interest rate
21.24% cash advance interest rate
Earn 2 points per dollar on purchases
Includes travel insurance
Includes Master Card and American Express


Platinum awards card
$249 annual fee first year
20.24% interest rate
21.24% cash advance interest rate
Earn 2.5 points per dollar on purchases
Includes travel insurance
Includes Master Card and American Express




Diamond awards card
$349 annual fee first year
20.24% interest rate
21.24% cash advance interest rate
Earn 3 points per dollar on purchases
Includes travel insurance
Includes lounge access
Includes Master Card and American Express

**Figure 1.** Credit card offerings from CBA during the time of the experiment. The bank offered cards in three families: 1) low rate cards, 2) low fee cards, and 3) awards cards. Because the credit card offerings were designed to serve customers with varying needs and preferences, the value proposition of each credit card included both benefits and tradeoffs, making it more appropriate for some prospective customers and less appropriate for others.

| | **Benefits** (included in treatment and control) | **Tradeoffs** (included in the treatment only) |
|---|---|---|
| **Low Rate** | • Our lowest interest rate on purchases, currently 13.24% p.a.<br>• Minimum credit limit of $500<br>• Free additional card holder | • The annual fee of $59 is higher than our Low Fee credit card<br>• Does not earn awards points<br>• Does not include travel insurance<br>• International purchases may incur international transaction fees |
| **Low Rate Gold** | • Our lowest interest rate on purchases, currently 13.24% p.a.<br>• International travel insurance included<br>• Minimum credit limit of $4,000<br>• Free additional card holder | • There's an annual fee of $89<br>• Does not earn awards points<br>• International purchases may incur international transaction fees |
| **Low Fee** | • $0 annual fee in the first year* and following years when you spend at least $1,000 in the previous year<br>• Free additional card holder | • Does not earn awards points<br>• Does not include travel insurance<br>• There's an annual fee of $29 in the second and later years if you spend less than $1,000* in the previous year<br>• International purchases may incur international transaction fees |
| **Low Fee Gold** | • $0 annual fee in the first year* and following years when you spend at least $10,000 in the previous year<br>• International travel insurance included<br>• Free additional card holder | • The purchase interest rate of 19.74% p.a. is higher than Low Rate credit cards<br>• Does not earn awards points<br>• There's an annual fee of $89 in the second and later years if you spend less than $10,000* in the previous year<br>• International purchases may incur international transaction fees |
| **Student** | • $0 annual fee in the first year**<br>• Free additional card holder | • The purchase interest rate of 19.74% p.a. is higher than Low Rate credit cards<br>• Does not earn awards points<br>• Does not include travel insurance<br>• International purchases may incur international transaction fees |
| **Awards** | • Earn up to 1.5 Awards points for every dollar you spend<br>• Access to the largest rewards program of any bank in Australia* | • The purchase interest rate of 20.24% p.a. is higher than Low Rate and Low Fee credit cards<br>• There's an annual fee of $59<br>• Does not include travel insurance<br>• International purchases may incur international transaction fees<br>• There's a $10 p.a. additional cardholder fee |
| **Gold Awards** | • Earn up to 2 Awards points for every dollar you spend<br>• Access to the largest rewards program of any bank in Australia*<br>• International travel insurance included when you activate cover within [our online banking platform] | • The purchase interest rate of 20.24% p.a. is higher than Low Rate and Low Fee credit cards<br>• There's an annual fee of $119<br>• International purchases may incur international transaction fees<br>• There's a $10 p.a. additional cardholder fee |
| **Platinum Awards** | • Earn up to 2.5 Awards points for every dollar you spend<br>• Access to the largest rewards program of any bank in Australia*<br>• International travel insurance included when you activate cover within [our online banking platform]<br>• No international transaction fees on overseas purchases made in-store and online using your CommBank Platinum American Express card | • The purchase interest rate of 20.24% p.a. is higher than Low Rate and Low Fee credit cards<br>• There's an annual fee of $249<br>• There's a $10 p.a. additional cardholder fee |
| **Diamond Awards** | • Earn up to 3 Awards points for every dollar you spend<br>• Access to the largest rewards program of any bank in Australia*<br>• International travel insurance included when you activate cover within [our online banking platform]<br>• No international transaction fees on overseas purchases made in-store and online using your CommBank Diamond American Express card | • The purchase interest rate of 20.24% p.a. is higher than Low Rate and Low Fee credit cards<br>• There's an annual fee of $349<br>• There's a $10 p.a. additional cardholder fee |

**Figure 2.** Website copy used in the treatment and control conditions for each card. Customers randomly assigned to the control condition only experienced the marketing of each credit card's benefits, which is common practice in the industry. Customers randomly assigned to the treatment condition were additionally presented with transparency into the tradeoffs of each offering.

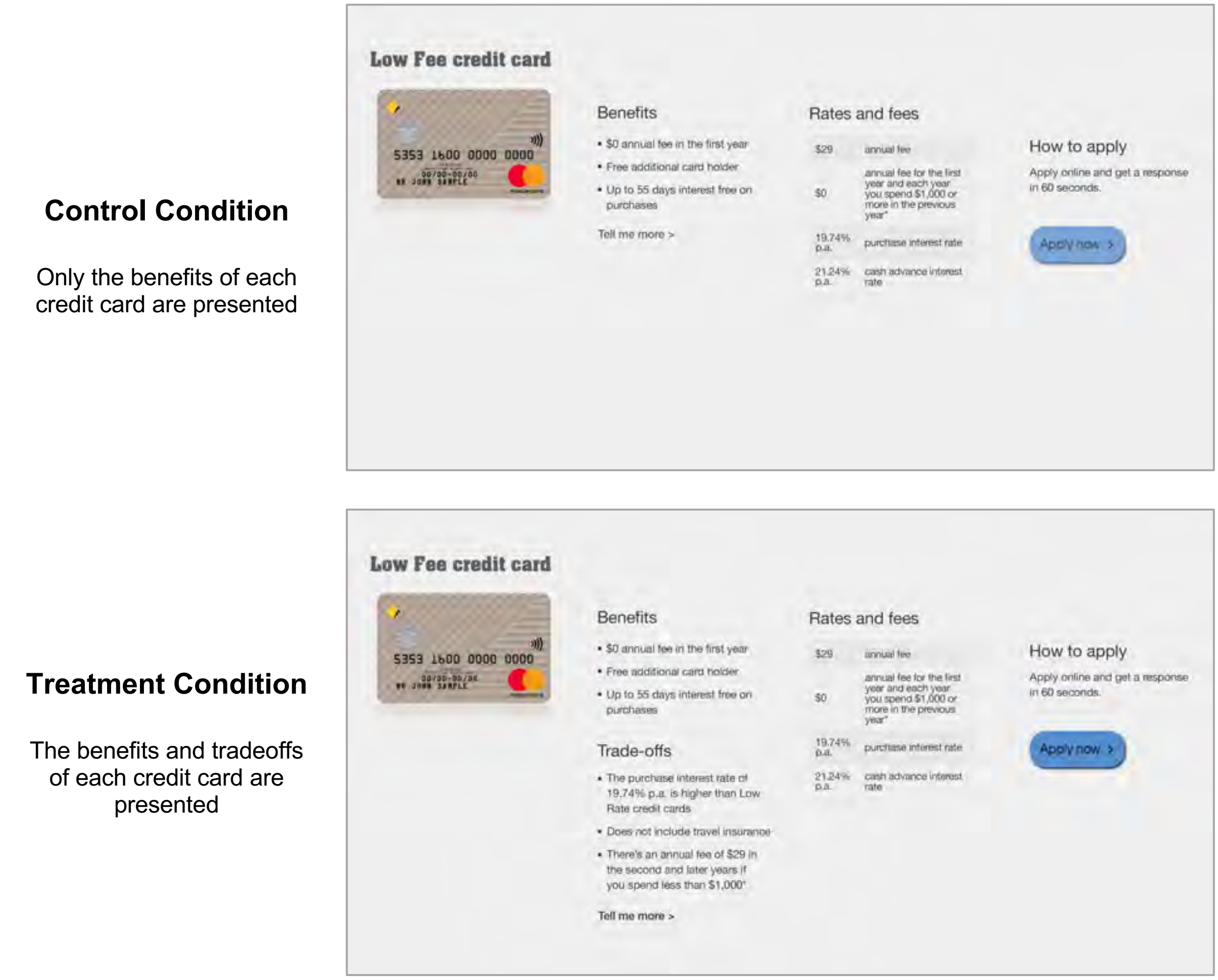



**Figure 3.** Example credit card product detail pages in the control and treatment conditions. The experimental manipulation involved more than 50 blocks of content on more than 20 pages of CBA's public-facing and secure online banking websites, such that every credit card marketing webpage where the features and benefits of a credit card were described, so too were its tradeoffs for customers randomly assigned to the treatment.

We note that in some cases, tradeoffs were presented in absolute terms, such as "there's a $10 p.a. additional cardholder fee," whereas in other cases, when a feature compared unfavorably with one or more cards, the copy was written to reveal the relative comparison, such as "the annual fee of $59 is higher than our Low Fee credit card." Importantly, as in many other developed economies, the Australian financial services industry is highly regulated. Banks are required by law to disclose the full terms and conditions of each offering in product disclosure statements, which are continuously monitored and assessed by the Australian Securities and Investments Commission (ASIC), the country's integrated corporate, markets, financial services, and consumer credit regulator. These statements, which appeared toward the bottom of each page of CBA's credit card website, disclosed information about the tradeoffs of each offering to customers in both conditions. However, the experimental treatment promoted its salience by moving its messaging into the page's primary copy. In this way, the experiment was designed to cleanly identify the effects of increasing the transparency of information about an offering's tradeoffs, on the acquisition rates, choices, and subsequent engagement, of prospective customers (**Figure 3**).

During the time of our experiment, the bank offered three channels through which customers could access information about its credit cards: a public-facing website, a secure online banking website, and a mobile banking application. Our experiment manipulated the presentation of information on the two websites, which historically had accounted for roughly 80% of new credit card applications that came in through digital channels. The mobile application, in which it was infeasible to experimentally manipulate content, contained an abbreviated version of the bank's traditional marketing messaging, emphasizing the features and benefits of each credit card in its primary copy. This content was held constant during the period of our experiment, and as such, we assessed it would not serve as a confound in our analyses. Furthermore, an ex-post analysis revealed very low average visit counts to the mobile credit card marketing content among customers in the experiment, with insignificant differences among customers in the treatment ($M$=0.0205, $SD$=0.1670) and control ($M$=0.0203, $SD$=0.1676) conditions, ($t$=0.3386, $p$=0.734).

For each customer, assignment to an experimental condition was randomly determined at the beginning of their first visit to the credit card marketing section of one of the bank's two websites and was stored in the bank's database. To ensure the consistency of each customer's experience across multiple visits, a cookie was passed from the bank's website to each customer's browser, which identified the customer and could be used to determine whether they should be shown content in the treatment or control condition. Using this approach facilitated tracking and analyzing the behavior of customers as they shopped for credit cards, even for those whose personal identities were unknown. If customers subsequently logged into their online banking accounts, or initiated a credit card application, their randomly-assigned conditions were associated with their personal identities in the bank's database. Although this design largely ensured consistent presentation of content across multiple sessions, 44,765 customers were identified to have experienced multiple conditions and were dropped from the analysis. Experiencing multiple conditions could take place, for example, if a customer visited the credit card website from multiple computers, if their browser's privacy settings prevented the use of cookies, or if they cleared their web browser's cookies midway through the experiment. We also dropped an additional 20,008 customer observations that were missing relevant data elements – such as the date the customer visited the credit card website, or the types of credit cards for which they browsed – that prevented tracking the customer's progression through the acquisition funnel. We additionally dropped 1,514 customers from the analysis who ultimately applied outside the experimental period. Moreover, midway through the experiment, the bank detected inconsistent benefits information across the treatment and control conditions for one type of low fee card on one page of the bank's website. To distinctly identify the effect of the treatment on the treated customers, for our primary analysis, we withhold observations from the 6,999 customers in both conditions who visited this specific page, though we note in an analysis in the online appendix that the results are substantively similar with these customers included. Our primary analysis, therefore, includes observations from a total of 393,036 customers, 195,438 of whom were randomly assigned to the control condition, and 197,598 of whom were randomly assigned to the treatment condition. **Table 1** presents summary statistics for customers in both experimental conditions on a variety of observable dimensions. We control for these customer-level factors in our empirical models, though we note that customers were well-balanced on both dimensions prior to random assignment, and as we show in analyses presented in the online appendix, results are substantively similar if controls are not included. These control variables were captured as of each customer's first documented arrival on the credit card website, and thus, were unaffected by the experimental treatment. Customer-level data captured temporally after the random assignment and presentation of the experimental treatment, such as which credit card each customer selected, are themselves outcomes in the notional experiment, and thus, would be considered "bad controls" if included in our analysis (Angrist and Pischke 2008). Accordingly, we refrain from using control variables captured after random assignment, and thus, all treatment effects presented in this paper can be interpreted as causal.

| | All | | | Control | | | Treatment | | | Diff. |
|---|---|---|---|---|---|---|---|---|---|---|
| | N | Mean | SD | N | Mean | SD | N | Mean | SD | P-value |
| Arrived during the promo period | 393,036 | 63.00% | 0.48 | 195,438 | 62.89% | 0.48 | 197,598 | 63.11% | 0.48 | 0.16 |
| Customer demographics | | | | | | | | | | |
| Customer age | 392,942 | 40.12387 | 15.59 | 195,386 | 40.12955 | 15.62 | 197,556 | 40.11824 | 15.57 | 0.82 |
| Customer tenure (months) | 392,357 | 198.5407 | 138.62 | 195,092 | 198.3877 | 138.74 | 197,265 | 198.692 | 138.50 | 0.49 |
| Male indicator | 392,863 | 45.03% | 0.50 | 195,350 | 45.11% | 0.50 | 197,513 | 44.95% | 0.50 | 0.29 |
| Product indicators | | | | | | | | | | |
| Transaction product | 392,874 | 89.73% | 0.30 | 195,355 | 89.72% | 0.30 | 197,519 | 89.73% | 0.30 | 0.91 |
| Savings product | 392,874 | 67.72% | 0.47 | 195,355 | 67.71% | 0.47 | 197,519 | 67.74% | 0.47 | 0.89 |
| Home loan product | 392,874 | 22.03% | 0.41 | 195,355 | 22.06% | 0.41 | 197,519 | 22.01% | 0.41 | 0.72 |
| Home insurance policy | 392,874 | 11.49% | 0.32 | 195,355 | 11.51% | 0.32 | 197,519 | 11.47% | 0.32 | 0.68 |
| Personal loan product | 392,874 | 10.01% | 0.30 | 195,355 | 9.96% | 0.30 | 197,519 | 10.06% | 0.30 | 0.30 |
| Retirement product | 392,874 | 4.75% | 0.21 | 195,355 | 4.77% | 0.21 | 197,519 | 4.73% | 0.21 | 0.53 |
| Motor insurance policy | 392,874 | 4.04% | 0.20 | 195,355 | 4.01% | 0.20 | 197,519 | 4.07% | 0.20 | 0.34 |
| Term deposit product | 392,874 | 3.89% | 0.19 | 195,355 | 3.90% | 0.19 | 197,519 | 3.88% | 0.19 | 0.71 |
| Credit card product | 392,874 | 62.84% | 0.48 | 195,355 | 62.90% | 0.48 | 197,519 | 62.78% | 0.48 | 0.45 |

**Table 1.** Summary statistics for customer characteristics in the control and treatment conditions. Customer characteristics were captured as of the time of each customer's first documented arrival on the credit card website, and thus, are unaffected by the experimental treatment. We control for all of these customer-level factors in our empirical models, though we note that customers in both conditions were quite well-balanced on these observable dimensions prior to their random assignment, and that uncontrolled analyses, presented in the online appendix, reveal substantively similar results. Differing numbers of observations for different variables reflect missing data. Analyses presented in the online appendix reveal that missing observations did not vary significantly across the treatment and control conditions.

Furthermore, on November 1, 2017 (midway through our period of experimentation), the bank publicly announced a promotion, offering \$300 cash back to customers who opened a credit card in the Low Rate family and spent \$1,000 in purchases with the new card within the first 90 days of activation. The announcement was broadly supported by advertising (e.g., television, radio, and print). Random assignment to the experimental conditions proceeded during this promotional period, affording an opportunity to assess whether providing customers with incentives to choose particular offerings moderates the effects of tradeoff transparency on customer acquisition and engagement. The promotional period ran from November 1 through the end of the experiment. We identified 145,412 customers who initially visited the credit card website prior to the November 1 promotion announcement to not be treated with the promotion, $PROMO_i$ =0, and 247,624 customers who initially visited on November 1 or after to be treated with the promotion, $PROMO_i$=1. We cannot directly observe whether customers who initially visited during this period were actually exposed to the promotion, so results from this variable should be interpreted according to the intention-to-treat principle. Our analysis yields an unbiased estimate of the effect of the promotion on the behavior of prospective customers, at the average level of exposure to the promotion.

*3.1.2 Acquisition measures.* In order to assess the impact of providing transparency into the tradeoffs of a service offering on rates of customer acquisition, we tracked: (1) whether the customer progressed through each stage of the acquisition funnel, $\Pr\left(ACQ_i\right)$, and (2) which types of credit card accounts customers opened, $\Pr\left(CHOICE_i\right)$. At the time of our study, the credit card acquisition funnel for our partner bank had six stages. In order to gain a detailed understanding of how the experimental treatment affected customer acquisition, we used six binary variables to capture each customer's progress through the acquisition funnel. In particular, we coded each stage of the acquisition funnel to be 1 if the customer passed the stage, and 0 if she did not.

The first stage of the acquisition funnel occurred when the customer *started an application*. During the application process, customers of the partner bank answered a series of 10 to 40 questions, providing their demographic information as well as their financial information (such as assets, liabilities, income, and expenses). For the more than 95% of prospective customers who had existing accounts with the bank, this stage of the process was streamlined considerably; by simply logging into their online banking accounts, these customers would be able to transfer their information from their existing accounts to their credit card applications. Answering these questions typically took customers 5-15 minutes, depending on how much information was required to complete their financial profile.

Submitting the answers to these questions triggered the second stage of the acquisition funnel, which was known as a *soft submission*. In this phase, the bank parsed through the information provided using an automated process to conduct a risk assessment and determine whether the customer's financial profile warranted approval for the credit card for which she had applied, as well as her eligible credit limit. After the results of this process were shared with the customer, the customer could choose to move on to the third stage, providing a *hard submission.* The application submitted at this stage constituted a formal application submission for a credit card; at this stage, customers provided verification documents required by the bank to substantiate their income as well as financial holdings and obligations that resided outside the bank. This stage could last two weeks or longer, depending on the amount of information provided by the customer and the amount of time the customer took to provide all the necessary information.

If the bank's process during the hard submission stage substantiated the information provided in the application, the customer was then moved into the fourth stage, wherein her account was *opened.* Here, the bank created and issued a new credit card for the customer and mailed it to her preferred address. This process typically took about a week. To test whether providing transparency influenced customer choices, we take a snapshot at this stage, comparing which types of card accounts were opened by customers randomly assigned to the treatment and control conditions.

In the fifth stage, the customer *activated* the card either by phone, online, through the mobile application, or by inserting the card into one of the bank's ATM machines. Finally, in the sixth stage, the customer *used* the card to make her first purchase; customers that reached the final stage of the acquisition funnel comprised 4.05% of all customers who shopped for a credit card. For our analyses, we denote a customer to have reached this final stage if a purchase was made with the card within two billing cycles of its activation. Imposing this constraint facilitates comparability among customers acquired early and late in our experimental period, as failure to do so would afford those who opened their accounts early more time to demonstrate usage. We note that 95.42% of all cards that were eventually used to make a purchase in our dataset were used within two billing cycles of their initial activation, and that all results presented in this manuscript are substantively similar if usage is instead defined by whether any transaction behavior is ever observed in our data.

Customer progress through the acquisition funnel could stall during any of these six stages, as each stage was comprised of different dynamics with respect to the customer's experience and the bank's operating costs. Hence, analyzing each stage separately provides important insights about the costs and benefits of providing prospective customers with transparency into the tradeoffs inherent in a service offering.

*3.1.3 Engagement measures.* For customers who made it all the way through the acquisition funnel, we tracked customer engagement with the card in two ways. First, we tracked product usage by examining logged monthly spending during the first nine months the customer held the card, $\ln\left(SPEND_{it}\right)$. Spend is a relevant engagement metric, both because it is a direct measure of usage that is an important behavioral indicator of a card's utility for the customer, and because it is an important driver of profitability for the bank. Bank-issued credit cards generate revenue through interest spreads (interest paid by customers who don't fully pay off their balances at the end of the month), interchange fees paid by merchants (small percentages of each transaction), and customer fees (usage fees, service fees, and penalty fees). Customer spend directly influences the bank's first two revenue streams.

Second, we tracked customer retention, i.e., whether the customer closed the credit card account during the first nine months it was open, $\Pr\left(RETAIN_i\right)$. As with spend, retention is a relevant measure of customer engagement, because loyalty is an important behavioral indicator of the card's utility to the customer, and because it is an important driver of profitability (Levesque and McDougall 1996). Previous research conducted in banking demonstrated that reducing defections by just 5% can increase firm profitability by as much as 100% (Reichheld and Sasser 1990).

We note that these engagement measures were queried on August 28, 2018, which means that six full months of account data are available for any customer who activated their credit cards before February 28, 2018, and nine full months of account data are available for any customer who activated their credit cards before November 30, 2017. Importantly, all customers who were denoted to have used their cards in our analysis had at least six months of available account data. Consistently, we analyze spend during any observed month for which a credit card account was activated, and we analyze retention six and nine months after the credit card account was activated.

*3.1.4 Control measures.* Although customers randomly assigned to the treatment and control conditions appear quite well balanced on observable dimensions (**Table 1**), to get the cleanest possible estimates of the effects of the experimental treatment on treated customers, we control for a vector of customer-level factors which were observable prior to each customer's random assignment. These factors include the non-linear effect of the customer's age, $AGE_i$ and $AGE_i^2$, and tenure with the bank (in months), $TENURE_i$, and $TENURE_i^2$, their gender $GENDER_i$ (operationalized as an indicator for whether the customer was male), and indicators for whether they held a home loan product, $HLOAN_i$, a personal loan product, $PLOAN_i$, a transaction product, $TRANS_i$, a savings product, $SAV_i$, a home insurance policy, $HINS_i$, a vehicle insurance policy, $VINS_i$, or a retirement product with the bank, $RET_i$. For the acquisition analyses, we also include a vector of date indicator variables, $X_i$, signifying the first week the customer visited the credit card pages on the bank's website, to control for time-varying factors that may have affected each customer's initial motivation for seeking a credit card. For the engagement analyses, we include a similar vector of date indicator variables, $Z_i$, signifying the first week the customer activated their credit card, to control for time-varying factors that may influence the way customers interact with the credit card (e.g., a credit card activated before the holidays may be used more intensively than one activated after, etc.)

*3.1.5 Category experience and financial incentives.* Lastly, to test Hypotheses 5-8, that the effects of providing transparency into the tradeoffs inherent in a service offering will accrue disproportionately to customers with more experience with the product category (H5-H6), and will have a greater effect on customers who were not attracted by a financial incentive (H7-H8), we incorporate data on proxies for both factors into our analysis. First, since we lack complete data on the degree of credit card experience for each prospective customer (e.g., because not every customer provides their financial history, and many customers may have accounts with financial institutions that are not our partner bank), we use the prospective customer's age as the primary proxy in our analysis. In the market where CBA operated during the time of our experiment, the minimum age to apply for a credit card was 18. As we demonstrate graphically in the online appendix, data captured by CBA reveals that the probability a prospective customer has at least one credit card increases concavely in age. As such, data on the ages of credit card browsers and applicants during our study was right skewed. The average credit card browser was 40.12 years old and the median browser was 37 years old. The average credit card applicant was 31.27 years old, and the median applicant was 28 years old. To test hypotheses 5 and 6 – that the effects of providing prospective customers with tradeoff transparency on product usage and retention will be most acute for customers with more prior category experience – we complement our analyses on the full sample of data with split sample analyses of customers who were 28 and younger (less experienced with credit cards), and customers who were older than 28 (more experienced with credit cards). Using the median applicant age to conduct a split sample analysis yields a comparable number of more-experienced and less-experienced customers. However, as we document in the online appendix, our results are substantively similar with different cutoffs, for example, splitting the bottom quartile of the credit card applicant data (24 and younger) from the top three quartiles.

Second, to test whether our experimental treatment had differential effects for customers who were motivated to shop for a credit card by a promotion, we include an interaction term in all of our empirical specifications, $PROMO_i$, denoting whether the customer first visited the credit card website or applied for a credit card on or after November 1, 2017, the day the promotion was publicly announced. This empirical approach enables us to directly test the significance of the effect of the treatment on customers who conducted their credit card searches organically (e.g., before November 1, in the absence of a promotion), and also to test whether the treatment had a differential effect among customers who are more likely to have been influenced by the promotion (e.g., on or after November 1, initiating their search or applying for a credit card while the promotion was live). Since we cannot observe whether customers who first began their search on or after November 1, 2017 were directly influenced by the promotion, this empirical approach constitutes an intention to treat design, and hence, should be interpreted as a conservative estimate of the contingent effect of the promotion on the transparency treatment. We note that this empirical design is consistent with prior research in the marketing literature that has analyzed customers who were acquired by means of a promotion differently from those who were acquired organically (Anderson and Simester 2004, Krishnamurthi and Raj 2006, Lewis 2006).

## 3.2 Empirical approach

We analyze rates of acquisition, card choice, spend, and customer retention as a part of our analysis. This section outlines our empirical approach for each analysis.

*3.2.1 Acquisition funnel.* To estimate the effects of the experimental treatment on the probability that customer *i* reached each stage of the acquisition funnel (application started, soft submission completed, submission completed, account opened, card activated, card used), $ACQ_i$, we use the following logistic regression model, with standard errors clustered by the date she first visited the credit card marketing site. Clustering standard errors in this way is intended to account for the possibility that customers who first arrive on the same day may be related to each other, for example in the driver of their motivation to consider a credit card (e.g., were attracted to the site by the same advertisement, were motivated to seek a credit card for the same upcoming holiday or event, etc.). We note that $ACQ_i$ for any particular customer refers to the furthest acquisition funnel stage she reached for any credit card for which she applied during the experimental period.

$$\Pr\left(ACQ_i\right) = f\left(\begin{array}{c} \alpha_0 + \alpha_1 TREAT_i + \alpha_2 PROMO_i + \alpha_3 TREAT_i \times PROMO_i + \\ \alpha_4 AGE_i + \alpha_5 AGE_i^2 + \alpha_6 TENURE_i + \alpha_7 TENURE_i^2 + \alpha_8 GENDER_i + \\ \alpha_9 HLOAN_i + \alpha_{10} PLOAN_i + \alpha_{11} TRANS_i + \alpha_{12} SAV_i + \alpha_{13} HINS_i + \\ \alpha_{14} VINS_i + \alpha_{15} RET_i + X_i + \epsilon_i \end{array}\right) \quad (1)$$

*3.2.2 Card choice.* To estimate the effects of the experimental treatment on the probability that customer *i* opened each type of card (awards, diamond, gold low fee low fee gold, low rate, low rate gold, platinum, or student), $CHOICE_i$, we use the following logistic regression specification with standard errors clustered by the date she first visited the credit card marketing site:

$$\Pr\left(CHOICE_i\right) = f\left(\begin{array}{c} \beta_0 + \beta_1 TREAT_i + \beta_2 PROMO_i + \beta_3 TREAT_i \times PROMO_i + \\ \beta_4 AGE_i + \beta_5 AGE_i^2 + \beta_6 TENURE_i + \beta_7 TENURE_i^2 + \beta_8 GENDER_i + \\ \beta_9 HLOAN_i + \beta_{10} PLOAN_i + \beta_{11} TRANS_i + \beta_{12} SAV_i + \beta_{13} HINS_i + \\ \beta_{14} VINS_i + \beta_{15} RET_i + X_i + \epsilon_i \end{array}\right) \quad (2)$$

*3.2.3 Spend.* To estimate the effects of the experimental treatment on the spend of customer *i*, in month *t,* observed in months after the activation of her credit card, we use the following fixed effects panel model, with standard errors clustered by the date she activated the card. Clustering standard errors in this way is intended to account for the possibility that customers who activate their cards on the same day may exhibit similar spending patterns (e.g., customers who activate immediately before a holiday may spend more than those who activate immediately after, etc.).

$$\ln\left(SPEND_{it}\right) = f\left(\begin{array}{c} \gamma_0 + \gamma_1 TREAT_i + \gamma_2 PROMO_i + \gamma_3 TREAT_i \times PROMO_i + \\ \gamma_4 AGE_i + \gamma_5 AGE_i^2 + \gamma_6 TENURE_i + \gamma_7 TENURE_i^2 + \gamma_8 GENDER_i + \\ \gamma_9 HLOAN_i + \gamma_{10} PLOAN_i + \gamma_{11} TRANS_i + \gamma_{12} SAV_i + \gamma_{13} HINS_i + \\ \gamma_{14} VINS_i + \gamma_{15} RET_i + Z_i + \epsilon_{it} \end{array}\right) \quad (3)$$

*3.2.4 Retention.* To estimate the effects of the experimental treatment on the retention of customer *i,* six and nine months after the activation of her card, we use the following logistic regression, with standard errors clustered by the date she activated the card.

$$\Pr\left(RETAIN_i\right) = f\left(\begin{array}{c} \delta_0 + \delta_1 TREAT_i + \delta_2 PROMO_i + \delta_3 TREAT_i \times PROMO_i + \\ \delta_4 AGE_i + \delta_5 AGE_i^2 + \delta_6 TENURE_i + \delta_7 TENURE_i^2 + \delta_8 GENDER_i + \\ \delta_9 HLOAN_i + \delta_{10} PLOAN_i + \delta_{11} TRANS_i + \delta_{12} SAV_i + \delta_{13} HINS_i + \\ \delta_{14} VINS_i + \delta_{15} RET_i + Z_i + \epsilon_t \end{array}\right) \quad (4)$$

## 3.3 Analysis and results

*3.2.1 Acquisition funnel (H1).* **Figure 4** graphically displays the marginal effects of the treatment on the acquisition funnel of the treated customers. The results show three features of consequence. First, rates of conversion were higher during the promotion period, likely owing to the efficacy of the promotion in attracting interested customers to the website and in enhancing the appeal of the promoted products. Second, during the promotion period, when offered transparency into the tradeoffs inherent in each credit card, significantly fewer customers chose to start the application process (17.78% vs 17.33%, p<0.01) or soft submit their application (14.53% vs 14.17%, p<0.01). Interestingly, these differences did not emerge among customers who shopped for credit cards outside the promotion period. Finally, and most importantly, during both periods, there were no significant differences in rates of conversion through the various phases of the acquisition funnel during the submission phase and beyond. As we document in the online appendix, we note that these patterns are substantively similar among customers with high and low experience levels with credit cards. Consequently, we fail to reject H1. Taken together, these results suggest that from an acquisition perspective, voluntarily providing prospective customers with transparency into the tradeoffs of the firm's offerings had an insignificant effect on the overall rate of customer acquisition.

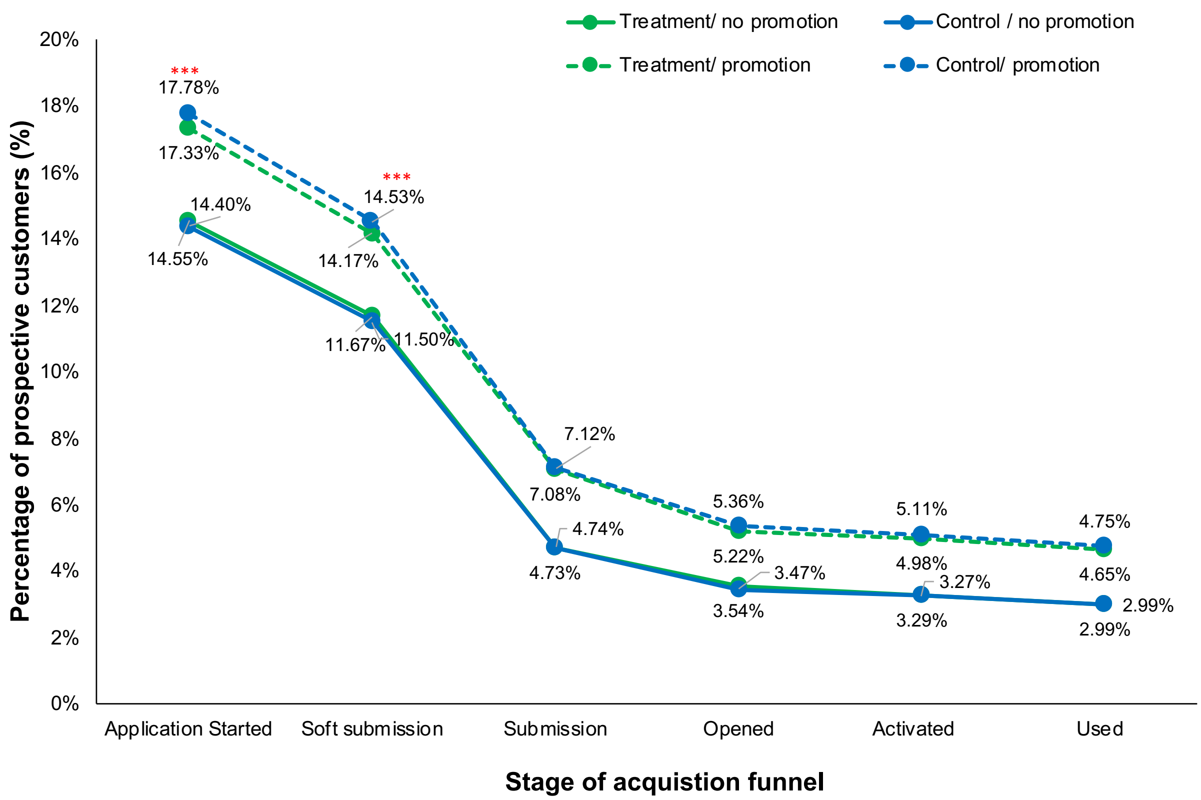



**Figure 4:** Acquisition funnel conversion percentages during the promotion and non-promotion periods. Marginal effects are from logistic regression models, estimated with robust standard errors clustered at the first website visit date level. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. The results indicate that conversion rates are significantly higher during the promotion period. Moreover, although fewer customers who are exposed to the treatment during the promotion period start or soft submit an application, the treatment ultimately had an insignificant effect on rates of customer acquisition, neither during the promotion nor the non-promotion periods. Consequently, we fail to reject H1.

*3.2.2 Card choice (H2).* **Table 2**, Panel A documents the marginal effects of Model (2), regressed first on the full sample of customers who opened credit cards during the entire experimental period, and then on a split sample, median-split by age, with customers at and below the median age presented first and customers above the median age presented second. This analysis is replicated on the subsample of customers who arrived during the non-promotion period and promotion period in Panels B and C, respectively. Each table directly compares the probabilities that applicants in the control and treatment conditions would open different types of cards. The results demonstrate that, consistent with H2, the treatment did affect the credit card choices of customers who opened credit card accounts. Panel A demonstrates that, controlling for other factors, customers in the treatment condition were less likely to open the low rate card ($\beta$ = -0.144, $p$<0.05) and more likely to open the low rate gold card ($\beta$ = 0.212, $p$<0.05) and platinum card ($\beta$ = 0.266, $p$<0.05) than customers in the control condition.

A. **Entire period** (n=18,152)

| | **All customers (n=18,152)** | | | | **Customer age ≤ 28 (n=9,090)** | | | | **Customer age > 28 (n=9,062)** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Control | Treatment | | P-value | Control | Treatment | | P-value | Control | Treatment | | P-value |
| Awards card | 4.81% | 4.97% | | 0.908 | 4.48% | 5.16% | | 0.992 | 5.14% | 4.76% | | 0.915 |
| Diamond card | 2.16% | 2.26% | | 0.396 | 1.66% | 2.12% | | 0.883 | 2.64% | 2.46% | | 0.239 |
| Gold card | 2.75% | 2.77% | | 0.195 | 2.83% | 2.99% | | 0.151 | 2.71% | 2.52% | | 0.664 |
| Low fee card | 14.78% | 14.54% | | 0.324 | 16.08% | 15.11% | | 0.715 | 13.43% | 14.02% | | 0.234 |
| Low fee gold card | 2.76% | 3.07% | | 0.977 | 2.85% | 3.09% | | 0.974 | 2.68% | 3.04% | | 0.936 |
| Low rate card | 51.75% | 49.81% | ** | 0.033 | 51.00% | 48.48% | * | 0.093 | 52.57% | 51.11% | * | 0.098 |
| Low rate gold card | 11.96% | 12.96% | ** | 0.013 | 10.60% | 11.50% | | 0.111 | 13.32% | 14.42% | * | 0.092 |
| Platinum card | 5.41% | 6.01% | ** | 0.013 | 3.98% | 4.89% | ** | 0.045 | 6.85% | 7.14% | | 0.278 |
| Student card | 3.56% | 3.65% | | 0.938 | 6.73% | 6.99% | | 0.818 | 0.78% | 0.71% | | 0.418 |

B. **Non-promotion period** (n=5,087)

| | **All customers (n=5,087)** | | | | **Customer age ≤ 28 (n=2,632)** | | | | **Customer age > 28 (n=2,455)** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Control | Treatment | | P-value | Control | Treatment | | P-value | Control | Treatment | | P-value |
| Awards card | 6.14% | 6.09% | | 0.944 | 5.71% | 5.62% | | 0.912 | 6.75% | 6.72% | | 0.979 |
| Diamond card | 3.15% | 2.81% | | 0.437 | 2.55% | 2.44% | | 0.852 | 3.81% | 3.25% | | 0.341 |
| Gold card | 3.85% | 3.21% | | 0.209 | 4.03% | 2.88% | | 0.148 | 3.88% | 3.54% | | 0.641 |
| Low fee card | 18.56% | 19.48% | | 0.342 | 18.48% | 19.09% | | 0.652 | 18.54% | 20.22% | | 0.219 |
| Low fee gold card | 4.19% | 4.17% | | 0.977 | 3.90% | 3.88% | | 0.974 | 4.51% | 4.56% | | 0.952 |
| Low rate card | 43.29% | 39.72% | ** | 0.027 | 43.88% | 40.67% | * | 0.094 | 42.71% | 38.51% | * | 0.068 |
| Low rate gold card | 10.54% | 12.79% | ** | 0.010 | 9.15% | 11.15% | * | 0.099 | 12.17% | 14.79% | * | 0.062 |
| Platinum card | 5.80% | 7.47% | ** | 0.011 | 4.37% | 6.35% | ** | 0.040 | 7.52% | 8.86% | | 0.265 |
| Student card | 5.41% | 5.42% | | 0.989 | 9.18% | 9.55% | | 0.745 | 1.31% | 0.87% | | 0.365 |

C. **Promotion period** (n=13,065)

| | **All customers (n=13,065)** | | | | **Customer age ≤ 28 (n=6,458)** | | | | **Customer age > 28 (n=6,607)** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Control | Treatment | | P-value | Control | Treatment | | P-value | Control | Treatment | | P-value |
| Awards card | 4.29% | 4.56% | | 0.438 | 4.02% | 5.00% | * | 0.065 | 4.57% | 4.11% | | 0.358 |
| Diamond card | 1.80% | 2.06% | | 0.298 | 1.37% | 2.04% | ** | 0.047 | 2.20% | 2.19% | | 0.957 |
| Gold card | 2.34% | 2.62% | | 0.292 | 2.41% | 3.11% | * | 0.056 | 2.36% | 2.22% | | 0.702 |
| Low fee card | 13.33% | 12.66% | | 0.291 | 15.12% | 13.55% | * | 0.093 | 11.56% | 11.82% | | 0.738 |
| Low fee gold card | 2.24% | 2.66% | | 0.108 | 2.47% | 2.77% | | 0.456 | 2.04% | 2.54% | | 0.199 |
| Low rate card | 55.09% | 53.70% | | 0.130 | 53.91% | 51.66% | * | 0.094 | 56.31% | 55.65% | | 0.598 |
| Low rate gold card | 12.55% | 13.07% | | 0.377 | 11.23% | 11.69% | | 0.573 | 13.83% | 14.43% | | 0.537 |
| Platinum card | 5.27% | 5.51% | | 0.578 | 3.91% | 4.48% | | 0.239 | 6.65% | 6.58% | | 0.917 |
| Student card | 3.43% | 3.59% | | 0.618 | 5.87% | 6.15% | | 0.639 | 1.02% | 1.10% | | 0.834 |

**Table 2:** Effects of the treatment on card choice. Marginal effects from logistic regression models, estimated with robust standard errors clustered at the first website visit date level are presented. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. The results indicate that providing transparency into the tradeoffs of the service offering influenced credit card choices, which is consistent with H2.

Examining the split sample analyses provides evidence that transparency into the tradeoffs of each credit card influenced the behavior of customers with varying levels of experience differently. For example, less-experienced customers exposed to the treatment were more likely to open the platinum card ($\beta = 0.393$, $p<0.05$) and were marginally less likely to open the low rate card ($\beta = -0.133$, $p<0.10$). More-experienced customers exposed to the treatment were marginally less likely to open the low rate card ($\beta = -0.162$, $p<0.10$), and were marginally more likely to open the low rate gold card ($\beta = 0.208$, $p<0.10$). These differences are interesting, since they suggest that different types of customers with different needs evaluated the tradeoffs differently, which in turn, differentially affected their choices. Moreover, these diverging effects of the treatment among customers with varying experience levels further suggest that the results were not merely driven by tradeoff copy that made some credit cards appear clearly dominant to others.

As additional evidence, it is revealing to consider the differential effects of the treatment on the choices of customers who applied for credit cards in the absence and presence of a promotion, which are presented in Panels B and C, respectively. For customers who arrived during the non-promotion period, the patterns of results among all, less-experienced, and more-experienced customers were consistent with those described above across the entire experimental period, with the addition that less-experienced customers were marginally more likely to select the low rate gold card ($\beta = 0.225$, $p<0.10$) when provided transparency into each card's tradeoffs. By contrast, during the promotion period, which likely attracted customers with different needs and preferences, the treatment dissuaded less-experienced customers from the low rate ($\beta = -0.092$, $p<0.10$) and low fee cards ($\beta = -0.129$, $p<0.10$), and increased their interest in the awards ($\beta = 0.229$, $p<0.10$), diamond ($\beta = 0.422$, $p<0.05$), and gold cards ($\beta = 0.265$, $p<0.10$). Interestingly, during the promotion period, the treatment exhibited no effect on the credit card choices of more-experienced customers. In sum, these results suggest that, consistent with H2, transparency influenced customers' credit card choices.

As noted previously, an alternative question is whether tradeoff transparency influenced customers' choices for the better. In the online appendix, we provide some evidence based on observable customer characteristics that the choices customers made in the presence of tradeoff transparency may have been more compatible with their banking needs. For example, customers with fewer banking products, who may have been more likely to use their credit cards for short-term borrowing, may have been best matched with the low rate card, which offers the most favorable interest rate. We find that such customers were marginally more likely to choose the low rate card in the presence of tradeoff transparency ($\gamma$=-0.069, $p<0.10$); a tendency that was especially strong during the non-promotion period ($\gamma$=-0.179, $p<0.01$).

Interestingly, although younger customers are more likely to choose the student card in general ($\beta$ =-0.225, $p$<0.01), we do not find that their propensity to do so is influenced by the presence of tradeoff transparency ($\beta$ =0.008, $p$=0.758). These results may suggest that when the intended segment for an offering is already clearly communicated (e.g., the student card is for students), the addition of tradeoff transparency may be less impactful. A complementary potential explanation is that, consistent with the arguments in Section 2.2, customers with less prior category experience may be less impacted by tradeoff transparency. We believe that a more reliable test of whether tradeoff transparency helps customers make better choices is to analyze how they engage with their chosen products after acquiring them, which is our objective in the next two subsections.

*3.2.3 Product usage (H3, H5, H7).* In **Table 3**, Columns 1-3, we analyze the average monthly spending of all, less-experienced, and more-experienced customers. Average monthly spending captures the usage behavior of customers in all months during which they held an active credit card, including months when they had it, but didn't use it (e.g., monthly spend is $0), and excluding months after which they may have cancelled it (e.g., monthly spend is missing, because the customer no longer held the card). As such, monthly spend, as it is estimated in Columns (1-3), is a direct measure of contemporary customer engagement, in that it captures how the card was used while the card was held. Column (1) shows that the treatment led to a nominal, though insignificant, increase in average monthly spending ($\gamma$=0.080, $p$=0.122), that the promotion had no effect on spend ($\gamma$=-0.104, $p$=0.314), and that there was an insignificant interaction ($\gamma$=-0.097, $p$=0.111), when the effects were considered across all customers.

Similarly, Column (2) shows that the monthly spending of less-experienced customers was neither influenced by the treatment ($\gamma$=0.035, $p$=0.675), nor by the promotion ($\gamma$=0.019, $p$=0.885), and that there was no interaction between the two ($\gamma$=-0.054, $p$=0.599). However, Column (3) illustrates that among customers who arrived during the non-promotion period, the monthly spending of more-experienced customers was 14.74% higher among those who were provided with transparency into the offerings' tradeoffs ($\gamma$=0.138, $p$<0.05). Monthly spending was nominally lower among more-experienced customers who arrived during the promotion period in the control condition ($\gamma$=-0.230, $p$=0.127), and there was a negative interaction, such that in aggregate, the treatment had no effect on monthly spending among more-experienced customers during the promotion period ($\gamma$=-0.162, $p$<0.05).

Columns (4-6) present the same specifications, regressed on monthly spend where missing values due to account cancellations were filled with zero. As such, monthly spend, as it is estimated in Columns (4-6), is a direct measure of cumulative customer engagement, in that it captures how the card was used inclusive of the customer's choice of whether to retain it.

| | (1) Ln(Spend) | (2) Ln(Spend) | (3) Ln(Spend) | (4) Ln(Spend) | (5) Ln(Spend) | (6) Ln(Spend) |
|---|---|---|---|---|---|---|
| Treatment | 0.080 | 0.035 | 0.138** | 0.095* | 0.033 | 0.176*** |
| | (0.052) | (0.085) | (0.069) | (0.054) | (0.092) | (0.062) |
| Promotion | -0.104 | 0.019 | -0.230 | -0.371*** | -0.150 | -0.593*** |
| | (0.103) | (0.131) | (0.151) | (0.114) | (0.125) | (0.166) |
| Treatment x promotion | -0.097 | -0.054 | -0.162** | -0.109* | -0.045 | -0.199** |
| | (0.061) | (0.103) | (0.082) | (0.064) | (0.110) | (0.080) |
| Customer age | 0.072*** | 0.077 | 0.009 | 0.046*** | 0.137 | 0.027 |
| | (0.008) | (0.129) | (0.018) | (0.009) | (0.136) | (0.019) |
| Customer age² | -0.001*** | -0.001 | -0.000 | -0.001*** | -0.002 | -0.000 |
| | (0.000) | (0.003) | (0.000) | (0.000) | (0.003) | (0.000) |
| Customer tenure | 0.002*** | 0.005*** | -0.001 | 0.000 | 0.004*** | -0.002*** |
| | (0.000) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| Customer tenure² | -0.000 | -0.000*** | 0.000* | 0.000 | -0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.050 | 0.100** | 0.001 | 0.072** | 0.106** | 0.035 |
| | (0.034) | (0.046) | (0.041) | (0.034) | (0.045) | (0.044) |
| Retirement product | 0.045 | 0.147 | 0.027 | 0.038 | 0.173 | 0.015 |
| | (0.105) | (0.160) | (0.123) | (0.108) | (0.164) | (0.122) |
| Home loan product | -0.038 | 0.028 | -0.045 | -0.044 | 0.142 | -0.090 |
| | (0.076) | (0.162) | (0.088) | (0.085) | (0.172) | (0.097) |
| Personal loan product | 0.496*** | 0.184 | 0.595*** | 1.043*** | 1.120*** | 0.999*** |
| | (0.105) | (0.231) | (0.121) | (0.101) | (0.228) | (0.123) |
| Savings product | 0.198* | 0.406** | 0.072 | -0.023 | 0.253 | -0.170 |
| | (0.120) | (0.161) | (0.154) | (0.133) | (0.189) | (0.168) |
| Term deposit product | 0.277*** | 0.403*** | 0.182*** | 0.247*** | 0.369*** | 0.156*** |
| | (0.040) | (0.062) | (0.050) | (0.040) | (0.062) | (0.050) |
| Transaction product | -0.324*** | -0.437*** | -0.230*** | -0.209*** | -0.376*** | -0.071 |
| | (0.039) | (0.063) | (0.054) | (0.043) | (0.065) | (0.057) |
| Home insurance policy | 0.509*** | 0.736*** | 0.468*** | 0.349*** | 0.500*** | 0.334*** |
| | (0.049) | (0.126) | (0.059) | (0.062) | (0.141) | (0.070) |
| Motor insurance policy | 0.067 | 0.019 | 0.179** | 0.105* | 0.026 | 0.270*** |
| | (0.058) | (0.077) | (0.087) | (0.058) | (0.078) | (0.095) |
| Constant | 3.348*** | 3.060* | 4.972*** | 3.455*** | 1.802 | 4.334*** |
| | (0.212) | (1.579) | (0.403) | (0.245) | (1.644) | (0.427) |
| Observations | 121,679 | 61,231 | 60,448 | 126,658 | 63,237 | 63,421 |
| Customers | 15,942 | 7,932 | 8,010 | 15,942 | 7,932 | 8,010 |
| Data treatment for closed accounts | Missing | Missing | Missing | Zero | Zero | Zero |
| Sample | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| R-squared | 0.0321 | 0.0461 | 0.0256 | 0.0271 | 0.0363 | 0.0302 |
| Pred(Y): Non-Promotion: Control | $196.64 | $158.84 | $245.45 | $192.67 | $151.58 | $246.66 |
| Pred(Y): Non-Promotion: Treatment | $212.98 | $164.56 | $281.63 | $211.79 | $156.60 | $294.06 |
| Pred(Y): Promotion: Control | $177.20 | $161.89 | $195.12 | $132.91 | $130.44 | $136.26 |
| Pred(Y): Promotion: Treatment | $174.20 | $158.90 | $190.30 | $130.99 | $128.84 | $133.13 |

**Table 3:** Effects of the treatment on monthly spend. Columns (1) – (3) show the results of panel data regression models for all customers as well as younger and older customers, with spend values for months after cancellation set to missing. Columns (4) – (6) show the same specifications with spend values for months after cancellation set to zero. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that customers exposed to the treatment spent more on their cards, that these effects were stronger for more-experienced customers, and that these effects were attenuated by the promotion, which is consistent with H3, H5, and H7.

Column (4) shows that during the non-promotion period, the treatment led to a marginal increase in spend ($\gamma$=0.095, $p$<0.10), increasing average monthly spending by 9.9% across all customers. It further shows that when cancellation dynamics are taken into account, the promotion decreased average monthly spend by 31.0% ($\gamma$=-0.371, $p$<0.01), and that there was a marginally significant interaction wherein the promotion attenuated the effects of the treatment to insignificance ($\gamma$=-0.109, $p$<0.10). Column (5) shows that the monthly spending of less-experienced customers was neither influenced by the treatment ($\gamma$=0.033, $p$=0.722), nor by the promotion ($\gamma$=-0.150, $p$=0.229), and that there was no interaction between the two ($\gamma$=-0.045, $p$=0.684). However, Column (6) illustrates that during the non-promotion period, the monthly spending of more-experienced customers was 19.2% higher among those who were provided with transparency into the offerings' tradeoffs ($\gamma$=0.176, $p$<0.01).

Monthly spending was 44.8% lower among more-experienced customers brought in by the promotion in the control condition ($\gamma$=-0.593, $p$<0.01), and there was a negative interaction, such that in aggregate, the treatment had no effect on monthly spending among more-experienced customers during the promotion period ($\gamma$=-0.199, $p$<0.05). With respect to monthly spending, taken together, these results suggest that although providing transparency to prospective customers can lead to greater engagement overall, which is consistent with H3, increases in engagement are most acute among more-experienced customers, which is consistent with H5. More-experienced customers who were provided transparency used their credit cards more intensively, spending 19.2% more on a monthly basis. The results further suggest that promotional efforts that provide incentives to encourage customers to choose one offering over another attenuate the effects of transparency on subsequent product usage, which is consistent with H7.

Importantly, although it may be reflective of firm profitability, we note that increased credit card spending, on its own, may not necessarily be emblematic of a better experience for customers. Indeed, higher spending rates could be detrimental to customers, if they meant customers were spending beyond their means. In a separate analysis, presented in **Table 4**, we analyze the probability that customers in different experimental conditions made late payments from month to month – the state of not having met the minimum required payment amount by the payment due date. Looking across the entire panel of data, in Columns (1-3), we find that the treatment had no effect on the probability a customer made a late payment – among all customers ($\gamma$=-0.117, $p$=0.124), less-experienced customers ($\gamma$=-0.114, $p$=0.332), and more-experienced customers ($\gamma$=-0.135, $p$=0.227). However, when in Columns (4-6) we consider the first six months of data for each customer – which balances the number of monthly observations among customers who opened accounts during the non-promotion (e.g., before November 1) and promotion (e.g., on or after November 1) periods – we observe that the probability of making a late payment was 10.8% lower among customers who were in the treatment condition ($\gamma$=-0.175, $p$<0.05), which suggests that the increased levels of engagement we observe do not come at the expense of customers' financial wellbeing.

| | (1) Pr(Late Payment) | (2) Pr(Late Payment) | (3) Pr(Late Payment) | (4) Pr(Late Payment) | (5) Pr(Late Payment) | (6) Pr(Late Payment) |
|---|---|---|---|---|---|---|
| Treatment | -0.117 | -0.114 | -0.135 | -0.175** | -0.200* | -0.157 |
| | (0.076) | (0.117) | (0.112) | (0.078) | (0.120) | (0.116) |
| Promotion | -0.471*** | -0.295 | -0.626*** | -0.447*** | -0.356 | -0.512** |
| | (0.139) | (0.206) | (0.186) | (0.168) | (0.230) | (0.228) |
| Treatment x promotion | 0.118 | 0.127 | 0.117 | 0.168 | 0.214 | 0.123 |
| | (0.102) | (0.147) | (0.143) | (0.106) | (0.152) | (0.147) |
| Customer age | -0.074*** | -0.170 | 0.027 | -0.071*** | -0.112 | 0.012 |
| | (0.012) | (0.212) | (0.030) | (0.013) | (0.232) | (0.031) |
| Customer age² | 0.001*** | 0.002 | -0.000 | 0.001*** | 0.001 | -0.000 |
| | (0.000) | (0.005) | (0.000) | (0.000) | (0.005) | (0.000) |
| Customer tenure | -0.005*** | -0.010*** | 0.000 | -0.005*** | -0.010*** | 0.001 |
| | (0.001) | (0.001) | (0.001) | (0.001) | (0.002) | (0.001) |
| Customer tenure² | 0.000*** | 0.000*** | -0.000 | 0.000*** | 0.000*** | -0.000 |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | -0.111** | -0.222*** | 0.008 | -0.141*** | -0.257*** | -0.013 |
| | (0.052) | (0.063) | (0.066) | (0.055) | (0.068) | (0.071) |
| Retirement product | -0.023 | -0.077 | 0.022 | -0.049 | -0.118 | 0.004 |
| | (0.095) | (0.116) | (0.150) | (0.101) | (0.123) | (0.161) |
| Home loan product | -0.752*** | -1.430*** | -0.603*** | -0.752*** | -1.383*** | -0.617*** |
| | (0.091) | (0.240) | (0.094) | (0.098) | (0.236) | (0.105) |
| Personal loan product | 0.942*** | 1.062*** | 0.866*** | 0.901*** | 1.066*** | 0.792*** |
| | (0.064) | (0.095) | (0.085) | (0.069) | (0.100) | (0.095) |
| Savings product | -0.772*** | -0.937*** | -0.624*** | -0.775*** | -0.904*** | -0.658*** |
| | (0.054) | (0.089) | (0.077) | (0.057) | (0.091) | (0.084) |
| Term deposit product | -0.749*** | -1.340*** | -0.468* | -0.793*** | -1.414*** | -0.493 |
| | (0.235) | (0.473) | (0.280) | (0.259) | (0.511) | (0.301) |
| Transaction product | 0.415** | 1.485*** | 0.085 | 0.326* | 1.336*** | 0.031 |
| | (0.169) | (0.407) | (0.182) | (0.175) | (0.417) | (0.187) |
| Home insurance policy | 0.105 | 0.308 | -0.020 | 0.126 | 0.290 | 0.005 |
| | (0.107) | (0.229) | (0.129) | (0.111) | (0.247) | (0.139) |
| Motor insurance policy | 0.184 | -0.014 | 0.213 | 0.241* | -0.103 | 0.332** |
| | (0.138) | (0.251) | (0.160) | (0.139) | (0.264) | (0.162) |
| Constant | -1.563*** | -1.007 | -3.792*** | -1.483*** | -1.573 | -3.434*** |
| | (0.295) | (2.459) | (0.648) | (0.325) | (2.711) | (0.695) |
| Observations | 112,641 | 56,266 | 56,375 | 92,988 | 46,253 | 46,735 |
| Customers | 15,942 | 7,932 | 8,010 | 15,942 | 7,932 | 8,010 |
| Account sample | All data | All data | All data | First six months | First six months | First six months |
| Sample | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| Wald Chi-Square | 1177.32 | 970.85 | 577.18 | 1031.43 | 841.47 | 461.97 |
| Pred(Y): Non-Promotion: Control | 9.24% | 9.68% | 8.70% | 8.82% | 9.56% | 7.96% |
| Pred(Y): Non-Promotion: Treatment | 8.57% | 9.02% | 7.96% | 7.87% | 8.43% | 7.15% |
| Pred(Y): Promotion: Control | 6.78% | 8.04% | 5.66% | 6.55% | 7.62% | 5.58% |
| Pred(Y): Promotion: Treatment | 6.78% | 8.11% | 5.58% | 6.52% | 7.69% | 5.44% |

**Table 4:** Effects of the treatment on the probability a customer would make a late payment in a given month. Columns (1-3) show the results of panel data regression models for all customers as well as younger and older customers, regressed on the full sample. Columns (4-6) show the same specifications regressed on the first six months of observations for each customer, which balances the sample across customers acquired during the non-promotion and promotion periods. All models are estimated with random effects panel logistic regression, with robust standard errors, clustered by activation date. All models additionally include indicator variables for the week during which the customer activated his or her card, which are withheld from the table for parsimony. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the table's bottom section. The results indicate that the treatment had an insignificant or negative effect on the probability a customer would make a late payment. Column (4) demonstrates that the probability that a customer in the treatment condition, who was acquired during the non-promotion period was 10.8% less likely to make a late payment in any given month than customers acquired in the control condition. This result suggests the treatment has the capacity to improve the financial wellbeing of customers.

In fact, the evidence is supportive of the idea that tradeoff transparency can help customers make better choices, by selecting products they're more likely to use, and that will improve their financial wellbeing. In the next section, we examine the effects of transparency on customer retention, another important measure of customer engagement, that is both emblematic of customer experiences (e.g., customers who are having more favorable experiences with a service offering are more likely to continue using it), and firm performance (e.g., customers who are retained are generally more profitable than those who defect).

*3.2.4 Retention (H4, H6, H8).* In **Table 5**, Columns (1-3) and (4-6) show the effects of tradeoff transparency on the retention of all customers, less-experienced customers, and more-experienced customers, after six and nine months, respectively. Columns (1) and (4) provide evidence that, looking across all customers, the treatment had an insignificant effect on retention after six months ($\delta$=0.259, $p$=0.198), but that it increased retention after nine months ($\delta$=0.249, $p$<0.05). After nine months, cancellations among customers in the treatment group were 20.5% lower, which is consistent with H4, and provides evidence that revealing the hidden tradeoffs to prospective customers can improve their decision making in ways that have positive implications for the long run trajectories of service relationships.

Interestingly, Columns (2) and (5) reveal that, consistent with the prior analysis on spend, there's no evidence that the positive implications of the treatment for retention accrued to customers with less category experience. Less-experienced customers in the treatment group were no more nor less likely to be retained after six ($\delta$=0.099, $p$=0.771) or nine months ($\delta$=0.111, $p$=0.546). In contrast, Columns (3) and (6) show that more-experienced customers who received the transparency treatment were more likely to have been retained after six ($\delta$=0.437, $p$<0.10) and nine months ($\delta$=0.447, $p$<0.01), which is consistent with H6. The positive effect of tradeoff transparency on product retention was most acute for customers with prior experience with the service category. Among these more-experienced customers, nine months into their service relationships, cancellation rates were 33.7% lower for those who had been randomly assigned to experience tradeoff transparency before making their product choice (falling from 6.62% to 4.39%). Finally, Column (6) demonstrates an interaction, wherein the effect of the treatment on retention among more-experienced customers was negatively moderated after nine months for customers who were attracted by a promotion ($\delta$=-0.569, $p$<0.05). Similarly, the negative coefficients on the interaction terms reduce the positive and significant effects of the treatment on retention to insignificance during the promotion period ($\chi^2s$<0.53, $ps$>0.469). These results are consistent with H8, in that they demonstrate how the positive effect of tradeoff transparency on product retention can be diminished among customers attracted to an offering by a promotion.

| | (1) Pr(Retain6) | (2) Pr(Retain6) | (3) Pr(Retain6) | (4) Pr(Retain9) | (5) Pr(Retain9) | (6) Pr(Retain9) |
|---|---|---|---|---|---|---|
| Treatment | 0.259 | 0.099 | 0.437* | 0.249** | 0.111 | 0.447*** |
| | (0.202) | (0.340) | (0.231) | (0.115) | (0.183) | (0.163) |
| Promotion | -1.736*** | -1.693*** | -1.758*** | -1.344*** | -0.882** | -1.647*** |
| | (0.256) | (0.633) | (0.275) | (0.231) | (0.422) | (0.294) |
| Treatment x promotion | -0.239 | -0.055 | -0.410 | -0.223 | 0.114 | -0.569** |
| | (0.220) | (0.365) | (0.258) | (0.165) | (0.258) | (0.226) |
| Customer age | -0.219*** | -0.038 | 0.060 | -0.094** | 0.469 | 0.010 |
| | (0.038) | (0.323) | (0.044) | (0.041) | (0.320) | (0.055) |
| Customer age² | 0.003*** | -0.002 | -0.000 | 0.001*** | -0.011 | 0.000 |
| | (0.001) | (0.007) | (0.001) | (0.001) | (0.007) | (0.001) |
| Customer tenure | -0.009*** | -0.011*** | -0.009*** | -0.005*** | -0.007** | -0.006*** |
| | (0.001) | (0.002) | (0.001) | (0.002) | (0.003) | (0.002) |
| Customer tenure² | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000* | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.046 | 0.011 | 0.065 | 0.078 | 0.039 | 0.118 |
| | (0.074) | (0.110) | (0.095) | (0.091) | (0.100) | (0.133) |
| Retirement product | 0.245 | 0.110 | 0.414* | -0.107 | -0.255 | 0.036 |
| | (0.154) | (0.221) | (0.229) | (0.155) | (0.176) | (0.235) |
| Home loan product | -0.595*** | -0.775*** | -0.539*** | -0.564*** | -0.545* | -0.538*** |
| | (0.105) | (0.192) | (0.124) | (0.167) | (0.278) | (0.189) |
| Personal loan product | 0.767*** | 0.408** | 1.025*** | 0.246** | -0.007 | 0.497*** |
| | (0.110) | (0.159) | (0.164) | (0.106) | (0.171) | (0.164) |
| Savings product | -0.188** | -0.141 | -0.188** | -0.132 | -0.070 | -0.179 |
| | (0.080) | (0.165) | (0.092) | (0.096) | (0.168) | (0.119) |
| Term deposit product | -0.617*** | -0.574 | -0.618*** | -0.662*** | -0.309 | -0.796*** |
| | (0.167) | (0.352) | (0.203) | (0.235) | (0.484) | (0.294) |
| Transaction product | 1.603*** | 2.235*** | 1.312*** | 1.402*** | 2.034*** | 1.074*** |
| | (0.125) | (0.269) | (0.159) | (0.222) | (0.413) | (0.282) |
| Home insurance policy | 0.084 | 0.504 | -0.012 | -0.068 | 0.777 | -0.233 |
| | (0.154) | (0.383) | (0.162) | (0.170) | (0.668) | (0.199) |
| Motor insurance policy | 0.082 | 0.585 | -0.045 | 0.027 | 0.744 | -0.109 |
| | (0.198) | (0.492) | (0.224) | (0.202) | (0.561) | (0.235) |
| Constant | 7.509*** | 5.895 | 1.963** | 3.739*** | -2.343 | 1.345 |
| | (0.711) | (4.006) | (0.914) | (0.773) | (3.835) | (1.219) |
| Observations | 15,942 | 7,864 | 8,010 | 6,314 | 3,231 | 3,083 |
| Customers | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| Pseudo R2 | 0.0845 | 0.0916 | 0.0977 | 0.0551 | 0.0509 | 0.0948 |
| Pred(Y): Non-Promotion: Control | 98.07% | 98.32% | 97.76% | 93.40% | 92.99% | 93.38% |
| Pred(Y): Non-Promotion: Treatment | 98.50% | 98.47% | 98.53% | 94.75% | 93.66% | 95.61% |
| Pred(Y): Promotion: Control | 90.50% | 92.14% | 88.99% | 79.43% | 84.98% | 75.00% |
| Pred(Y): Promotion: Treatment | 90.67% | 92.43% | 89.23% | 79.83% | 87.54% | 72.87% |

**Table 5:** Effects of the treatment on customer retention rates. Columns (1) – (3) and columns (4) – (6) show the results of cross-sectional data regression models of all, younger, and older customers for retention six and nine months after card activation, respectively. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that transparency can lead to increased customer retention, which is consistent with H4, that the effects are strongest for more-experienced customers, which is consistent with H6, and that the effects are dampened during the promotion, which is consistent with H8. Marginal effect estimates for retention after six and nine months are provided for each experimental condition.

## 4. General discussion

Through a large-scale field experiment conducted with 393,036 customers of a nationwide retail bank, we have investigated the impact of providing prospective customers with transparency into a service offering's tradeoffs on rates of acquisition, product choice, and subsequent long-term engagement. Our results suggest that although providing prospective customers with tradeoff transparency may have an insignificant effect on overall rates of acquisition, it can affect customers' choices – causing them to select different offerings than they would in the absence of tradeoff transparency. Moreover, we find evidence that the information tradeoff transparency provides can help customers make better choices, leading to long-term outcomes that are improved for customers and firms alike. Customers who were provided tradeoff transparency and chose to move forward by opening an account exhibited higher levels of engagement during their first nine months of service, effects that were especially pronounced among customers who had prior experience with the service category. Monthly spending was 9.9% higher, and after nine months, cancellation rates were 20.5% lower, among all customers who received the tradeoff transparency treatment. Among more-experienced customers, monthly spending was 19.2% higher, and rates of cancellation after nine months were 33.7% lower for those who received the tradeoff transparency treatment. Importantly, we find that increased engagement levels do not come at the expense of customers' financial wellbeing. During the initial six months of their relationships, customers who experienced tradeoff transparency were 10.8% less likely to make late payments on a monthly basis. These results provide evidence that being more transparent with customers – not just about an offering's advantages, but also about its tradeoffs – can help customers make choices that lead to better long-run outcomes for customers and firms alike. Finally, we find that the effects of transparency on engagement and retention are attenuated in the presence of a promotion. The results are consistent with the idea that customers who organically seek out an offering are more likely to benefit from tradeoff transparency than those who are attracted or influenced by a promotion.

We note that the results documented in this paper arose from a field experiment conducted in a highly-regulated service environment, where providers are legally required to fully disclose the terms and conditions of their offerings to prospective customers. In this context, tradeoff transparency merely increased the salience of information that was already being provided, and its effects were substantial. It stands to reason that the effects of tradeoff transparency may be even stronger in less-regulated contexts, where such disclosures are not required and customers face even greater information asymmetries.

### 4.1 Managerial implications

Taken together, these results have important implications for practice, while also raising additional questions that will serve as fruitful opportunities for future research.

*4.1.1 Tradeoff transparency may not harm acquisition rates.* Although research has shown how the revelation of unflattering information can harm demand, and conventional wisdom has long suggested the fallacy of marketing an offering's weaknesses, our results demonstrate that voluntarily providing customers with transparency into an offering's tradeoffs may not harm rates of acquisition. These results are important, because they suggest that the costs of being fully open with prospective customers that many managers perceive (e.g., that prospective customers who are shown the tradeoffs in an offering may seek service elsewhere) may be misplaced.

*4.1.2 Tradeoff transparency may lead to better long-run outcomes for customers and firms.* Although the provision of transparency in our study had an insignificant effect on overall rates of customer acquisition, we find that it caused customers to make different product choices than they would make in the absence of transparency. Specifically, we observed that customers in different segments, who were given the same tradeoff information, selected different credit cards. This pattern of results is consistent with the idea that the revelation of information about a product's tradeoffs can help shape customers' decisions. Furthermore, we find that transparency improves customer engagement – both by increasing product usage and rates of retention over the longer-term. Customers who were randomly-selected to experience transparency into each offering's tradeoffs, and who moved forward in opening an account, used their cards more intensively, spending 9.9% more per month, and were 10.8% less likely to make late payments on a monthly basis. Furthermore, customers who experienced transparency were 20.5% less likely to cancel their credit cards during the first nine months of their relationships. To the extent that heightened engagement and retention are behavioral indicators of a better customer experience, and a more profitable service relationship, these results suggest that providing transparency to prospective customers may be mutually beneficial.

*4.1.3 There may be a tradeoff between acquisition and retention-based strategies.* Midway through the experiment, our research partner launched a promotion in which they offered a financial incentive to prospective customers who opened a credit card in the Low Rate family, and who spent $1,000 during the first three months of their service relationships. From an acquisition perspective, the efficacy of this promotional strategy is evident. Traffic to the credit card website increased, and the conversion rate of browsers to buyers was enhanced, such that the number of new credit card accounts opened per day went up by 47.3% during the promotion period. Furthermore, the probability a customer would open a promoted credit card went up – during the promotion the probability of opening a low rate or low rate gold credit card increased by 28.1% and 18.0%, respectively – highlighting the promotion's capacity to increase sales and influence customer choices. However, customers acquired during the promotion spent a third less on a monthly basis, due in part to the fact that they were 3.8 times more likely to cancel their cards during the initial months of their relationships than those who were not acquired through a promotion. These results highlight a potential tradeoff for managers between marketing-based promotional strategies that are optimized around increasing rates of customer acquisition, and operations-based transparency strategies that are optimized around increasing customer compatibility and long-term engagement. Future research could delve more deeply into this tradeoff – for example, by examining whether over a longer timeframe, the gains in engagement and retention brought about by tradeoff transparency might outweigh the improvement in acquisition rates promotions can foster.

*4.1.4 Tradeoff transparency is especially effective for customers who have prior category experience.* Consistent with prior research, our results provide evidence that transparency may be particularly effective for improving relationship quality and engagement among more-experienced customers. In our experiment, more-experienced customers (operationalized as customers older than 28 years of age), who received the transparency treatment, used the credit cards they chose more intensively, exhibiting cumulative spending that was 19.2% higher after nine months, than those who did not experience transparency. They also were 33.7% less likely to cancel their credit cards during the first nine months of their relationship. These results underscore the promise of tradeoff transparency, for customers and companies alike, in helping customers make more well-informed decisions. However, it's interesting that less-experienced customers did not exhibit similar gains in subsequent engagement and retention. Less-experienced customers (those 28 years of age and younger), who were treated with tradeoff transparency, spent 3.3% more during a typical month, and were 9.6% less likely to cancel their accounts after nine months – results that are statistically indistinguishable from zero. We note, however, that although the benefits of tradeoff transparency for relationship quality and engagement are especially pronounced among more-experienced customers, supplementary interaction models presented in full in the online appendix do not find age (our proxy for experience) to significantly moderate the effects of tradeoff transparency on spending (*ps*>0.16) or retention (*ps*>0.21). Nevertheless, our results highlight an opportunity to explore strategies for helping to better inform less-experienced prospective customers who do not significantly benefit from tradeoff transparency in our analyses.

*4.1.5 Promotions may crowd-out the benefits of providing prospective customers with tradeoff transparency.* Finally, our results suggest that the effects of providing prospective customers with transparency into an offering's tradeoffs may be attenuated by promotions, which themselves are designed to influence customer choices. More-experienced customers, in particular, who exhibited the greatest increases in product usage and retention in response to tradeoff transparency, also exhibited an interaction effect, wherein those gains were mitigated, though not reversed, in the presence of a promotion. Importantly, in our experiment, the promotion in question offered financial incentives for only one family of credit cards, which may have created a conflict for some consumers between the card that offered a bonus and the card that best fit their needs and preferences.

Future research could more holistically disentangle whether promotions and tradeoff transparency are necessarily in conflict by treating every offering with the same promotion. Such a design would afford a better understanding of the mechanisms that drive the crowding-out effect we observe – whether it's driven by the conflict between fit and financial incentives some customers may have experienced, or whether it's driven by promotions attracting customers who are less sensitive to the effects of tradeoff transparency. Owing to the demonstrated efficacy of promotions to attract new customers, and tradeoff transparency to foster decision making that boosts long-run engagement, innovations that achieve the best of both worlds could enhance customer experiences while unlocking considerable value for firms.

### 4.2 Limitations and opportunities for future research

An important caveat of this work is that since the focal firm offered a portfolio of products, many of the tradeoffs presented for particular cards could be overcome by choosing a different card from the bank. Indeed, as shown in **Figure 2**, some of the tradeoffs presented specifically compare service attributes with those of other cards, such as "the annual fee of $59 is higher than our Low Fee credit card," or "the purchase interest rate of 20.24% p.a. is higher than the Low Rate and Low Fee credit cards." It is possible that one of the reasons acquisition rates didn't fall in this study was that tradeoff transparency helped facilitate comparisons among the bank's own offerings. Future research could examine the effects of tradeoff transparency in contexts where particular tradeoffs cannot be overcome by selecting another offering from the firm. Relatedly, future research might investigate how tradeoff transparency that varies in valence has differential effects on consumers. In the present research, benefits and tradeoffs were quite evenly-balanced, but consumers may draw fundamentally different inferences as the ratio of good to bad changes. Indeed, it may be the case that companies need to improve the quality of their underlying offerings to a particular level before being genuinely transparent about tradeoffs becomes a viable strategy.

These results are consistent with prior research that has shown how the act of being voluntarily transparent with information that customers perceive to be sensitive can engender higher levels of trust in the firm and attraction to its offerings (Mohan et al. 2020). Indeed, in an online experiment conducted with 201 participants (62.2% male, $M$=39.18 years old), and presented in full in the online appendix, we find evidence that tradeoff transparency can increase trust and brand preference. 71% of participants ($p<0.01$, one-sided) presented with credit card webpages from competing banks reported the bank that provided tradeoff transparency to be more trustworthy, and 69% ($p<0.01$, one-sided) reported being more likely to apply for a credit card with the transparent bank. An opportunity for future research would be to delve more deeply into the mechanisms that underlie these effects, exploring the conditions under which tradeoff transparency is more and less likely to engender trust, and how it affects the relationship with the firm more holistically (e.g., does it affect peoples' willingness to engage with other offerings).

Likewise, future research could explore whether the provision of tradeoff transparency is generally appealing, or if there are individual differences among appetites for transparency, such that transparent organizations attract different types of customers. In the online appendix, we present an analysis that shows that customers who chose to open credit cards in the treatment and control conditions of our experiment were substantively similar on observable characteristics. These results suggest that in this setting, improved engagement arose from tradeoff transparency's capacity to help otherwise-similar customers make choices that were better aligned with their needs and preferences, rather than by attracting fundamentally different customers. We note, however, that results may differ in other settings – particularly when the marketed offering primarily targets new customers, rather than extends an ongoing service relationship, as may be the case with bank-issued credit cards. Similarly, research on the psychology of choice has demonstrated how providing more information that complicates a choice can lead a customer to opt out of choosing (Gourville and Soman 2005, Iyengar and Lepper 2000). Future research could investigate how tradeoffs could be most effectively communicated to help customers make better-informed decisions as easily as possible.

## 4.3 Conclusion

Conventional wisdom and common practice dictate that service firms should emphasize the advantages of their offerings and downplay the tradeoffs when marketing to prospective customers. We suggest that taking a different approach – providing transparency into both an offering's advantages and its tradeoffs, in order to help customers make more well-informed decisions – can lead to better outcomes over the long run for everyone involved. We hope that the present work will lead to more research in this area, and influence practice, in order to foster better customer experiences, and more engaging and profitable service relationships among customers and the organizations that serve them.

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**Online appendix for "Improving Customer Compatibility with Tradeoff Transparency"**

## Visualizing Prior Category Experience

We conduct an analysis of all customers visiting the CBA credit card marketing website, irrespective of assignment to experimental condition, for which information about whether they had a credit card was known. As the information on whether customers had credit cards was collected after their visit, and thus may have included cards opened during the period of the experiment, we do not use this variable as a control in our analysis. However, it enables us to model how the probability of having a credit card varies with age. We use the following logistic regression specification, where $AGE_i$ is entered into the specification as a categorical variable.

$$\Pr\left(CARD_i\right) = f\left(\alpha_0 + \alpha_1 AGE_i + \epsilon_i\right) \quad (1)$$

Next, we plot the marginal effects for each age category, calculating the 95% confidence interval band, as shown in Figure A1. The results indicate that the probability of having a credit card increases non-linearly in customer age.

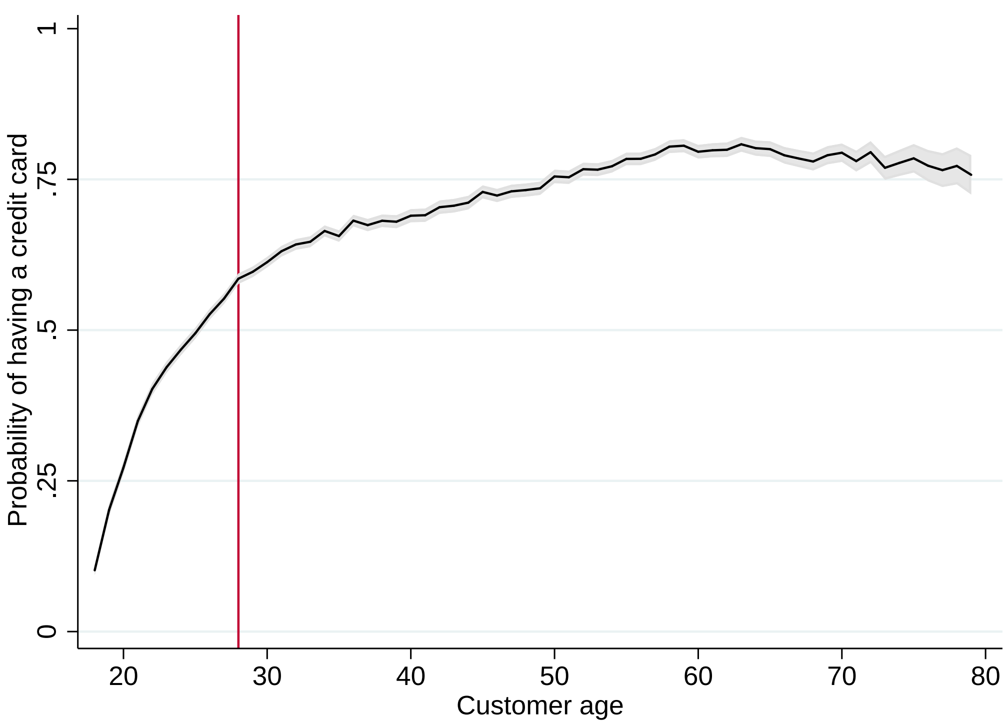



**Figure A1:** The probability a prospective customer shopping for a credit card already has a credit card, as a function of their age (n=389,085). Predicted plots are based on a logistic regression of the probability of having a credit card as a function of indicator variables for the prospective customer's age, regressed on the full sample of prospective customers aged 18-79 (within support of 99% of the data). A 95% confidence interval band is shown in light grey. For our primary analyses, we denote customers aged 28 and younger (the median age for a credit card applicant) as less experienced, and customers 28 and older to be more experienced with credit cards. This cutoff is shown with a red vertical line, and it demonstrates that the probability a 28-year old prospective customer had a credit card at the time of this study was 58.5%.

## Supplemental Summary Statistics Tables

In Table 1 of the paper, we present summary statistics for the control variables used in all regressions. Below, we present complementary summary statistics for the dependent measures. **Table A1** presents summary statistics for the full dataset. **Table A2** presents summary statistics during the non-promotion period. **Table A3** presents summary statistics during the promotion period. We have also included a correlation table in **Table A4**. All statistics represent each variable, presented at the equivalent level of analysis as the primary regressions presented and described in the paper.

The acquisition analysis sample (n=393,036) includes all prospective customers who arrived on the CBA credit card marketing website during the period of our study, minus noted exclusions (e.g., people who were assigned to both experimental conditions). The card application analysis sample (n=18,157) includes all prospective customers who applied for a credit card, so we could examine how the treatment affected the types of cards for which people applied. The engagement analyses sample (n=15,947) includes all customers who used their credit cards. Within the engagement analyses, we do look at different levels of data aggregation, depending on what best suits the hypothesis being tested:

- Spend analysis – Data are analyzed at the customer/month level, either during months where the customer still held their account (n=121,679) or with months after closing filled with zeroes, to determine no spend (n=126,658).

- Late payment analysis – Data are analyzed at the customer/month level, just including months for which a payment was due (n=112,641) or all months for which a payment was due within the first six months of having the card (n=92,988)

- Retention analysis – Data are analyzed at the customer level, either after six months (n=15,942) or after nine months (n=6,314). For the latter analysis, we only included customers in the analysis who opened their credit cards in time to have used them for nine months during the period of our study.

| | All | | | Control | | | Treatment | | | Diff. |
|---|---|---|---|---|---|---|---|---|---|---|
| | N | Mean | SD | N | Mean | SD | N | Mean | SD | P-value |
| Dependent Variables | | | | | | | | | | |
| App start | 393,036 | 16.39% | 0.37 | 195,438 | 16.66% | 0.37 | 197,598 | 16.13% | 0.37 | 0.00 |
| Soft submission | 393,036 | 13.31% | 0.34 | 195,438 | 13.50% | 0.34 | 197,598 | 13.12% | 0.34 | 0.00 |
| Submission | 393,036 | 6.18% | 0.24 | 195,438 | 6.20% | 0.24 | 197,598 | 6.16% | 0.24 | 0.59 |
| Opened | 393,036 | 4.62% | 0.21 | 195,438 | 4.66% | 0.21 | 197,598 | 4.58% | 0.21 | 0.21 |
| Activated | 393,036 | 4.38% | 0.20 | 195,438 | 4.43% | 0.21 | 197,598 | 4.34% | 0.20 | 0.15 |
| Used | 393,036 | 4.06% | 0.20 | 195,438 | 4.09% | 0.20 | 197,598 | 4.02% | 0.20 | 0.24 |
| Got Awards card | 18,157 | 4.89% | 0.22 | 9,111 | 4.80% | 0.21 | 9,046 | 4.97% | 0.22 | 0.58 |
| Got Diamond card | 18,157 | 2.21% | 0.15 | 9,111 | 2.15% | 0.15 | 9,046 | 2.28% | 0.15 | 0.56 |
| Got Gold card | 18,157 | 2.76% | 0.16 | 9,111 | 2.74% | 0.16 | 9,046 | 2.77% | 0.16 | 0.90 |
| Got Low Fee card | 18,157 | 14.67% | 0.35 | 9,111 | 14.73% | 0.35 | 9,046 | 14.61% | 0.35 | 0.83 |
| Got Low Fee Gold card | 18,157 | 29.13% | 0.17 | 9,111 | 27.55% | 0.16 | 9,046 | 30.73% | 0.17 | 0.20 |
| Got Low Rate card | 18,157 | 50.78% | 0.50 | 9,111 | 51.84% | 0.50 | 9,046 | 49.71% | 0.50 | 0.00 |
| Got Low Rate Gold card | 18,157 | 12.46% | 0.33 | 9,111 | 12.00% | 0.32 | 9,046 | 12.92% | 0.34 | 0.06 |
| Got Platinum card | 18,157 | 5.72% | 0.23 | 9,111 | 5.42% | 0.23 | 9,046 | 6.01% | 0.24 | 0.09 |
| Got Student card | 18,157 | 3.60% | 0.19 | 9,111 | 3.57% | 0.19 | 9,046 | 3.64% | 0.19 | 0.80 |
| Monthly spend (Missing=.) | 121,714 | $ 187.52 | 17.17 | 60,918 | $ 186.16 | 17.25 | 60,796 | $ 188.89 | 17.08 | 0.37 |
| Monthly spend (Missing=0) | 126,696 | $ 152.64 | 19.42 | 63,466 | $ 150.92 | 19.55 | 63,230 | $ 154.38 | 19.29 | 0.17 |
| Open (6 months) | 15,947 | 93.61% | 0.24 | 8,003 | 93.53% | 0.25 | 7,944 | 93.69% | 0.24 | 0.67 |
| Open (9 months) | 6,314 | 90.51% | 0.29 | 3,137 | 89.99% | 0.30 | 3,177 | 91.03% | 0.29 | 0.16 |
| Late pay (6 months) | 93,016 | 7.03% | 0.26 | 46,674 | 7.15% | 0.26 | 46,342 | 6.90% | 0.25 | 0.14 |
| Late pay (9 months) | 112,674 | 7.51% | 0.26 | 56,455 | 7.61% | 0.27 | 56,219 | 7.40% | 0.26 | 0.18 |

**Table A1.** Summary statistics of the dependent variables for customers in the control and treatment conditions (full sample).

| | All | | | Control | | | Treatment | | | Diff. |
|---|---|---|---|---|---|---|---|---|---|---|
| | N | Mean | SD | N | Mean | SD | N | Mean | SD | P-value |
| Dependent Variables | | | | | | | | | | |
| App start | 145,412 | 14.46% | 0.3516526 | 72,521 | 14.64% | 0.35 | 72,891 | 14.27% | 0.35 | 0.05 |
| Soft submission | 145,412 | 11.56% | 0.32 | 72,521 | 11.64% | 0.32 | 72,891 | 11.49% | 0.32 | 0.38 |
| Submission | 145,412 | 4.64% | 0.21 | 72,521 | 4.62% | 0.21 | 72,891 | 4.66% | 0.21 | 0.72 |
| Opened | 145,412 | 3.50% | 0.18 | 72,521 | 3.46% | 0.18 | 72,891 | 3.54% | 0.18 | 0.44 |
| Activated | 145,412 | 3.27% | 0.18 | 72,521 | 3.25% | 0.18 | 72,891 | 3.29% | 0.18 | 0.69 |
| Used | 145,412 | 2.98% | 0.17 | 72,521 | 2.97% | 0.17 | 72,891 | 2.99% | 0.17 | 0.78 |
| Got Awards card | 5,089 | 6.07% | 0.24 | 2,511 | 6.13% | 0.24 | 2,578 | 6.01% | 0.24 | 0.86 |
| Got Diamond card | 5,089 | 2.95% | 0.17 | 2,511 | 3.19% | 0.18 | 2,578 | 2.72% | 0.16 | 0.32 |
| Got Gold card | 5,089 | 3.48% | 0.18 | 2,511 | 3.82% | 0.19 | 2,578 | 3.14% | 0.17 | 0.18 |
| Got Low Fee card | 5,089 | 18.94% | 0.39 | 2,511 | 18.48% | 0.39 | 2,578 | 19.39% | 0.40 | 0.40 |
| Got Low Fee Gold card | 5,089 | 4.11% | 0.20 | 2,511 | 4.10% | 0.20 | 2,578 | 4.11% | 0.20 | 0.99 |
| Got Low Rate card | 5,089 | 41.48% | 0.49 | 2,511 | 43.13% | 0.50 | 2,578 | 39.88% | 0.49 | 0.02 |
| Got Low Rate Gold card | 5,089 | 11.57% | 0.32 | 2,511 | 10.43% | 0.31 | 2,578 | 12.68% | 0.33 | 0.01 |
| Got Platinum card | 5,089 | 6.56% | 0.25 | 2,511 | 5.77% | 0.23 | 2,578 | 7.33% | 0.26 | 0.03 |
| Got Student card | 5,089 | 4.83% | 0.21 | 2,511 | 4.94% | 0.22 | 2,578 | 4.73% | 0.21 | 0.73 |
| Monthly spend (Missing=.) | 40,701 | 170.66 | 18.30 | 20,174 | 165.49 | 18.57 | 20,527 | 175.90 | 18.02 | 0.03 |
| Monthly spend (Missing=0) | 41,802 | 149.05 | 19.38 | 20,770 | 142.92 | 20.15 | 21,032 | 155.37 | 19.38 | 0.00 |
| Open (6 months) | 4,331 | 97.37% | 0.16 | 2,151 | 97.02% | 0.17 | 2,180 | 97.71% | 0.15 | 0.16 |
| Open (9 months) | 4,194 | 93.09% | 0.25 | 2,086 | 92.28% | 0.27 | 2,108 | 93.88% | 0.24 | 0.04 |
| Late pay (6 months) | 25,902 | 8.12% | 0.27 | 12,871 | 8.51% | 0.28 | 13,031 | 7.73% | 0.27 | 0.02 |
| Late pay (9 months) | 38,080 | 8.93% | 0.29 | 18,897 | 9.20% | 0.29 | 19,183 | 8.66% | 0.28 | 0.07 |

**Table A2.** Summary statistics of the dependent variables for customers in the control and treatment conditions (non-promotion period).

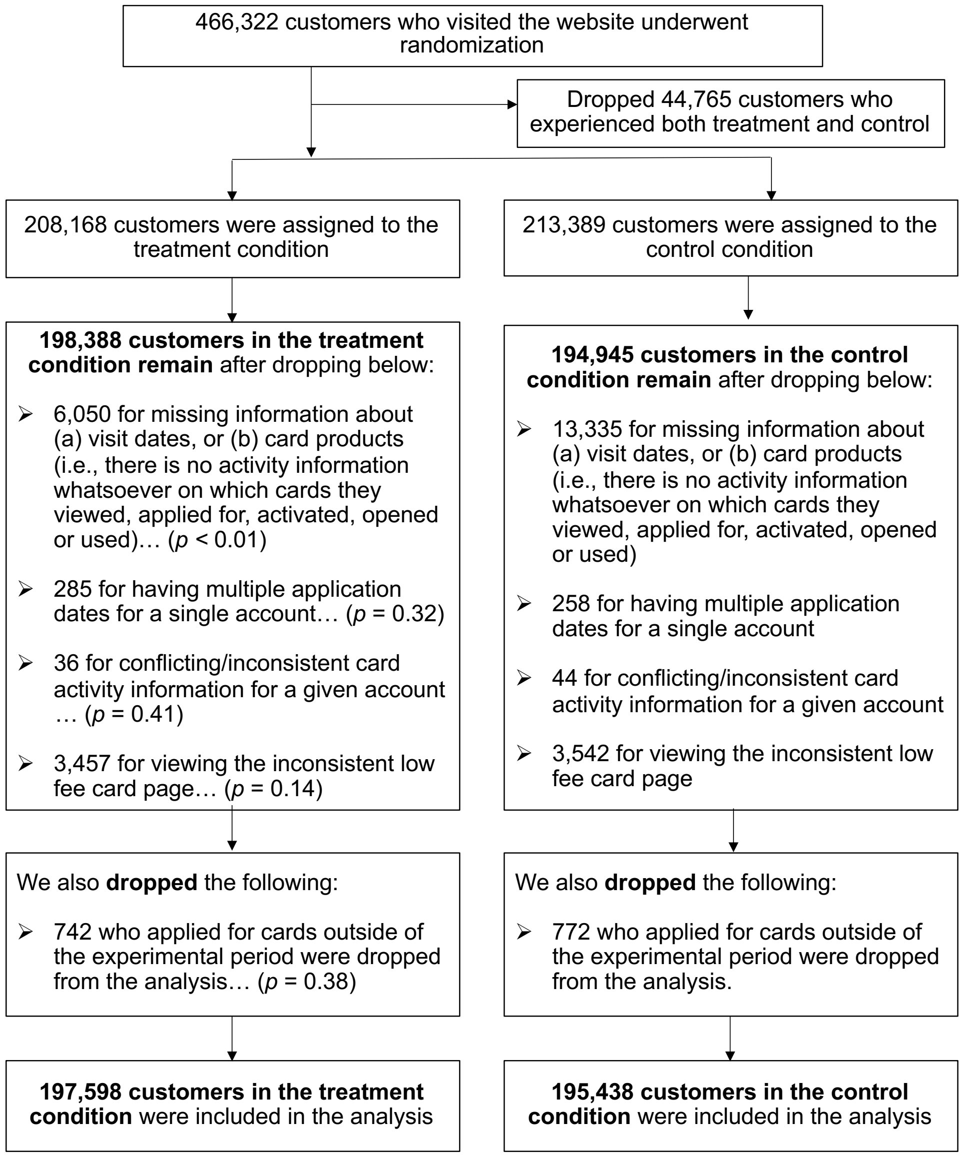
| | All | | | Control | | | Treatment | | | Diff. |
|---|---|---|---|---|---|---|---|---|---|---|
| | N | Mean | SD | N | Mean | SD | N | Mean | SD | P-value |
| Dependent Variables | | | | | | | | | | |
| App start | 247,624 | 17.53% | 0.38 | 122,917 | 17.86% | 0.38 | 124,707 | 17.21% | 0.38 | 0.00 |
| Soft submission | 247,624 | 14.33% | 0.35 | 122,917 | 14.59% | 0.35 | 124,707 | 14.07% | 0.35 | 0.00 |
| Submission | 247,624 | 7.09% | 0.26 | 122,917 | 7.14% | 0.26 | 124,707 | 7.04% | 0.26 | 0.34 |
| Opened | 247,624 | 5.28% | 0.22 | 122,917 | 5.37% | 0.23 | 124,707 | 5.19% | 0.22 | 0.04 |
| Activated | 247,624 | 5.03% | 0.22 | 122,917 | 5.12% | 0.22 | 124,707 | 4.95% | 0.22 | 0.04 |
| Used | 247,624 | 4.69% | 0.21 | 122,917 | 4.76% | 0.21 | 124,707 | 4.62% | 0.21 | 0.10 |
| Got Awards card | 13,068 | 4.42% | 0.21 | 6,600 | 4.29% | 0.20 | 6,468 | 4.56% | 0.21 | 0.45 |
| Got Diamond card | 13,068 | 1.93% | 0.14 | 6,600 | 1.76% | 0.13 | 6,468 | 2.10% | 0.14 | 0.15 |
| Got Gold card | 13,068 | 2.48% | 0.16 | 6,600 | 2.33% | 0.15 | 6,468 | 2.63% | 0.16 | 0.28 |
| Got Low Fee card | 13,068 | 13.01% | 0.34 | 6,600 | 13.30% | 0.34 | 6,468 | 12.71% | 0.33 | 0.31 |
| Got Low Fee Gold card | 13,068 | 2.45% | 0.15 | 6,600 | 2.24% | 0.15 | 6,468 | 2.66% | 0.16 | 0.12 |
| Got Low Rate card | 13,068 | 54.40% | 0.50 | 6,600 | 55.15% | 0.50 | 6,468 | 53.63% | 0.50 | 0.08 |
| Got Low Rate Gold card | 13,068 | 12.80% | 0.33 | 6,600 | 12.59% | 0.33 | 6,468 | 13.02% | 0.34 | 0.47 |
| Got Platinum card | 13,068 | 5.39% | 0.23 | 6,600 | 5.29% | 0.22 | 6,468 | 5.49% | 0.23 | 0.61 |
| Got Student card | 13,068 | 3.12% | 0.17 | 6,600 | 3.05% | 0.17 | 6,468 | 3.20% | 0.18 | 0.61 |
| Monthly spend (Missing=.) | 81,013 | $ 196.61 | 16.60 | 40,744 | $ 197.34 | 16.59 | 40,269 | $ 195.87 | 16.60 | 0.71 |
| Monthly spend (Missing=0) | 84,894 | $ 154.44 | 19.25 | 42,696 | $ 154.98 | 19.25 | 42,198 | $ 153.89 | 19.25 | 0.73 |
| Open (6 months) | 11,616 | 92.21% | 0.27 | 5,852 | 92.24% | 0.27 | 5,764 | 92.18% | 0.27 | 0.89 |
| Open (9 months) | 2,120 | 85.42% | 0.35 | 1,051 | 85.44% | 0.35 | 1,069 | 85.41% | 0.35 | 0.98 |
| Late pay (6 months) | 67,114 | 6.61% | 0.25 | 33,803 | 6.63% | 0.25 | 33,311 | 6.58% | 0.25 | 0.79 |
| Late pay (9 months) | 74,594 | 6.78% | 0.25 | 37,558 | 6.82% | 0.25 | 37,036 | 6.75% | 0.25 | 0.73 |

**Table A3.** Summary statistics of the dependent variables for customers in the control and treatment conditions (promotion period).

| | Customer age | Customer tenure | Male indicator | Transaction product | Savings product | Home loan product | Home insurance policy | Personal loan product | Retirement product | Motor insurance policy | Term deposit product | Got Awards | Got Diamond |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Customer age | 1 | | | | | | | | | | | | |
| Customer tenure | 0.55*** | 1 | | | | | | | | | | | |
| Male indicator | 0.0030* | 0.040*** | 1 | | | | | | | | | | |
| Transaction product | -0.29*** | -0.12*** | 0.0082*** | 1 | | | | | | | | | |
| Savings product | -0.092*** | -0.0034** | 0.031*** | 0.16*** | 1 | | | | | | | | |
| Home loan product | 0.14*** | 0.15*** | -0.0052*** | 0.13*** | -0.13*** | 1 | | | | | | | |
| Home insurance policy | 0.16*** | 0.15*** | -0.0012 | 0.057*** | -0.029*** | 0.37*** | 1 | | | | | | |
| Personal loan product | -0.097*** | 0.0054*** | 0.0015 | 0.098*** | 0.054*** | -0.045*** | -0.0024 | 1 | | | | | |
| Retirement product | -0.14*** | -0.16*** | -0.037*** | 0.062*** | 0.064*** | -0.071*** | -0.036*** | 0.016*** | 1 | | | | |
| Motor insurance policy | 0.069*** | 0.050*** | -0.0014 | 0.038*** | 0.0040** | 0.13*** | 0.29*** | 0.061*** | -0.0029* | 1 | | | |
| Term deposit product | 0.16*** | 0.12*** | 0.027*** | -0.029*** | 0.078*** | -0.019*** | 0.031*** | -0.052*** | -0.022*** | 0.0090*** | 1 | | |
| Awards | -0.027*** | -0.017*** | 0.00049 | 0.013*** | 0.014*** | -0.013*** | -0.0079*** | 0.012*** | 0.011*** | -0.0013 | -0.0029* | 1 | |
| Diamond | -0.017*** | -0.014*** | -0.012*** | 0.0085*** | 0.00047 | 0.014*** | 0.0039** | 0.0028* | 0.0045*** | 0.00071 | -0.0023 | -0.0015 | 1 |
| Gold | -0.021*** | -0.012*** | -0.0045*** | 0.0076*** | 0.012*** | -0.0076*** | -0.0055*** | 0.0054*** | 0.0031* | 0.0021 | -0.0017 | -0.0017 | -0.0011 |
| Low Fee | -0.047*** | -0.040*** | 0.0021 | 0.021*** | 0.017*** | -0.025*** | -0.018*** | 0.0080*** | 0.014*** | -0.0059*** | -0.0049*** | -0.0039** | -0.0026* |
| Low Fee Gold | -0.021*** | -0.024*** | -0.0077*** | 0.0088*** | 0.0075*** | -0.011*** | -0.0076*** | 0.0039** | 0.0071*** | -0.00084 | -0.0045*** | -0.0017 | -0.0012 |
| Low Rate | -0.087*** | -0.065*** | 0.0047*** | 0.035*** | 0.027*** | -0.046*** | -0.031*** | 0.061*** | 0.019*** | -0.010*** | -0.017*** | -0.0074*** | -0.0050*** |
| Low Rate Gold | -0.034*** | -0.034*** | -0.0062*** | 0.020*** | 0.012*** | -0.019*** | -0.012*** | 0.033*** | 0.012*** | -0.0025 | -0.0099*** | -0.0036** | -0.0024 |
| Platinum | -0.023*** | -0.020*** | -0.013*** | 0.016*** | 0.0050*** | -0.0037** | -0.0047*** | 0.033*** | 0.011*** | 0.002 | -0.0086*** | -0.0024 | -0.0016 |
| Student | -0.046*** | -0.033*** | 0.0029* | 0.013*** | 0.017*** | -0.022*** | -0.014*** | -0.0074*** | 0.011*** | -0.0065*** | -0.0050*** | -0.0019 | -0.0013 |
| Spend 6 month | 0.062*** | 0.045*** | -0.027*** | 0.022** | -0.012 | 0.13*** | 0.037*** | -0.033*** | -0.0058 | 0.019** | 0.002 | 0.081*** | 0.23*** |
| Spend 9 month | 0.058*** | 0.040*** | -0.034** | 0.01 | 0.00039 | 0.13*** | 0.023 | -0.050*** | -0.0025 | 0.0066 | 0.0069 | 0.059*** | 0.23*** |
| Retention 6 month | -0.018** | -0.032*** | 0.0061 | 0.11*** | 0.00011 | -0.064*** | -0.013* | 0.057*** | 0.029*** | 0.0019 | -0.037*** | 0.037*** | 0.024*** |
| Retention 9 month | -0.0096 | -0.032*** | 0.0065 | 0.095*** | 0.00024 | -0.063*** | -0.015* | 0.052*** | 0.026*** | 0.0052 | -0.038*** | 0.031*** | 0.021*** |
| Late pay 6 month | -0.046*** | -0.039*** | -0.025*** | 0.015* | -0.054*** | -0.050*** | -0.020** | 0.075*** | 0.013 | 0.0016 | -0.030*** | -0.027*** | 0.017** |
| Late pay 9 month | -0.062*** | -0.045*** | -0.0064 | 0.014 | -0.024 | -0.038** | -0.022 | 0.064*** | 0.018 | -0.012 | -0.014 | -0.041** | 0.0066 |

| | Got Gold | Got Low Fee | Got Low Fee Gold | Got Low Rate | Got Low Rate Gold | Got Platinum | Got Student | Spend 6 month | Spend 9 month | Retention 6 month | Retention 9 month | Late pay 6 month | Late pay 9 month |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Customer age | | | | | | | | | | | | | |
| Customer tenure | | | | | | | | | | | | | |
| Male indicator | | | | | | | | | | | | | |
| Transaction product | | | | | | | | | | | | | |
| Savings product | | | | | | | | | | | | | |
| Home loan product | | | | | | | | | | | | | |
| Home insurance policy | | | | | | | | | | | | | |
| Personal loan product | | | | | | | | | | | | | |
| Retirement product | | | | | | | | | | | | | |
| Motor insurance policy | | | | | | | | | | | | | |
| Term deposit product | | | | | | | | | | | | | |
| Awards | | | | | | | | | | | | | |
| Diamond | | | | | | | | | | | | | |
| Gold | 1 | | | | | | | | | | | | |
| Low Fee | -0.0030* | 1 | | | | | | | | | | | |
| Low Fee Gold | -0.0013 | -0.0030* | 1 | | | | | | | | | | |
| Low Rate | -0.0055*** | -0.013*** | -0.0057*** | 1 | | | | | | | | | |
| Low Rate Gold | -0.0027* | -0.0063*** | -0.0028* | -0.012*** | 1 | | | | | | | | |
| Platinum | -0.0018 | -0.0043*** | -0.0019 | -0.0080*** | 0.39** | 1 | | | | | | | |
| Student | -0.0015 | -0.0034** | -0.0015 | -0.0063*** | 0.31* | -0.0021 | 1 | | | | | | |
| Spend 6 month | 0.100*** | -0.027*** | 0.021** | -0.15*** | 0.79 | 0.077*** | -0.040*** | 1 | | | | | |
| Spend 9 month | 0.063*** | -0.026* | 0.022 | -0.12*** | 0.66 | 0.060*** | -0.037*** | 0.99*** | 1 | | | | |
| Retention 6 month | 0.023*** | 0.063*** | 0.026*** | -0.15*** | 0.46*** | 0.042*** | 0.038*** | 0.091*** | 0.067*** | 1 | | | |
| Retention 9 month | 0.028*** | 0.056*** | 0.024*** | -0.13*** | 0.36*** | 0.035*** | 0.029*** | 0.10*** | 0.085*** | 0.79*** | 1 | | |
| Late pay 6 month | -0.021** | -0.0055 | -0.000054 | 0.0063 | 0.8 | 0.019** | -0.0089 | -0.057*** | -0.071*** | 0.074*** | 0.089*** | 1 | |
| Late pay 9 month | 0.00041 | -0.041** | -0.0059 | 0.037** | 0.85 | 0.015 | 0.0046 | -0.042** | -0.051** | 0.053*** | 0.077*** | 0.30*** | 1 |

**Table A4.** Correlation Matrix for Independent and Dependent Variables.




| | All | | | Control | | | Treatment | | | Diff |
|---|---|---|---|---|---|---|---|---|---|---|
| | Missing | Total | % Missing | Missing | Total | % Missing | Missing | Total | % Missing | P-value |
| Customer age | 94 | 393,036 | 0.02% | 52 | 195,438 | 0.03 | 42 | 197,598 | 0.02 | 0.28 |
| Customer tenure (months) | 679 | 393,036 | 0.17% | 346 | 195,438 | 0.18 | 333 | 197,598 | 0.17 | 0.52 |
| Male indicator | 173 | 393,036 | 0.04% | 88 | 195,438 | 0.05 | 85 | 197,598 | 0.04 | 0.76 |
| Transaction product | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Savings product | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Home loan product | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Home insurance policy | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Personal loan product | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Retirement product | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Motor insurance policy | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |
| Term deposit product | 162 | 393,036 | 0.04% | 83 | 195,438 | 0.04 | 79 | 197,598 | 0.04 | 0.70 |

**Figure A2.** Randomization and Dropped Observations. P-values in parentheses indicate the statistical significance of the proportional difference in the number of dropped observations. For example, p=0.32 indicates that 258 dropped observations among 213,389 customers in the control condition is not significant from 285 dropped observations among 208,168 treated customers. Moreover, table reveals the number of control variables missing in the control and treatment conditions.

## Supplemental Acquisition Analyses

In the primary analyses presented in the paper, we present the controlled analyses for all customers who visited the credit card website during the experimental period, as described in Section 3.2.1. Here, we present two additional analyses. First, **Figures A3** and **A4** present split-sample analyses for customers with low experience levels (ages 28 and under) and high experience levels (over 28 years of age). Consistent with the primary results presented, we find no differences in net acquisition rates among customers in the treatment and control condition. Second, **Figure A5** replicates the primary results without control variables. This is important since controlled analyses will exclude some customers who did not start the application process, owing to missing control variables. Again, we note that the results are substantively similar if control variables are excluded from the analysis.

Finally, **Tables A5-A8** present the marginal effects and significance values for the acquisition analyses presented in figures. Rather than presenting the results from the interaction model shown as Equation 1 in the paper, we formulate the analyses in **Tables A5-A8** from conducting split sample analyses during the promotion and non-promotion periods. The results obtained are equivalent to the fully-interacted model, but they afford a clearer picture of the relative number of observations collected during the non-promotion and promotion periods.

| | No Promo | | | | | Promo | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Stage** | **n** | **Treatment/ no promotion** | **Control / no promotion** | **sig** | **p-val** | **n** | **Treatment/ promotion** | **Control/ promotion** | **sig** | **p-val** |
| Application Started | 145,030 | 14.55% | 14.40% | | 0.491 | 245,244 | 17.33% | 17.78% | *** | 0.001 |
| Soft submission | 145,030 | 11.67% | 11.50% | | 0.375 | 245,244 | 14.17% | 14.53% | *** | 0.007 |
| Submission | 142,319 | 4.73% | 4.74% | | 0.979 | 245,244 | 7.08% | 7.12% | | 0.711 |
| Opened | 145,030 | 3.54% | 3.47% | | 0.436 | 245,244 | 5.22% | 5.36% | | 0.122 |
| Activated | 145,030 | 3.29% | 3.27% | | 0.876 | 245,244 | 4.98% | 5.11% | | 0.122 |
| Used | 144,843 | 2.99% | 2.99% | | 0.963 | 245,244 | 4.65% | 4.75% | | 0.221 |

**Table A5.** Marginal Effects Results Table Corresponding with the Acquisition Funnel presented in Figure 4 in the manuscript. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively.

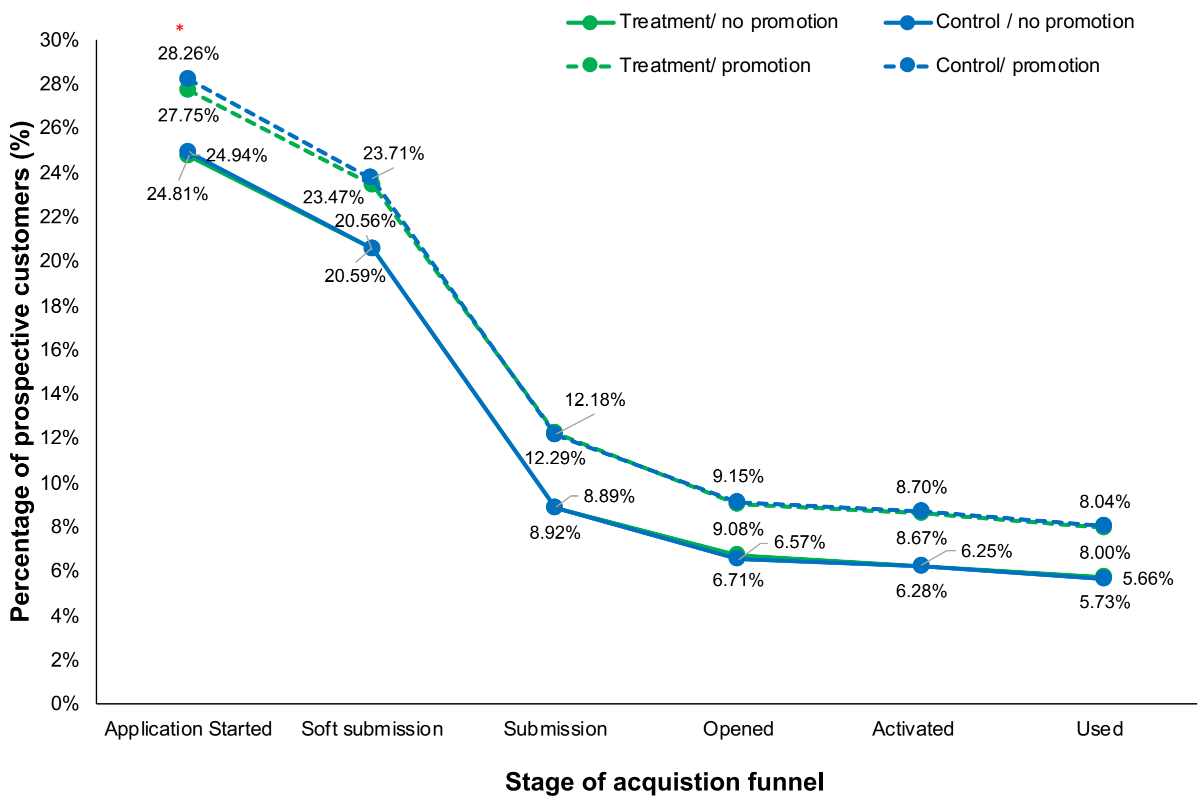



**Figure A3**: Acquisition funnel conversion percentages during the promotion and non-promotion periods for less-experienced customers (28 years of age and younger). Marginal effects are from logistic regression models, estimated with robust standard errors clustered at the first website visit date level. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. The results indicate that, consistent with Figure 5, conversion rates are significantly higher during the promotion period. Moreover, although fewer customers start the application process in the treatment condition during the promotion period, the treatment ultimately had no effect on rates of customer acquisition, neither during the promotion nor the non-promotion periods.

| | No Promo | | | | | Promo | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Stage** | **n** | **Treatment/ no promotion** | **Control / no promotion** | **sig** | **p-val** | **n** | **Treatment/ promotion** | **Control/ promotion** | **sig** | **p-val** |
| Application Started | 39,661 | 24.81% | 24.94% | | 0.760 | 70,831 | 27.75% | 28.26% | * | 0.078 |
| Soft submission | 39,661 | 20.59% | 20.56% | | 0.956 | 70,831 | 23.47% | 23.71% | | 0.371 |
| Submission | 38,661 | 8.92% | 8.89% | | 0.914 | 70,831 | 12.29% | 12.18% | | 0.609 |
| Opened | 39,661 | 6.71% | 6.57% | | 0.540 | 70,831 | 9.08% | 9.15% | | 0.732 |
| Activated | 39,538 | 6.28% | 6.25% | | 0.878 | 70,831 | 8.67% | 8.70% | | 0.894 |
| Used | 39,538 | 5.73% | 5.66% | | 0.728 | 70,831 | 8.00% | 8.04% | | 0.860 |

**Table A6.** Marginal Effects Results Table Corresponding with the Acquisition Funnel presented in Figure A3 in the Appendix. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively.

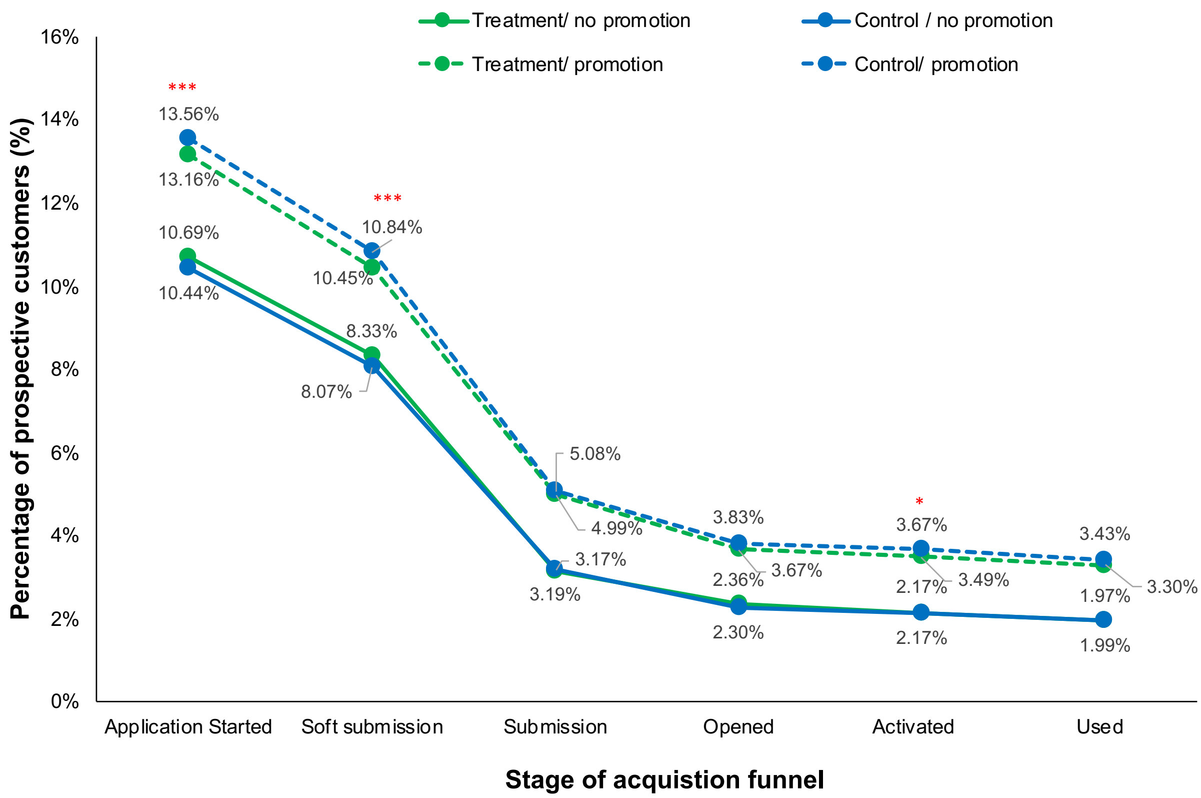



**Figure A4**: Acquisition funnel conversion percentages during the promotion and non-promotion periods for more-experienced customers (over 28 years of age). Marginal effects are from logistic regression models, estimated with robust standard errors clustered at the first website visit date level. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. The results indicate that, consistent with Figure 5, conversion rates are significantly higher during the promotion period. Moreover, although fewer customers start the application process or soft submit their application in the treatment condition during the promotion period, the treatment ultimately had no effect on rates of customer acquisition, neither during the promotion nor the non-promotion periods.

| | No Promo | | | | | Promo | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Stage** | **n** | **Treatment/ no promotion** | **Control / no promotion** | **sig** | **p-val** | **n** | **Treatment/ promotion** | **Control/ promotion** | **sig** | **p-val** |
| Application Started | 105,369 | 10.69% | 10.44% | | 0.267 | 176,276 | 13.16% | 13.56% | *** | 0.009 |
| Soft submission | 105,369 | 8.33% | 8.07% | | 0.167 | 176,276 | 10.45% | 10.84% | *** | 0.007 |
| Submission | 103,658 | 3.17% | 3.19% | | 0.900 | 176,276 | 4.99% | 5.08% | | 0.405 |
| Opened | 105,369 | 2.36% | 2.30% | | 0.542 | 176,276 | 3.67% | 3.83% | | 0.104 |
| Activated | 105,040 | 2.17% | 2.17% | | 0.957 | 176,276 | 3.49% | 3.67% | * | 0.064 |
| Used | 104,911 | 1.97% | 1.99% | | 0.787 | 176,276 | 3.30% | 3.43% | | 0.160 |

**Table A7.** Marginal Effects Results Table Corresponding with the Acquisition Funnel presented in Figure A4 in the Appendix. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively.

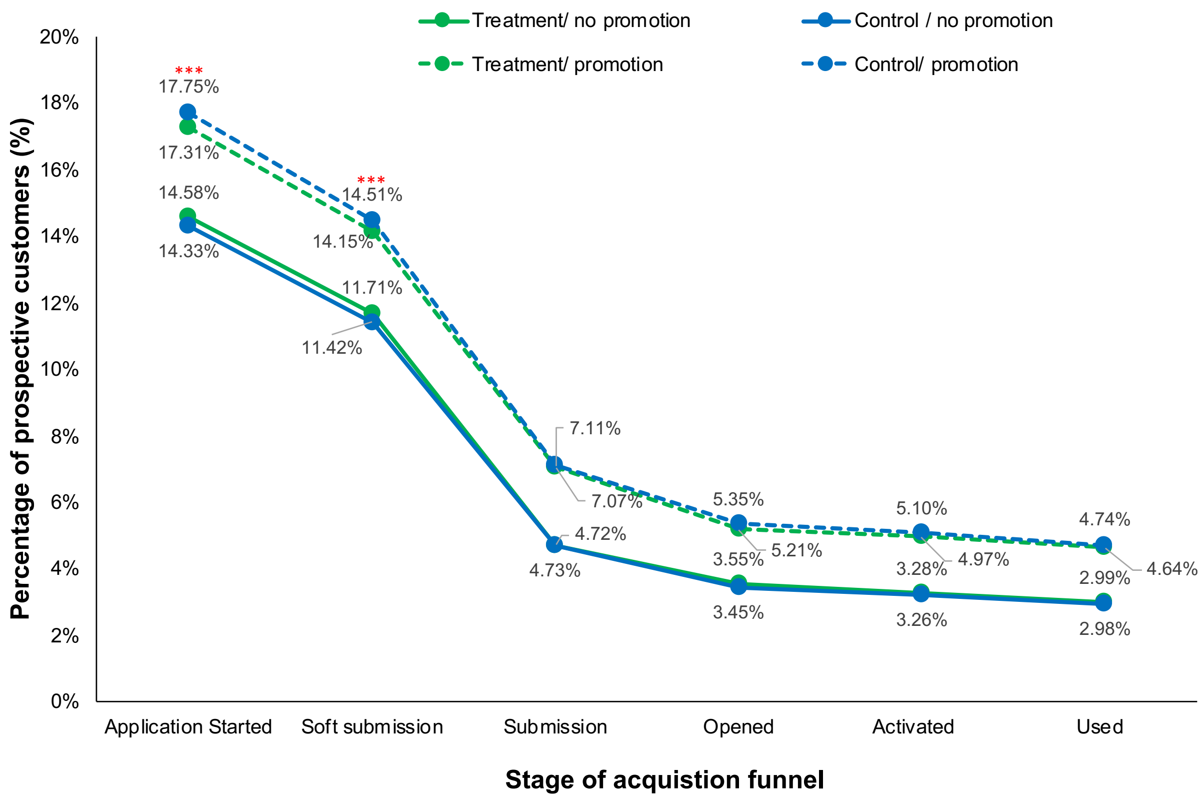



**Figure A5:** Acquisition funnel conversion percentages during the promotion (n=245,756) and non-promotion (n=145,412) periods excluding control variables. Date fixed effects are retained. This analysis enables the inclusion of observations for which control variables were missing. Beyond the acquisition phase, prospective customers were obliged to provide information about themselves, which entered into the controlled analyses. Marginal effects are from logistic regression models, estimated with robust standard errors clustered at the first website visit date level. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. The results indicate that conversion rates are significantly higher during the promotion period. Moreover, although fewer customers who are exposed to the treatment during the promotion period start or soft submit an application, the treatment ultimately had no effect on rates of customer acquisition, neither during the promotion nor the non-promotion periods.

| | No Promo | | | | | Promo | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Stage | n | Treatment/ no promotion | Control / no promotion | sig | p-val | n | Treatment/ promotion | Control/ promotion | sig | p-val |
| Application Started | 145,412 | 14.58% | 14.33% | | 0.297 | 245,756 | 17.31% | 17.75% | *** | 0.001 |
| Soft submission | 145,412 | 11.71% | 11.42% | | 0.168 | 245,756 | 14.15% | 14.51% | *** | 0.006 |
| Submission | 142,694 | 4.72% | 4.73% | | 0.959 | 245,756 | 7.07% | 7.11% | | 0.702 |
| Opened | 145,412 | 3.55% | 3.45% | | 0.343 | 245,756 | 5.21% | 5.35% | | 0.118 |
| Activated | 145,412 | 3.28% | 3.26% | | 0.833 | 245,756 | 4.97% | 5.10% | | 0.116 |
| Used | 145,224 | 2.99% | 2.98% | | 0.928 | 245,756 | 4.64% | 4.74% | | 0.215 |

**Table A8.** Marginal Effects Results Table Corresponding with the Acquisition Funnel presented in Figure A5 in the Appendix. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively.

## Supplemental Card Choice Analyses

In the primary analysis presented in the paper, **Table 2** shows the marginal effects that emerge from the results of the logistic regression models specified in **Equation (2)** in the paper and presented in regression table form in **Table A9**. The results demonstrate that, consistent with H2, customers make different choices in the presence of tradeoff transparency.

| | (1) Awards | (2) Diamond | (3) Gold | (4) Low Fee | (5) Low Fee Gold | (6) Low Rate | (7) Low Rate Gold | (8) Platinum | (9) Student |
|---|---|---|---|---|---|---|---|---|---|
| Treatment | -0.014 | -0.133 | -0.193 | 0.064 | 0.005 | -0.144** | 0.212** | 0.266** | -0.011 |
| | (0.125) | (0.157) | (0.149) | (0.064) | (0.166) | (0.067) | (0.086) | (0.108) | (0.136) |
| Promotion | -0.150 | -0.826*** | -0.437** | -0.517*** | -0.593*** | 0.498*** | 0.170 | -0.084 | -0.325** |
| | (0.130) | (0.273) | (0.179) | (0.087) | (0.175) | (0.062) | (0.107) | (0.136) | (0.154) |
| Treatment x promotion | 0.075 | 0.286 | 0.310* | -0.126 | 0.170 | 0.088 | -0.162 | -0.221 | 0.061 |
| | (0.150) | (0.210) | (0.188) | (0.084) | (0.206) | (0.077) | (0.101) | (0.137) | (0.173) |
| Customer age | 0.024 | 0.060 | 0.000 | -0.035*** | -0.002 | 0.035*** | 0.039*** | 0.116*** | -0.400*** |
| | (0.023) | (0.037) | (0.027) | (0.011) | (0.022) | (0.009) | (0.012) | (0.021) | (0.027) |
| Customer age² | -0.000 | -0.001** | -0.000 | 0.000*** | 0.000 | -0.000*** | -0.000 | -0.001*** | 0.003*** |
| | (0.000) | (0.001) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Customer tenure | 0.004*** | -0.005*** | 0.002 | -0.001* | -0.004*** | 0.002*** | -0.001 | -0.000 | -0.008*** |
| | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.000) | (0.001) | (0.001) | (0.001) |
| Customer tenure² | -0.000** | 0.000*** | -0.000 | 0.000 | 0.000 | -0.000*** | 0.000 | 0.000 | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.027 | -0.724*** | -0.280*** | 0.097** | -0.373*** | 0.216*** | -0.135*** | -0.498*** | 0.202** |
| | (0.069) | (0.123) | (0.096) | (0.038) | (0.092) | (0.033) | (0.047) | (0.065) | (0.080) |
| Retirement product | 0.263** | 0.061 | -0.201 | -0.010 | -0.047 | -0.090 | 0.080 | 0.236** | -0.124 |
| | (0.115) | (0.180) | (0.179) | (0.073) | (0.172) | (0.058) | (0.085) | (0.107) | (0.144) |
| Home loan product | -0.099 | 1.794*** | 0.233 | -0.106 | -0.101 | -0.299*** | -0.075 | 0.510*** | -2.804*** |
| | (0.118) | (0.128) | (0.152) | (0.081) | (0.177) | (0.052) | (0.088) | (0.110) | (1.005) |
| Personal loan product | -0.257*** | -0.456*** | -0.470*** | -0.576*** | -0.419*** | 0.295*** | 0.235*** | 0.518*** | -1.487*** |
| | (0.097) | (0.147) | (0.144) | (0.068) | (0.128) | (0.045) | (0.052) | (0.081) | (0.171) |
| Savings product | 0.296*** | -0.030 | 0.445*** | 0.034 | 0.100 | -0.112*** | 0.002 | -0.073 | 0.261** |
| | (0.083) | (0.110) | (0.132) | (0.051) | (0.097) | (0.038) | (0.059) | (0.075) | (0.130) |
| Term deposit product | 0.309 | 0.507* | 0.433 | 0.457*** | -0.148 | -0.159 | -0.412** | -0.942*** | 0.197 |
| | (0.224) | (0.302) | (0.285) | (0.148) | (0.347) | (0.099) | (0.172) | (0.365) | (0.343) |
| Transaction product | 0.169 | -0.162 | -0.447* | 0.167 | 0.010 | -0.438*** | 0.405*** | 0.928*** | 1.017* |
| | (0.240) | (0.364) | (0.245) | (0.127) | (0.267) | (0.091) | (0.140) | (0.305) | (0.597) |
| Home insurance policy | 0.114 | 0.342* | -0.035 | -0.144 | -0.072 | -0.040 | 0.077 | 0.056 | -1.381** |
| | (0.147) | (0.201) | (0.197) | (0.089) | (0.250) | (0.067) | (0.108) | (0.138) | (0.700) |
| Motor insurance policy | 0.122 | -0.175 | 0.614*** | -0.042 | 0.258 | -0.152* | -0.037 | 0.174 | -0.063 |
| | (0.208) | (0.283) | (0.206) | (0.135) | (0.258) | (0.082) | (0.127) | (0.171) | (0.413) |
| Constant | -3.724*** | -3.251*** | -2.830*** | -0.810*** | -2.639*** | -0.746*** | -3.327*** | -5.929*** | 3.941*** |
| | (0.498) | (0.872) | (0.583) | (0.283) | (0.495) | (0.205) | (0.277) | (0.578) | (0.662) |
| Observations | 18,152 | 18,152 | 18,152 | 18,152 | 18,152 | 18,152 | 18,152 | 18,152 | 18,152 |
| Sample | All | All | All | All | All | All | All | All | All |
| R-squared | 0.200 | 0.200 | 0.200 | 0.200 | 0.200 | 0.200 | 0.200 | 0.200 | 0.200 |

**Table A9:** Effects of the treatment on the credit card choices of customers. Columns (1)-(9) show the results of the logistic regression models with robust standard errors clustered by customers' first website visit dates. All models additionally include indicator variables for the week during which the customer first visited the credit card marketing website, which are withheld from the table for parsimony. Indicator variables for *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. Low rate and low rate gold cards were part of the publicly announced promotion campaign beginning on November 1, 2017. The results indicate that providing transparency into the tradeoffs of the service offering influenced credit card choices, which is consistent with H2.

An alternative question is whether tradeoff transparency affected customers' choices for the better. In the paper, we present results demonstrating that customers use the cards more, are less likely to make late payments, and are more likely to keep the cards longer, if they are provided with tradeoff transparency when they are choosing credit cards. These results serve as our primary evidence that tradeoff transparency affects customers' choices for the better, since these are behavioral indicators that suggest that customers find the credit cards they choose in the presence of tradeoff transparency to be more valuable, and less burdensome to their financial health. An alternative way of assessing the quality of customer choices is to examine whether, based on observable customer-level characteristics, the cards customers select in the presence of tradeoff transparency appear to be in better alignment with their presumed financial needs than the cards customers select in the absence of tradeoff transparency.

In this supplemental analysis, we considered two observable demographic variables – customer age and total count of financial products. We presume that:

1. The student card might be disproportionately more aligned with the needs and preferences of younger customers, who are more likely to be students and have basic financial needs.

2. The low rate card might be disproportionately more aligned with the needs and preferences of customers with a less-developed financial position, who have fewer banking products, and who in turn may wish to use their credit card for borrowing, carrying a balance from month to month.

To test these hypotheses, we consider the interaction of the treatment variable with continuous variables measuring the customer's age, and their total count of financial products at the time they first visited the bank's credit card marketing website. In particular, we use separate regression specifications for analyzing the moderating effects of tradeoff transparency on influencing the choices of customers who vary in age and the breadth of their financial relationship with the bank.

In the first specification presented below, $\beta_3$ captures the degree to which tradeoff transparency has a differential effect on older customers to choose the student card.

$$\Pr\left(STUDENT_i\right) = f\left(\begin{array}{c} \beta_0 + \beta_1 TREAT_i + \beta_2 AGE + \beta_3 TREAT_i \times AGE_i + \\ \beta_4 TENURE_i + \beta_5 TENURE_i^2 + \beta_6 GENDER_i + \\ \beta_7 HLOAN_i + \beta_8 PLOAN_i + \beta_9 TRANS_i + \beta_{10} SAV_i + \beta_{11} HINS_i + \\ \beta_{12} VINS_i + \beta_{13} RET_i + X_i + \epsilon_i \end{array}\right) \tag{2}$$

If tradeoff transparency increases the probability that younger customers will select the student card, as predicted above, then $\beta_3$ will be negative and significant. In the second specification, $\gamma_3$ captures the degree to which tradeoff transparency has a differential effect on customers with broader financial relationships to select the low rate card.

$$\Pr\left(LOWRATE_i\right) = f\left(\begin{array}{c} \gamma_0 + \gamma_1 TREAT_i + \gamma_2 PRODUCTS_i + \gamma_3 TREAT_i \times PRODUCTS_i + \\ \gamma_4 AGE_i + \gamma_5 AGE_i^2 + \gamma_6 TENURE_i + \gamma_7 TENURE_i^2 + \gamma_8 GENDER_i + \\ \gamma_9 HLOAN_i + \gamma_{10} PLOAN_i + \gamma_{11} TRANS_i + \gamma_{12} SAV_i + \gamma_{13} HINS_i + \\ \gamma_{14} VINS_i + \gamma_{15} RET_i + X_i + \epsilon_i \end{array}\right) \tag{3}$$

If tradeoff transparency increases the probability that customers with narrower financial relationships with the bank will select the low rate card, then we would predict $\gamma_3$ will be negative and significant.

| | (1) Pr(Student) | (2) Pr(Student) | (3) Pr(Student) | (4) Pr(Low rate) | (5) Pr(Low rate) | (6) Pr(Low rate) |
|---|---|---|---|---|---|---|
| Treatment | -0.163 | 0.295 | -0.363 | 0.067 | 0.257 | -0.014 |
| | (0.603) | (1.003) | (0.779) | (0.085) | (0.180) | (0.094) |
| Customer age | -0.225*** | -0.219*** | -0.226*** | 0.037*** | 0.001 | 0.047*** |
| | (0.018) | (0.030) | (0.023) | (0.009) | (0.017) | (0.010) |
| Treatment x customer age | 0.008 | -0.013 | 0.018 | | | |
| | (0.026) | (0.044) | (0.033) | | | |
| Product count | | | | -0.086 | -0.434 | -0.059 |
| | | | | (0.100) | (0.278) | (0.113) |
| Treatment x product count | | | | -0.069* | -0.179*** | -0.021 |
| | | | | (0.035) | (0.068) | (0.039) |
| Customer age$^2$ | | | | -0.000*** | -0.000 | -0.001*** |
| | | | | (0.000) | (0.000) | (0.000) |
| Tenure | -0.008*** | -0.007*** | -0.009*** | 0.002*** | 0.000 | 0.002*** |
| | (0.002) | (0.002) | (0.002) | (0.000) | (0.001) | (0.001) |
| Tenure$^2$ | 0.000*** | 0.000** | 0.000*** | -0.000*** | 0.000 | -0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Gender | 0.203** | 0.257* | 0.169* | 0.204*** | 0.259*** | 0.197*** |
| | (0.079) | (0.136) | (0.099) | (0.033) | (0.069) | (0.038) |
| Home loan indicator | -2.806*** | -1.611 | | -0.169 | 0.263 | -0.239* |
| | (1.004) | (1.003) | | (0.114) | (0.295) | (0.132) |
| Home insurance indicator | -1.371* | | -0.815 | 0.073 | 0.503* | 0.021 |
| | (0.702) | | (0.692) | (0.114) | (0.298) | (0.129) |
| Motor insurance indicator | -0.054 | 0.210 | -0.269 | -0.038 | 0.435 | -0.093 |
| | (0.410) | (0.614) | (0.574) | (0.138) | (0.357) | (0.156) |
| Personal loan indicator | -1.449*** | -1.681*** | -1.342*** | 0.374*** | 1.174*** | 0.201* |
| | (0.170) | (0.266) | (0.219) | (0.107) | (0.277) | (0.119) |
| Transaction product indicator | 1.017* | | 0.681 | -0.347*** | 0.604* | -0.532*** |
| | (0.585) | | (0.585) | (0.132) | (0.334) | (0.146) |
| Savings product indicator | 0.259** | 0.465** | 0.141 | 0.012 | 0.322 | -0.018 |
| | (0.126) | (0.217) | (0.156) | (0.103) | (0.265) | (0.119) |
| Retirement product indicator | -0.102 | -0.062 | -0.164 | 0.011 | 0.347 | 0.010 |
| | (0.143) | (0.244) | (0.174) | (0.120) | (0.312) | (0.140) |
| Term deposit indicator | 0.163 | 0.534 | 0.043 | | | |
| | (0.344) | (0.679) | (0.399) | | | |
| Constant | 1.751*** | 2.701*** | 2.098*** | -0.292 | -0.611* | -0.177 |
| | (0.663) | (0.732) | (0.722) | (0.187) | (0.325) | (0.227) |
| Sample | All | Non-promo | Promo | All | Non-promo | Promo |
| Observations | 18,152 | 4,678 | 11,632 | 18,152 | 5,087 | 13,065 |

**Table A10:** Moderating effects of tradeoff transparency on customers who vary by age and count of financial products. Models estimated with logistic regression with robust standard errors clustered by customers' first website visit dates. All models additionally include indicator variables for the week during which the customer first visited the credit card marketing website, which are withheld from the table for parsimony. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively. Models are estimated on the full sample of observations, and on subsamples of customer observations for customers who applied outside and inside the promotion period. Due to the logistic regression models, observations for which results were perfectly predicted by particular covariates were dropped. The results provide evidence that customers with fewer financial products were more likely to select the Low Rate card in the presence of tradeoff transparency, especially when a promotion was not taking place. Although younger customers were more likely to select the Student card, their propensity to do so did not depend upon the presence of tradeoff transparency.

As shown in **Table A10**, we do not find that younger customers are disproportionately likely to select the student card in the presence of tradeoff transparency, but we do find that customers with fewer financial products are more likely to select the low rate card in the presence of tradeoff transparency, especially during the non-promotion period.

Our intuition is that the student card is much more clearly branded than the other two cards as being “for students,” and hence, the addition of tradeoff transparency has no effect on its relative appeal to younger customers. It may also be the case that since we find that tradeoff transparency is more effective for older customers who have more prior experience with credit cards, younger customers didn’t get as much benefit from the tradeoff messaging. Nevertheless, the fact that the predicted results hold for customers with fewer products is directionally consistent with the idea that tradeoff transparency may help customers make *better* choices. We note these results are also consistent with the results of our tests of H3 and H4.

Finally, consistent with prior research demonstrating how customer compatibility can lead to more satisfying and profitable customer relationships, it is our account that through its capacity to inform customer decisions, tradeoff transparency drives profitability by enhancing customer engagement. However, an alternative account is that tradeoff transparency may lead companies to attract more profitable customers in general. Since all of the primary analyses presented in this paper control for observable customer characteristics, the results are consistent with the idea that otherwise-similar customers will exhibit higher levels of long-term engagement with they are provided with tradeoff transparency as a part of the customer acquisition process. To test whether tradeoff transparency may also lead to the attraction of more or less profitable customers, we analyze all customers who opened credit card accounts during the experiment, analyzing how the observable characteristics (age, tenure, and product ownership) differ among customers who were motivated to open accounts in the treatment and control conditions.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Age | Tenure | Male | Home loan | Home insurance | Motor insurance | Personal loan | Transaction product | Savings product | Retirement product | Term deposit |
| Treatment | -0.132 | -3.799 | 0.057 | -0.038 | 0.045 | -0.003 | 0.092 | 0.217 | 0.012 | 0.176* | 0.305 |
| | (0.254) | (3.039) | (0.057) | (0.076) | (0.136) | (0.159) | (0.059) | (0.189) | (0.060) | (0.103) | (0.263) |
| Promo | 0.575** | -4.681* | -0.031 | -0.015 | -0.094 | -0.182 | -0.254*** | -0.297** | -0.016 | -0.040 | 0.418** |
| | (0.233) | (2.686) | (0.045) | (0.075) | (0.120) | (0.150) | (0.065) | (0.141) | (0.056) | (0.088) | (0.206) |
| Treatment x promo | -0.401 | 1.831 | -0.033 | 0.106 | 0.083 | 0.083 | -0.167** | 0.029 | -0.019 | -0.239** | -0.138 |
| | (0.330) | (3.699) | (0.065) | (0.097) | (0.164) | (0.187) | (0.076) | (0.212) | (0.077) | (0.121) | (0.291) |
| Constant | 31.107*** | 141.064*** | -0.234*** | -2.112*** | -2.822*** | -3.364*** | -1.221*** | 3.628*** | 1.192*** | -2.426*** | -4.382*** |
| | (0.182) | (2.191) | (0.039) | (0.062) | (0.097) | (0.129) | (0.054) | (0.123) | (0.045) | (0.075) | (0.180) |
| Observations | 18,157 | 18,152 | 18,157 | 18,157 | 18,157 | 18,157 | 18,157 | 18,157 | 18,157 | 18,157 | 18,157 |
| Model | OLS | OLS | Logistic | Logistic | Logistic | Logistic | Logistic | Logistic | Logistic | Logistic | Logistic |
| R-squared (Pseudo R2) | 0.001 | 0.000 | 0.000145 | 0.000174 | 0.000441 | 0.000568 | 0.00412 | 0.00316 | 2.55e-05 | 0.00114 | 0.00309 |

**Table A11:** Comparison of Characteristics of Customers Attracted in the Presence and Absence of Tradeoff Transparency. Columns (1-2) are estimated with OLS regression and Columns (3-11) are estimated with logistic regression, all with robust standard errors, clustered by date of first website visit, shown in parentheses. models additionally include indicator variables for the week during which the customer first visited the credit card marketing website, which are withheld from the table for parsimony. *, **, and ***, signify significance at the 10%, 5%, and 1% levels respectively.

As demonstrated in **Table A11**, we observe that the bank attracted customers that were insignificantly different in the presence and absence of tradeoff transparency, with the exception that customers attracted in the presence of tradeoff transparency were marginally more likely to have a retirement product – the baseline probability increasing from 8.12% to 9.54% ($\beta$=0.176, p<0.10). However, we do observe that in the presence of a promotion, customers attracted tended to be older ($\beta$=0.575, p<0.05) and marginally newer to the bank ($\beta$=-4.681, p<0.10), and were less likely to have a personal loan account ($\beta$=-0.254, p<0.01 a transaction product ($\beta$ =-0.297, p<0.01), but were more likely to have a term deposit product ($\beta$ =0.418, p<0.05).

Customer Engagement Robustness Analysis: Considering Alternative Experience Delineations

In the primary analysis presented in the manuscript, a median split on age is used to distinguish customers with more and less prior credit card experience. **Tables A12-13** demonstrate that the results are substantively similar if the 25th percentile of age (24 years old) is instead used as the delineator.

| | (1) Ln(Spend) | (2) Ln(Spend) | (3) Ln(Spend) | (4) Ln(Spend) | (5) Ln(Spend) | (6) Ln(Spend) |
|---|---|---|---|---|---|---|
| Treatment | 0.080 | -0.008 | 0.128** | 0.095* | 0.007 | 0.149*** |
| | (0.052) | (0.138) | (0.060) | (0.054) | (0.144) | (0.057) |
| Promotion | -0.104 | 0.209 | -0.251** | -0.371*** | 0.107 | -0.588*** |
| | (0.103) | (0.185) | (0.118) | (0.114) | (0.172) | (0.129) |
| Treatment x promotion | -0.097 | -0.034 | -0.138* | -0.109* | -0.039 | -0.160** |
| | (0.061) | (0.153) | (0.071) | (0.064) | (0.158) | (0.070) |
| Customer age | 0.072*** | 0.556 | 0.050*** | 0.046*** | 0.646 | 0.042*** |
| | (0.008) | (0.408) | (0.014) | (0.009) | (0.409) | (0.014) |
| Customer age² | -0.001*** | -0.012 | -0.001*** | -0.001*** | -0.014 | -0.000*** |
| | (0.000) | (0.010) | (0.000) | (0.000) | (0.010) | (0.000) |
| Customer tenure | 0.002*** | 0.006*** | 0.000 | 0.000 | 0.005*** | -0.001** |
| | (0.000) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| Customer tenure² | -0.000 | -0.000*** | 0.000 | 0.000 | -0.000** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.050 | 0.069 | 0.031 | 0.072** | 0.081 | 0.055 |
| | (0.034) | (0.058) | (0.037) | (0.034) | (0.059) | (0.039) |
| Retirement product | 0.067 | -0.010 | 0.123 | 0.105* | -0.003 | 0.182** |
| | (0.058) | (0.095) | (0.076) | (0.058) | (0.096) | (0.077) |
| Home loan product | 0.509*** | 0.546** | 0.516*** | 0.349*** | 0.209 | 0.370*** |
| | (0.049) | (0.219) | (0.053) | (0.062) | (0.229) | (0.063) |
| Personal loan product | -0.324*** | -0.365*** | -0.308*** | -0.209*** | -0.369*** | -0.151*** |
| | (0.039) | (0.082) | (0.048) | (0.043) | (0.084) | (0.052) |
| Savings product | 0.277*** | 0.402*** | 0.235*** | 0.247*** | 0.376*** | 0.208*** |
| | (0.040) | (0.084) | (0.043) | (0.040) | (0.080) | (0.043) |
| Term deposit product | 0.198* | 0.558*** | 0.100 | -0.023 | 0.503** | -0.150 |
| | (0.120) | (0.196) | (0.136) | (0.133) | (0.218) | (0.151) |
| Transaction product | 0.496*** | 0.105 | 0.527*** | 1.043*** | 0.990** | 1.026*** |
| | (0.105) | (0.445) | (0.111) | (0.101) | (0.455) | (0.108) |
| Home insurance policy | -0.038 | 0.158 | -0.044 | -0.044 | 0.332 | -0.070 |
| | (0.076) | (0.235) | (0.080) | (0.085) | (0.235) | (0.091) |
| Motor insurance policy | 0.045 | 0.189 | 0.044 | 0.038 | 0.151 | 0.045 |
| | (0.105) | (0.239) | (0.114) | (0.108) | (0.245) | (0.117) |
| Constant | 3.348*** | -2.057 | 4.010*** | 3.455*** | -3.696 | 3.843*** |
| | (0.212) | (4.371) | (0.304) | (0.245) | (4.311) | (0.331) |
| Observations | 121,679 | 36,540 | 85,139 | 126,658 | 37,478 | 89,180 |
| Customers | 15,942 | 4,686 | 11,256 | 15,942 | 4,686 | 11,256 |
| Data treatment for closed accounts | Missing | Missing | Missing | Zero | Zero | Zero |
| Sample | All | Age≤24 | Age>24 | All | Age≤24 | Age>24 |
| R-squared | 0.0321 | 0.0463 | 0.0277 | 0.0271 | 0.0403 | 0.0292 |
| Pred(Y): Non-Promotion: Control | $196.64 | $132.62 | $235.76 | $192.67 | $124.92 | $235.17 |
| Pred(Y): Non-Promotion: Treatment | $212.98 | $131.62 | $268.03 | $211.79 | $125.79 | $272.87 |
| Pred(Y): Promotion: Control | $177.20 | $163.51 | $183.48 | $132.91 | $139.02 | $130.63 |
| Pred(Y): Promotion: Treatment | $174.20 | $156.93 | $181.69 | $130.99 | $134.57 | $129.23 |

**Table A12:** Effects of the treatment on monthly spend with alternative age cutoff (24 years). Columns (1) – (3) show the results of panel data regression models for all customers as well as younger and older customers, with spend values for months after cancellation set to missing. Columns (4) – (6) show the same specifications with spend values for months after cancellation set to zero. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. Consistent with H3, H5, and H7, the results indicate that customers exposed to the treatment spent more on their cards, that these effects were stronger for more-experienced customers, and that these effects were attenuated by the promotion.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | Pr(Retain6) | Pr(Retain6) | Pr(Retain6) | Pr(Retain9) | Pr(Retain9) | Pr(Retain9) |
| Treatment | 0.259 | 0.234 | 0.314 | 0.249** | 0.247 | 0.274** |
| | (0.202) | (0.389) | (0.208) | (0.115) | (0.212) | (0.131) |
| Promotion | -1.736*** | -1.321* | -1.807*** | -1.344*** | -0.277 | -1.658*** |
| | (0.256) | (0.758) | (0.248) | (0.231) | (0.490) | (0.237) |
| Treatment x promotion | -0.239 | -0.115 | -0.313 | -0.223 | -0.033 | -0.306 |
| | (0.220) | (0.426) | (0.229) | (0.165) | (0.293) | (0.186) |
| Customer age | -0.219*** | 0.634 | -0.072* | -0.094** | -0.573 | -0.065 |
| | (0.038) | (0.975) | (0.042) | (0.041) | (1.078) | (0.053) |
| Customer age² | 0.003*** | -0.018 | 0.001** | 0.001*** | 0.015 | 0.001* |
| | (0.001) | (0.023) | (0.001) | (0.001) | (0.026) | (0.001) |
| Customer tenure | -0.009*** | -0.011*** | -0.009*** | -0.005*** | -0.007 | -0.005*** |
| | (0.001) | (0.004) | (0.001) | (0.002) | (0.005) | (0.002) |
| Customer tenure² | 0.000*** | 0.000* | 0.000*** | 0.000*** | 0.000 | 0.000** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.046 | 0.042 | 0.049 | 0.078 | 0.288* | 0.005 |
| | (0.074) | (0.157) | (0.080) | (0.091) | (0.168) | (0.123) |
| Retirement product | 0.245 | 0.131 | 0.284* | -0.107 | -0.418 | 0.036 |
| | (0.154) | (0.322) | (0.162) | (0.155) | (0.258) | (0.194) |
| Home loan product | -0.595*** | -1.322*** | -0.556*** | -0.564*** | -1.118** | -0.534*** |
| | (0.105) | (0.334) | (0.111) | (0.167) | (0.464) | (0.175) |
| Personal loan product | 0.767*** | -0.005 | 0.968*** | 0.246** | -0.250 | 0.425*** |
| | (0.110) | (0.205) | (0.142) | (0.106) | (0.236) | (0.145) |
| Savings product | -0.188** | -0.173 | -0.163** | -0.132 | 0.160 | -0.194* |
| | (0.080) | (0.231) | (0.078) | (0.096) | (0.203) | (0.103) |
| Term deposit product | -0.617*** | -0.122 | -0.658*** | -0.662*** | -0.383 | -0.657** |
| | (0.167) | (0.621) | (0.181) | (0.235) | (0.637) | (0.265) |
| Transaction product | 1.603*** | 2.696*** | 1.468*** | 1.402*** | 1.976*** | 1.344*** |
| | (0.125) | (0.494) | (0.133) | (0.222) | (0.641) | (0.254) |
| Home insurance policy | 0.084 | 1.709* | 0.012 | -0.068 | | -0.206 |
| | (0.154) | (1.035) | (0.162) | (0.170) | | (0.179) |
| Motor insurance policy | 0.082 | 0.103 | 0.087 | 0.027 | 0.504 | 0.058 |
| | (0.198) | (0.630) | (0.209) | (0.202) | (0.932) | (0.222) |
| Constant | 7.509*** | -2.371 | 4.791*** | 3.739*** | 7.499 | 3.145*** |
| | (0.711) | (10.550) | (0.809) | (0.773) | (11.368) | (1.068) |
| Observations | 15,942 | 4,278 | 11,256 | 6,314 | 1,924 | 4,357 |
| Customers | All | Age≤24 | Age>24 | All | Age≤24 | Age>24 |
| Pseudo R2 | 0.0845 | 0.0666 | 0.0883 | 0.0551 | 0.0462 | 0.0760 |
| Pred(Y): Non-Promotion: Control | 98.07% | 98.19% | 97.90% | 98.07% | 98.19% | 97.90% |
| Pred(Y): Non-Promotion: Treatment | 98.50% | 98.56% | 98.45% | 98.50% | 98.56% | 98.45% |
| Pred(Y): Promotion: Control | 90.50% | 93.87% | 89.07% | 90.50% | 93.87% | 89.07% |
| Pred(Y): Promotion: Treatment | 90.67% | 94.49% | 89.08% | 90.67% | 94.49% | 89.08% |

**Table A13:** Effects of the treatment on customer retention rates with alternative age cutoff (24 years). Effects of the treatment on customer retention rates. Columns (1) – (3) and columns (4) – (6) show the results of cross-sectional data regression models of all, younger, and older customers for retention six and nine months after card activation, respectively. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that transparency can lead to increased customer retention, which is consistent with H4, that the effects are strongest for more-experienced customers, which is consistent with H6, and that the effects are dampened during the promotion, which is consistent with H8.

## Customer Engagement Robustness Analysis: Inconsistent Benefits Information

In the primary analysis presented in the manuscript, we excluded observations from customers who observed inconsistent benefits information during their deliberation process. **Tables A14-15** reveal that the results are substantively similar if those observations are included in the analysis.

| | (1)<br>Ln(Spend) | (2)<br>Ln(Spend) | (3)<br>Ln(Spend) | (4)<br>Ln(Spend) | (5)<br>Ln(Spend) | (6)<br>Ln(Spend) |
|---|---|---|---|---|---|---|
| Treatment | 0.093* | 0.051 | 0.148** | 0.108** | 0.047 | 0.190*** |
| | (0.051) | (0.088) | (0.061) | (0.053) | (0.094) | (0.054) |
| Promotion | -0.083 | 0.026 | -0.194 | -0.348*** | -0.139 | -0.556*** |
| | (0.099) | (0.131) | (0.146) | (0.111) | (0.125) | (0.162) |
| Treatment x promotion | -0.107* | -0.076 | -0.159** | -0.121* | -0.070 | -0.198*** |
| | (0.060) | (0.104) | (0.075) | (0.062) | (0.111) | (0.073) |
| Customer age | 0.070*** | 0.116 | 0.008 | 0.043*** | 0.164 | 0.027 |
| | (0.008) | (0.127) | (0.018) | (0.009) | (0.135) | (0.019) |
| Customer age² | -0.001*** | -0.002 | -0.000 | -0.000*** | -0.003 | -0.000 |
| | (0.000) | (0.003) | (0.000) | (0.000) | (0.003) | (0.000) |
| Customer tenure | 0.002*** | 0.005*** | -0.001 | 0.000 | 0.004*** | -0.002*** |
| | (0.000) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| Customer tenure² | -0.000* | -0.000*** | 0.000* | 0.000 | -0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.040 | 0.093** | -0.012 | 0.062* | 0.101** | 0.020 |
| | (0.033) | (0.045) | (0.042) | (0.033) | (0.044) | (0.044) |
| Retirement product | 0.066 | 0.027 | 0.162* | 0.094* | 0.032 | 0.229** |
| | (0.057) | (0.076) | (0.085) | (0.057) | (0.077) | (0.094) |
| Home loan product | 0.504*** | 0.730*** | 0.462*** | 0.344*** | 0.514*** | 0.323*** |
| | (0.048) | (0.127) | (0.060) | (0.061) | (0.140) | (0.070) |
| Personal loan product | -0.337*** | -0.459*** | -0.232*** | -0.223*** | -0.401*** | -0.073 |
| | (0.039) | (0.060) | (0.053) | (0.042) | (0.062) | (0.056) |
| Savings product | 0.277*** | 0.408*** | 0.178*** | 0.244*** | 0.376*** | 0.147*** |
| | (0.039) | (0.060) | (0.050) | (0.040) | (0.061) | (0.051) |
| Term deposit product | 0.180 | 0.382** | 0.059 | -0.025 | 0.240 | -0.165 |
| | (0.116) | (0.157) | (0.154) | (0.130) | (0.183) | (0.165) |
| Transaction product | 0.447*** | 0.155 | 0.541*** | 0.987*** | 1.081*** | 0.937*** |
| | (0.102) | (0.220) | (0.118) | (0.100) | (0.228) | (0.121) |
| Home insurance policy | -0.023 | 0.036 | -0.029 | -0.023 | 0.149 | -0.066 |
| | (0.076) | (0.160) | (0.088) | (0.085) | (0.169) | (0.098) |
| Motor insurance policy | 0.037 | 0.121 | 0.024 | 0.032 | 0.142 | 0.013 |
| | (0.106) | (0.164) | (0.124) | (0.109) | (0.168) | (0.122) |
| Constant | 3.448*** | 2.704* | 5.045*** | 3.561*** | 1.561 | 4.381*** |
| | (0.204) | (1.560) | (0.389) | (0.239) | (1.639) | (0.422) |
| Observations | 125,902 | 63,661 | 62,241 | 131,055 | 65,737 | 65,318 |
| Number of Customers | 16,487 | 8,243 | 8,244 | 16,487 | 8,243 | 8,244 |
| Data treatment for closed accounts | Missing | Missing | Missing | Zero | Zero | Zero |
| Sample | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| R-squared | 0.0316 | 0.0467 | 0.0244 | 0.026 | 0.037 | 0.028 |
| Pred(Y): Non-Promotion: Control | $196.15 | $161.56 | $240.16 | $191.83 | $154.04 | $240.16 |
| Pred(Y): Non-Promotion: Treatment | $215.36 | $170.07 | $278.58 | $213.75 | $161.40 | $290.41 |
| Pred(Y): Promotion: Control | $180.53 | $165.80 | $197.75 | $135.42 | $134.09 | $137.66 |
| Pred(Y): Promotion: Treatment | $178.11 | $161.69 | $195.70 | $133.74 | $130.96 | $136.63 |

**Table A14:** Effects of the treatment on monthly spend including customers who viewed inconsistent benefits information for a type of the low fee card. Columns (1) – (3) show the results of panel data regression models for all customers as well as younger and older customers, with spend values for months after cancellation set to missing. Columns (4) – (6) show the same specifications with spend values for months after cancellation set to zero. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that customers exposed to the treatment spent more on their cards, that these effects were stronger for more-experienced customers, and that these effects were attenuated by the promotion, which is consistent with H3, H5, and H7.

| | (1) Pr(Retain6) | (2) Pr(Retain6) | (3) Pr(Retain6) | (4) Pr(Retain9) | (5) Pr(Retain9) | (6) Pr(Retain9) |
|---|---|---|---|---|---|---|
| Treatment | 0.281 | 0.084 | 0.508** | 0.234** | 0.074 | 0.462*** |
| | (0.190) | (0.333) | (0.215) | (0.110) | (0.180) | (0.161) |
| Promotion | -1.777*** | -1.718*** | -1.791*** | -1.313*** | -0.902** | -1.566*** |
| | (0.254) | (0.628) | (0.265) | (0.231) | (0.421) | (0.318) |
| Treatment x promotion | -0.269 | -0.067 | -0.483** | -0.227 | 0.102 | -0.577** |
| | (0.208) | (0.356) | (0.242) | (0.158) | (0.245) | (0.226) |
| Customer age | -0.221*** | -0.094 | 0.065 | -0.100** | 0.328 | 0.019 |
| | (0.036) | (0.328) | (0.044) | (0.040) | (0.315) | (0.056) |
| Customer age² | 0.003*** | -0.001 | -0.000 | 0.001*** | -0.008 | 0.000 |
| | (0.001) | (0.007) | (0.001) | (0.001) | (0.007) | (0.001) |
| Customer tenure | -0.008*** | -0.010*** | -0.009*** | -0.005*** | -0.007** | -0.006*** |
| | (0.001) | (0.002) | (0.001) | (0.002) | (0.003) | (0.002) |
| Customer tenure² | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000* | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.048 | 0.028 | 0.059 | 0.075 | 0.040 | 0.111 |
| | (0.072) | (0.109) | (0.094) | (0.089) | (0.099) | (0.136) |
| Retirement product | 0.178 | 0.101 | 0.270 | -0.106 | -0.209 | -0.027 |
| | (0.141) | (0.215) | (0.208) | (0.151) | (0.178) | (0.223) |
| Home loan product | -0.617*** | -0.754*** | -0.575*** | -0.550*** | -0.439 | -0.558*** |
| | (0.106) | (0.189) | (0.125) | (0.170) | (0.281) | (0.193) |
| Personal loan product | 0.758*** | 0.389** | 1.022*** | 0.249** | 0.001 | 0.493*** |
| | (0.109) | (0.159) | (0.162) | (0.111) | (0.166) | (0.164) |
| Savings product | -0.201** | -0.114 | -0.221** | -0.139 | -0.078 | -0.182 |
| | (0.081) | (0.167) | (0.091) | (0.095) | (0.166) | (0.117) |
| Term deposit product | -0.569*** | -0.547 | -0.565*** | -0.619*** | -0.221 | -0.776*** |
| | (0.166) | (0.358) | (0.201) | (0.236) | (0.488) | (0.288) |
| Transaction product | 1.581*** | 2.239*** | 1.284*** | 1.326*** | 1.852*** | 1.034*** |
| | (0.125) | (0.261) | (0.161) | (0.224) | (0.432) | (0.283) |
| Home insurance policy | 0.106 | 0.527 | 0.017 | -0.046 | 0.736 | -0.186 |
| | (0.151) | (0.380) | (0.159) | (0.173) | (0.662) | (0.204) |
| Motor insurance policy | 0.101 | 0.597 | -0.029 | 0.031 | 0.738 | -0.120 |
| | (0.194) | (0.494) | (0.221) | (0.196) | (0.559) | (0.231) |
| Constant | 7.519*** | 6.340 | 1.907** | 3.945*** | -0.468 | 1.241 |
| | (0.683) | (4.068) | (0.893) | (0.775) | (3.809) | (1.229) |
| Observations | 16,487 | 8,173 | 8,244 | 6,565 | 3,373 | 3,192 |
| Sample | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| Pseudo R2 | 0.0823 | 0.0884 | 0.0965 | 0.0528 | 0.0461 | 0.0947 |
| Pred(Y): Non-Promotion: Control | 98.11% | 98.38% | 97.75% | 93.34% | 93.11% | 93.17% |
| Pred(Y): Non-Promotion: Treatment | 98.56% | 98.50% | 98.62% | 94.64% | 93.56% | 95.53% |
| Pred(Y): Promotion: Control | 90.33% | 92.22% | 88.66% | 79.73% | 84.93% | 75.75% |
| Pred(Y): Promotion: Treatment | 90.43% | 92.34% | 88.89% | 79.84% | 86.98% | 73.78% |

**Table A15:** Effects of the treatment on customer retention rates including customers who viewed inconsistent benefits information for a type of the low fee card. Effects of the treatment on customer retention rates. Columns (1) – (3) and columns (4) – (6) show the results of cross-sectional data regression models of all, younger, and older customers for retention six and nine months after card activation, respectively. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that transparency can lead to increased customer retention, which is consistent with H4, that the effects are strongest for more-experienced customers, which is consistent with H6, and that the effects are dampened during the promotion, which is consistent with H8.

## Customer Engagement Robustness Analysis: Uncontrolled Analyses

In the primary analysis presented in the manuscript, we included a series of customer-level control variables that were collected prior to the customer's exposure to the experimental manipulation. **Tables A16-17** demonstrate that the effects are substantively similar if these controls are excluded.

| | (1) Ln(Spend) | (2) Ln(Spend) | (3) Ln(Spend) | (4) Ln(Spend) | (5) Ln(Spend) | (6) Ln(Spend) |
|---|---|---|---|---|---|---|
| Treatment | 0.066 | 0.004 | 0.136* | 0.082 | 0.001 | 0.172*** |
| | (0.053) | (0.086) | (0.069) | (0.056) | (0.092) | (0.062) |
| Promotion | 0.194*** | 0.258*** | 0.113 | 0.091 | 0.189** | -0.021 |
| | (0.057) | (0.075) | (0.073) | (0.061) | (0.080) | (0.082) |
| Treatment x promotion | -0.076 | -0.008 | -0.150* | -0.094 | 0.001 | -0.197** |
| | (0.062) | (0.103) | (0.084) | (0.065) | (0.110) | (0.080) |
| Constant | 5.096*** | 4.927*** | 5.278*** | 4.968*** | 4.815*** | 5.133*** |
| | (0.047) | (0.063) | (0.059) | (0.047) | (0.066) | (0.061) |
| Observations | 121,714 | 61,231 | 60,483 | 126,696 | 63,237 | 63,459 |
| Customers | 15,947 | 7,932 | 8,015 | 15,947 | 7,932 | 8,015 |
| Data treatment for closed accounts | Missing | Missing | Missing | Zero | Zero | Zero |
| Sample | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| R-squared | 0.0015 | 0.0038 | 0.0004 | 0.0002 | 0.0018 | 0.001 |
| Pred(Y): Non-Promotion: Control | $163.37 | $137.94 | $195.96 | $143.78 | $123.32 | $169.51 |
| Pred(Y): Non-Promotion: Treatment | $174.48 | $138.49 | $224.51 | $156.04 | $123.49 | $201.42 |
| Pred(Y): Promotion: Control | $198.32 | $178.47 | $219.33 | $157.45 | $149.01 | $165.92 |
| Pred(Y): Promotion: Treatment | $196.35 | $177.81 | $216.18 | $155.62 | $149.32 | $161.94 |

**Table A16:** Effects of the treatment on monthly spend excluding control variables. Columns (1) – (3) show the results of panel data regression models for all customers as well as younger and older customers, with spend values for months after cancellation set to missing. Columns (4) – (6) show the same specifications with spend values for months after cancellation set to zero. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that customers exposed to the treatment spent more on their cards, that these effects were stronger for more-experienced customers, and that these effects were attenuated by the promotion, which is consistent with H3, H5, and H7.

| | (1) Pr(Retain6) | (2) Pr(Retain6) | (3) Pr(Retain6) | (4) Pr(Retain9) | (5) Pr(Retain9) | (6) Pr(Retain9) |
|---|---|---|---|---|---|---|
| Treatment | 0.267 | 0.094 | 0.451* | 0.248** | 0.103 | 0.416*** |
| | (0.191) | (0.320) | (0.232) | (0.114) | (0.183) | (0.157) |
| Promotion | -1.009*** | -0.846*** | -1.116*** | -0.713*** | -0.670*** | -0.751*** |
| | (0.128) | (0.198) | (0.174) | (0.123) | (0.153) | (0.192) |
| Treatment x promotion | -0.276 | -0.084 | -0.475* | -0.251 | 0.142 | -0.594*** |
| | (0.209) | (0.344) | (0.254) | (0.166) | (0.251) | (0.227) |
| Constant | 3.485*** | 3.586*** | 3.385*** | 2.482*** | 2.525*** | 2.439*** |
| | (0.114) | (0.177) | (0.160) | (0.080) | (0.120) | (0.128) |
| Observations | 15,947 | 7,932 | 8,015 | 6,316 | 3,231 | 3,085 |
| Customers | All | Age≤28 | Age>28 | All | Age≤28 | Age>28 |
| Pseudo R2 | 0.0221 | 0.0144 | 0.0284 | 0.0241 | 0.0129 | 0.0395 |
| Pred(Y): Non-Promotion: Control | 97.02% | 97.30% | 96.72% | 92.29% | 92.59% | 91.97% |
| Pred(Y): Non-Promotion: Treatment | 97.71% | 97.54% | 97.89% | 93.88% | 93.26% | 94.55% |
| Pred(Y): Promotion: Control | 92.24% | 93.93% | 90.63% | 85.44% | 86.47% | 84.39% |
| Pred(Y): Promotion: Treatment | 92.18% | 93.99% | 90.43% | 85.41% | 89.08% | 81.90% |

**Table A17:** Effects of the treatment on customer retention rates excluding control variables. Effects of the treatment on customer retention rates. Columns (1) – (3) and columns (4) – (6) show the results of cross-sectional data regression models of all, younger, and older customers for retention six and nine months after card activation, respectively. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively. Marginal effect estimates are provided for each condition in the bottom section of the table. The results indicate that transparency can lead to increased customer retention, which is consistent with H4, that the effects are strongest for more-experienced customers, which is consistent with H6, and that the effects are dampened during the promotion, which is consistent with H8.

## Customer Engagement Robustness Analysis: Truncated Spend Analysis

In the primary spend analysis, presented in the manuscript, we present panel analyses based on the full sample of data. In **Table A18**, below, we present the same analysis, regressed on the first six months of observations for each customer. This approach enables us to analyze a sample of equivalent length from every customer in our study. We note that the results are substantively similar to the results presented in the primary analysis. The sole difference is that with the diminished power afforded by having fewer observations, we see an insignificant main effect of spend on the full sample, but the effect emerges for the high-experience subgroup analysis.

| | (1) Ln(Spend) | (2) Ln(Spend) | (3) Ln(Spend) | (4) Ln(Spend) | (5) Ln(Spend) | (6) Ln(Spend) |
|---|---|---|---|---|---|---|
| Treatment | 0.071 | 0.035 | 0.120* | 0.076 | 0.030 | 0.137** |
| | (0.050) | (0.073) | (0.067) | (0.052) | (0.077) | (0.061) |
| Promotion | -0.159 | -0.005 | -0.316* | -0.361*** | -0.132 | -0.592*** |
| | (0.115) | (0.132) | (0.167) | (0.122) | (0.126) | (0.177) |
| Treatment x promotion | -0.088 | -0.040 | -0.158* | -0.093 | -0.032 | -0.176** |
| | (0.059) | (0.093) | (0.081) | (0.061) | (0.097) | (0.078) |
| Customer age | 0.071*** | 0.099 | 0.013 | 0.050*** | 0.136 | 0.027 |
| | (0.008) | (0.117) | (0.017) | (0.009) | (0.124) | (0.018) |
| Customer age² | -0.001*** | -0.001 | -0.000 | -0.001*** | -0.002 | -0.000* |
| | (0.000) | (0.002) | (0.000) | (0.000) | (0.003) | (0.000) |
| Customer tenure | 0.002*** | 0.005*** | -0.001 | 0.000 | 0.004*** | -0.002*** |
| | (0.000) | (0.001) | (0.001) | (0.000) | (0.001) | (0.001) |
| Customer tenure² | -0.000 | -0.000*** | 0.000* | 0.000 | -0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.070** | 0.112*** | 0.027 | 0.093*** | 0.115*** | 0.067* |
| | (0.032) | (0.043) | (0.040) | (0.032) | (0.042) | (0.041) |
| Retirement product | 0.082 | 0.033 | 0.195** | 0.120** | 0.049 | 0.273*** |
| | (0.056) | (0.072) | (0.084) | (0.056) | (0.073) | (0.090) |
| Home loan product | 0.475*** | 0.715*** | 0.432*** | 0.343*** | 0.514*** | 0.323*** |
| | (0.048) | (0.124) | (0.059) | (0.059) | (0.136) | (0.067) |
| Personal loan product | -0.221*** | -0.341*** | -0.123** | -0.120*** | -0.279*** | 0.010 |
| | (0.038) | (0.062) | (0.049) | (0.039) | (0.062) | (0.052) |
| Savings product | 0.249*** | 0.346*** | 0.173*** | 0.226*** | 0.318*** | 0.155*** |
| | (0.039) | (0.058) | (0.048) | (0.039) | (0.059) | (0.048) |
| Term deposit product | 0.137 | 0.320** | 0.031 | -0.049 | 0.189 | -0.172 |
| | (0.114) | (0.145) | (0.150) | (0.125) | (0.165) | (0.160) |
| Transaction product | 0.503*** | 0.249 | 0.586*** | 0.946*** | 0.954*** | 0.928*** |
| | (0.100) | (0.218) | (0.115) | (0.095) | (0.214) | (0.116) |
| Home insurance policy | -0.057 | 0.041 | -0.076 | -0.067 | 0.124 | -0.116 |
| | (0.068) | (0.150) | (0.080) | (0.077) | (0.158) | (0.088) |
| Motor insurance policy | 0.072 | 0.152 | 0.066 | 0.054 | 0.147 | 0.046 |
| | (0.097) | (0.153) | (0.117) | (0.101) | (0.159) | (0.117) |
| Constant | 3.540*** | 2.923** | 5.099*** | 3.592*** | 2.062 | 4.530*** |
| | (0.214) | (1.449) | (0.398) | (0.234) | (1.507) | (0.415) |
| Observations | 90,208 | 45,165 | 45,043 | 92,696 | 46,098 | 46,598 |
| Data treatment for closed accounts | Missing | Missing | Missing | Zero | Zero | Zero |
| Customers | 15,942 | 7,932 | 8,010 | 15,942 | 7,932 | 8,010 |

**Table A18:** Spend analysis, restricted to the first six months of usage. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. Columns (1) – (3) show the results of panel data regression models for all customers as well as younger and older customers, with spend values for months after cancellation set to missing. Columns (4) – (6) show the same specifications with spend values for months after cancellation set to zero. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively.

## Testing Experience as a Potential Moderator of the Effects of Tradeoff Transparency

In the primary analyses presented in the manuscript, we demonstrate positive main effects of tradeoff transparency on customer spending and retention, and we note that the effects are particularly pronounced among more-experienced customers. However, a different, yet related, question is whether experience moderates the relationships between tradeoff transparency and these measures of customer engagement.

To test these relationships, we create a moderator variable, called $OVER28_i$, which denotes a more-experienced customer. As noted earlier in the Appendix, the probability a 28-year old prospective customer had a credit card at the time of this study was 58.5%. Prospective customers who were older than 28 years old were more likely to have a credit card, and prospective customers who were younger were less likely to have one.

In the first specification presented below, $\gamma_5$ captures the degree to which tradeoff transparency has a differential effect on the spending of more-experienced customers during the non-promotion period:

$$\ln\left(SPEND_{it}\right) = f\begin{pmatrix} \gamma_0 + \gamma_1 TREAT_i + \gamma_2 PROMO_i + \gamma_3 TREAT_i \times PROMO_i + \gamma_4 OVER28_i + \\ \gamma_5 TREAT_i \times OVER28_i + \gamma_6 PROMO_i \times OVER28_i + \\ \gamma_7 TREAT_i \times OVER28_i \times PROMO_i + \gamma_8 AGE_i + \gamma_9 AGE_i^2 + \gamma_{10} TENURE_i + \\ \gamma_{11} TENURE_i^2 + \gamma_{12} GENDER_i + \gamma_{13} HLOAN_i + \gamma_{14} PLOAN_i + \gamma_{15} TRANS_i + \\ \gamma_{16} SAV_i + \gamma_{17} HINS_i + \gamma_{18} VINS_i + \gamma_{19} RET_i + X_i + \epsilon_{it} \end{pmatrix} \quad (4)$$

In the next specification presented below, $\delta_5$ captures the degree to which tradeoff transparency has a differential effect on the retention of more-experienced customers during the non-promotion period:

$$\Pr\left(RETAIN_i\right) = f\begin{pmatrix} \delta_0 + \delta_1 TREAT_i + \delta_2 PROMO_i + \delta_3 TREAT_i \times PROMO_i + \delta_4 OVER28_i + \\ \delta_5 TREAT_i \times OVER28_i + \delta_6 PROMO_i \times OVER28_i + \\ \delta_7 TREAT_i \times OVER28_i \times PROMO_i + \delta_8 AGE_i + \delta_9 AGE_i^2 + \delta_{10} TENURE_i + \\ \delta_{11} TENURE_i^2 + \delta_{12} GENDER_i + \delta_{13} HLOAN_i + \delta_{14} PLOAN_i + \delta_{15} TRANS_i + \\ \delta_{16} SAV_i + \delta_{17} HINS_i + \delta_{18} VINS_i + \delta_{19} RET_i + X_i + \epsilon_i \end{pmatrix} \quad (5)$$

We present the results of these analyses in **Table A19.** The results demonstrate that experience does not moderate the relationship between tradeoff transparency and spending, neither when spending is set to missing during months following cancellation ($\gamma$=0.117, $p$=0.31), or when spending is set to 0 during months following cancellation ($\gamma$=0.158, $p$=0.16). Similarly, the results reveal that experience does not moderate the relationship between tradeoff transparency, neither after six months ($\delta$=0.324, $p$=0.44), nor after nine months ($\delta$=0.328, $p$=0.21).

| | (1) Ln(Spend) | (2) Ln(Spend) | (3) Pr(Retain6) | (4) Pr(Retain9) |
|---|---|---|---|---|
| Treatment | 0.025 | 0.020 | 0.105 | 0.096 |
| | (0.085) | (0.092) | (0.327) | (0.180) |
| Promo | -0.035 | -0.271** | -1.573*** | -1.301*** |
| | (0.114) | (0.124) | (0.295) | (0.252) |
| Treatment x Promo | -0.034 | -0.021 | -0.065 | 0.165 |
| | (0.103) | (0.111) | (0.352) | (0.254) |
| Over 28 | 0.222** | 0.196** | 0.054 | -0.109 |
| | (0.087) | (0.094) | (0.274) | (0.229) |
| Treatment x Over 28 | 0.117 | 0.158 | 0.324 | 0.328 |
| | (0.115) | (0.113) | (0.422) | (0.262) |
| Promo x Over 28 | -0.138 | -0.197* | -0.271 | -0.090 |
| | (0.095) | (0.103) | (0.269) | (0.253) |
| Treatment x Promo x Over 28 | -0.134 | -0.186 | -0.353 | -0.734** |
| | (0.142) | (0.144) | (0.451) | (0.354) |
| Customer age | 0.053*** | 0.036*** | -0.186*** | -0.065 |
| | (0.012) | (0.012) | (0.050) | (0.045) |
| Customer age² | -0.001*** | -0.000*** | 0.003*** | 0.001* |
| | (0.000) | (0.000) | (0.001) | (0.001) |
| Customer tenure | 0.002*** | 0.000 | -0.009*** | -0.005*** |
| | (0.000) | (0.001) | (0.001) | (0.002) |
| Customer tenure² | -0.000 | 0.000 | 0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| Male indicator | 0.052 | 0.073** | 0.045 | 0.075 |
| | (0.034) | (0.034) | (0.074) | (0.090) |
| Retirement product | 0.067 | 0.104* | 0.241 | -0.121 |
| | (0.058) | (0.058) | (0.155) | (0.154) |
| Home loan product | 0.509*** | 0.349*** | -0.595*** | -0.565*** |
| | (0.049) | (0.062) | (0.105) | (0.166) |
| Personal loan product | -0.324*** | -0.209*** | 0.767*** | 0.240** |
| | (0.040) | (0.043) | (0.110) | (0.105) |
| Savings product | 0.279*** | 0.249*** | -0.187** | -0.124 |
| | (0.040) | (0.040) | (0.080) | (0.096) |
| Term deposit product | 0.192 | -0.029 | -0.614*** | -0.650*** |
| | (0.119) | (0.132) | (0.167) | (0.230) |
| Transaction product | 0.496*** | 1.041*** | 1.606*** | 1.402*** |
| | (0.105) | (0.101) | (0.126) | (0.225) |
| Home insurance policy | -0.039 | -0.046 | 0.080 | -0.084 |
| | (0.076) | (0.085) | (0.154) | (0.169) |
| Motor insurance policy | 0.043 | 0.036 | 0.078 | 0.029 |
| | (0.105) | (0.108) | (0.197) | (0.199) |
| Constant | 3.625*** | 3.554*** | 6.810*** | 3.238*** |
| | (0.250) | (0.265) | (0.911) | (0.818) |
| Observations | 121,679 | 126,658 | 15,942 | 6,314 |
| Customers | 15,942 | 15,942 | 15,942 | 6,314 |
| Data treatment for closed accounts | Missing | 0 | NA | NA |
| R-squared (Pseudo R-squared) | 0.0174 | 0.0143 | 0.0854 | 0.0580 |

**Table A19:** Experience does not moderate the relationship between tradeoff transparency and spending, nor between tradeoff transparency and retention. All models include indicator variables for the week the credit card was activated, and are estimated with robust standard errors clustered by activation date. *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively.

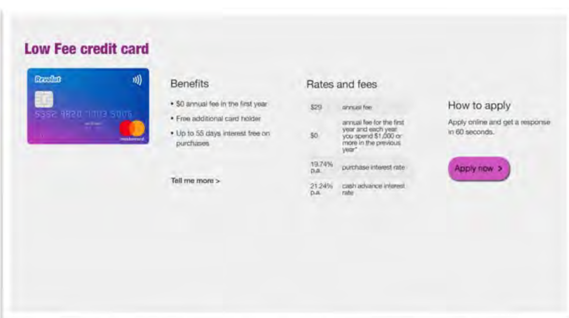
## Testing Trust as a Potential Mechanism Underlying the Effects of Tradeoff Transparency

We recruited 201 participants (62.2% male, *M*=39.18 years old) on the Amazon Mechanical Turk platform. After completing the informed consent process, participants were shown the credit card detail pages from two competing banks – Purple Bank and Yellow Bank – both of which offered a "Low Fee Credit Card." Participants were instructed to carefully review the two webpages, as if they were considering applying for a credit card from these two banks.

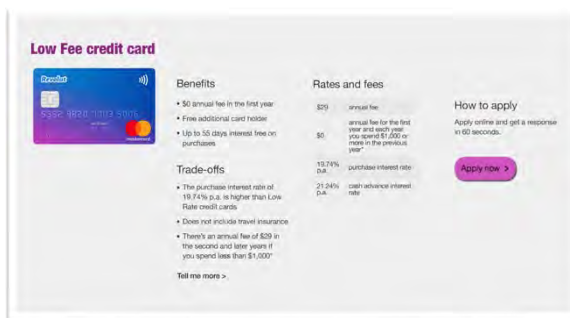
The details of the credit card offerings were identical to the Low Fee Credit Card offered by CBA in our field experiment. For each participant, we randomly assigned which of the two hypothetical banks offered tradeoff transparency, and we counterbalanced the order of each bank's presentation (**Figure A6**). After reviewing the two banks' webpages, every participant was asked a pair of forced choice questions – "Which bank is more trustworthy?" and "If you were given the opportunity, from which bank would you be more likely to apply for a credit card?"


Yellow Bank – Control (benefits only)


Yellow Bank – Tradeoff transparency




Purple Bank – Control (benefits only)




Purple Bank – Tradeoff transparency

**Figure A6:** Screenshots of Credit Card Websites Presented in the Online Experiment (n=201).

We test the impact of tradeoff transparency on trust and willingness to apply for a credit card using one-sided binomial tests of the null hypothesis that tradeoff transparency has no effect. If participant responses were the product of random chance, we would expect to see participants choose the bank leveraging tradeoff transparency as being more trustworthy and being their preferred bank half of the time. However, 71% of participants reported that the bank that exhibited tradeoff transparency was more trustworthy ($p$<0.01, one-sided). Moreover, 69% reported a higher intention to apply for a credit card from the bank that provided tradeoff transparency ($p$<0.01, one-sided). Finally, consistent with prior research, there was a high correlation between trust and preference, such that the bank that was identified as being more trustworthy was also selected disproportionately as the participant's choice as the bank from which they would be more likely to apply for a credit card ($\rho$=0.794, $p$<0.01) (**Figure A7**).

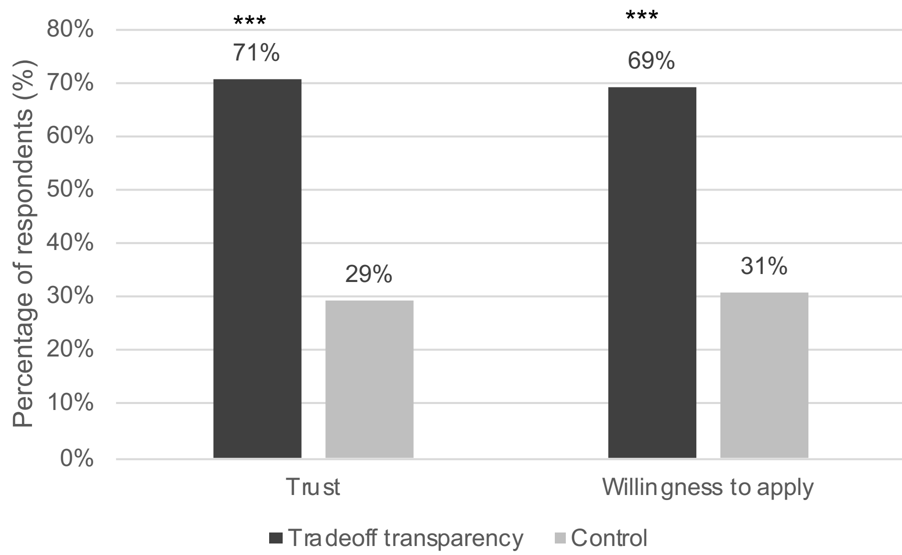


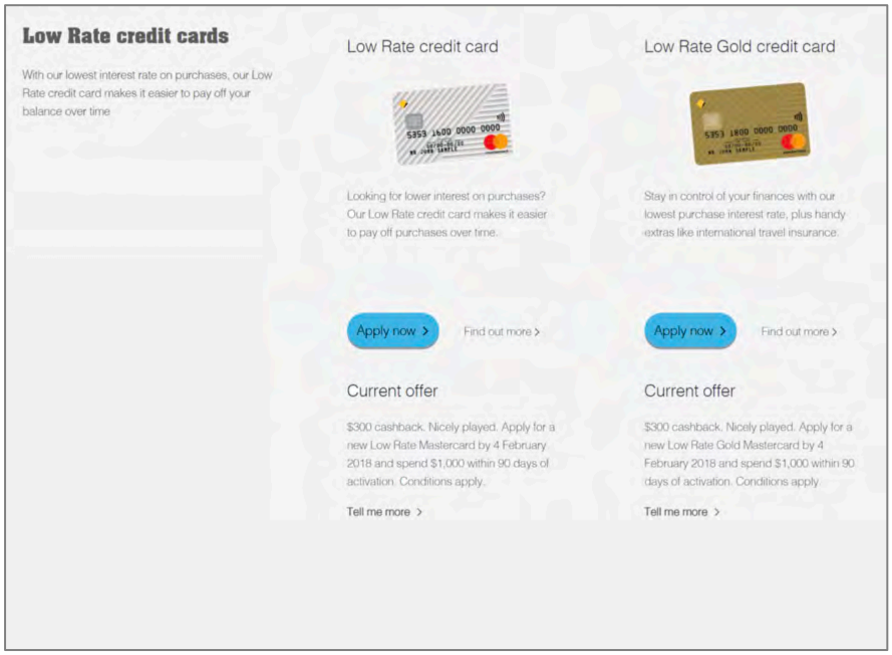

**Figure A7:** Effects of Tradeoff Transparency on Trust and Willingness to Apply for a Credit Card (n=201). *, **, and *** signify significance at the 10%, 5%, and 1% levels respectively, drawn from one-sided binomial tests of the null hypothesis that tradeoff transparency has no effect on trust and willingness to apply for a credit card. The results provide evidence that tradeoff transparency can engender trust and brand preference in customers. Moreover, consistent with prior literature, we note a positive and statistically significant correlation between trust and willingness to apply ($\rho$=0.794, $p$<0.01).

## Additional Screenshots from the Treatment and Control Conditions

**Figures A8-A9** present additional screenshots from the treatment and control condition of the primary experiment. The experimental manipulation involved more than 50 blocks of content on more than 20 pages of CBA's public-facing and secure online banking websites, such that on every credit card marketing webpage where the features and benefits of a credit card were described, so too were its tradeoffs for customers randomly assigned to the treatment. As shown below, the nature of the tradeoffs manipulation varied on different types of pages, to complement the ways in which benefits information was presented throughout the site, though the nature of these manipulations were consistent across credit card types.

**Control Condition**

Only the benefits of each credit card are presented on this page




**Treatment Condition**

The benefits and the tradeoffs, in terms of the annual fees, are presented

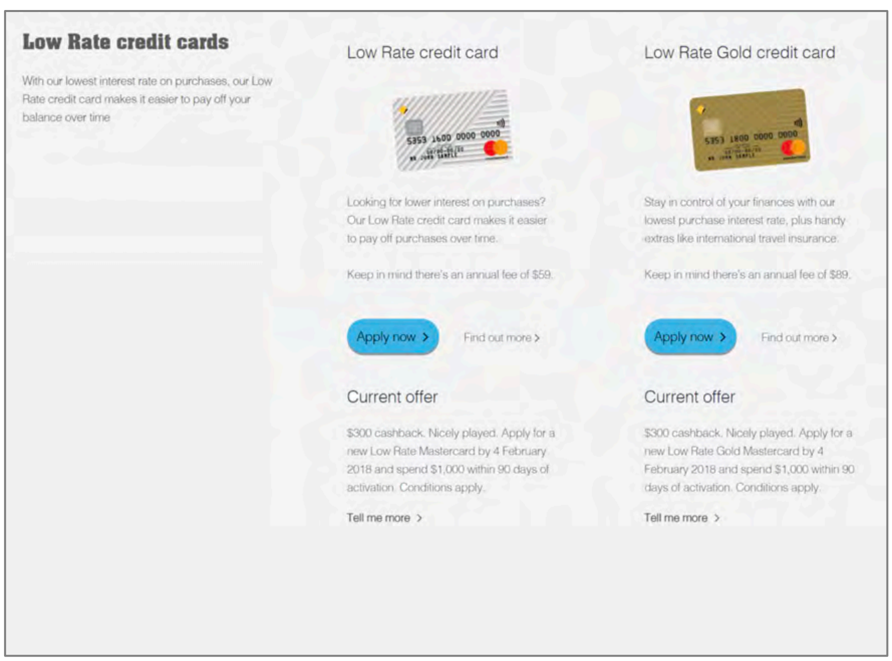
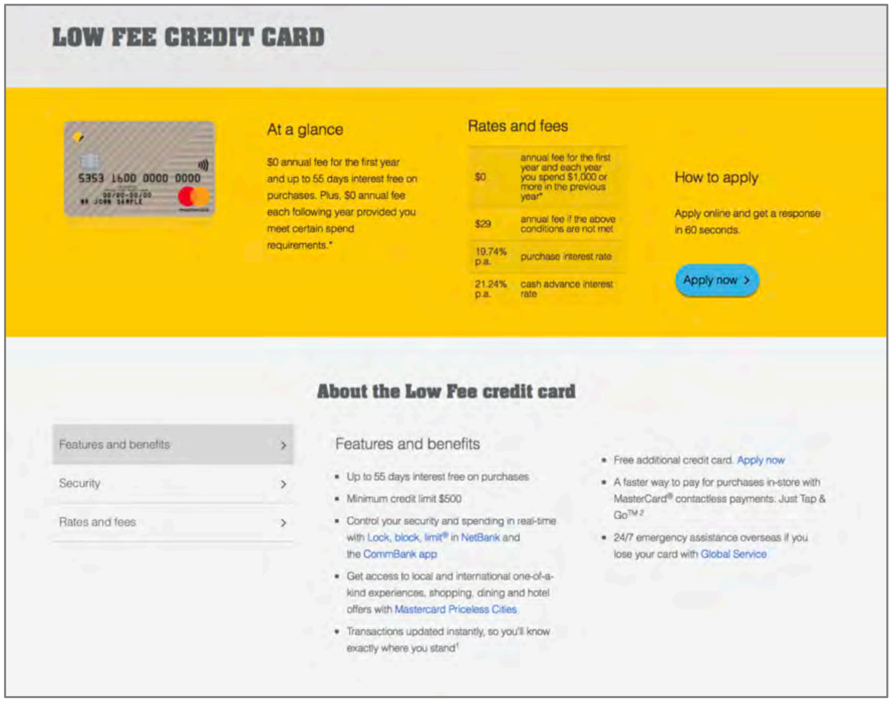



**Figure A8.** Additional example credit card marketing pages in the control and treatment conditions for a page presenting the two cards in the low rate credit card family.

## Control Condition

Only the benefits of each credit card are presented in the product detail pages




## Treatment Condition

In the treatment condition, a section was added beneath the Features and benefits tab, which presents the card's tradeoffs

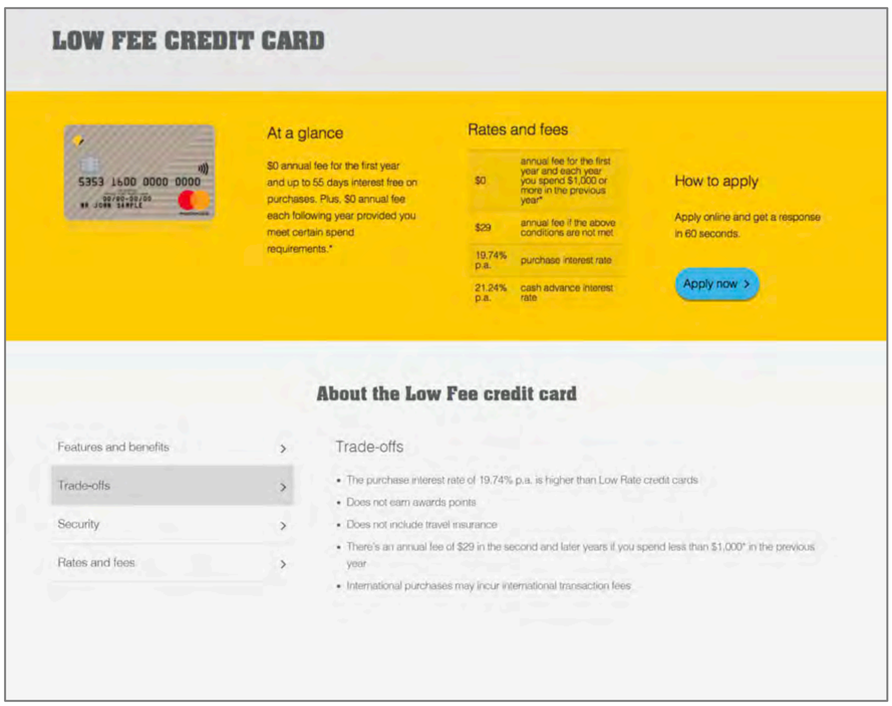



**Figure A9.** Additional example credit card product detail pages in the control and treatment conditions for a credit card detail page. For the low fee credit card detail page, in the treatment condition, an entire section was added to the left panel, including the tradeoffs, in addition to the features and benefits.